Face selection for slowly switching Markov chains

Arjun Pemmasani
Department of Mathematics, Harvey Mudd College
Claremont, CA 91711, USA
apemmasani@hmc.edu

Abstract

We consider a Markov chain on a finite state space that rarely switches states, and its occupation counts, the numbers of steps that it spends in each state up to time N. The most likely vector of counts uses only some set of states, and we find which set. Two effects compete. The chain must never switch to a state outside the set, and this costs less when the set is hard to leave. But a larger set has more count vectors to share its probability, and each single count vector loses a factor N for each extra state. Let the switch probabilities be T/N times fixed rates. When T is a fixed multiple of $\log N$, both effects are powers of N, and the winning set minimizes the sum of the two exponents. When the chain can pass between any two states of a set without leaving it, a local limit theorem inside the set gives the constant in front. With it we locate the value of T at which two such sets are equally likely, up to an error that tends to zero, and we find the next term. A chain that cycles through three states shows that the answer can change without this condition. We also apply the rule to sticky hidden Markov models, given the kernel of the switches and a common exit probability.

Keywords: occupation times of Markov chains; mode; principal Dirichlet eigenvalue; quasi-stationary distribution; local limit theorem; persistent random walk; correlated random walk; hidden Markov model.
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1 Introduction

Let $X_1, X_2, \dots$ be a Markov chain that rarely switches between its d states. At each step it moves from state i to state $j \neq i$ with probability $q_{ij}T/N$, so that it makes of order T switches in N steps. Its *composition* is the vector $n = (n_1, \dots, n_d)$ of occupation counts, where n_i is the number of times $t \leq N$ with $X_t = i$. The support of n is a set F of states, which we call a *face*, since it indexes a face of the simplex of visit fractions. We ask which face carries the most likely composition, and at which T the mode moves to another face. With $T = \tau \log N$ and N large, the mode sits on a face that minimizes $\tau\lambda_F + |F| - 1$ among the possible supports, where λ_F is the rate at which the chain leaves F (Theorem 3.3). When the chain can pass between any two states of a face without leaving it, a local limit theorem inside the face gives the constant in front, and with it the value of T at which two such faces of different sizes are equally likely, up to an error that tends to zero (Theorems 4.4 and 4.7). The rule and its second-order form recover the crossings of the two-state walk [2] and of the complete graph [3], with their constants (Corollaries 4.10 and 4.11).

Occupation measures of finite Markov chains have a long history and remain an active subject. The exit rate λ_F is the principal Dirichlet eigenvalue of $-Q$ on F, where Q is the generator with

the off-diagonal entries q_{ij}, and it has the variational formula of Donsker and Varadhan [8]. A chain conditioned to stay in F has quasi-stationary and quasi-ergodic laws [9, 10, 11]. Recent work treats large deviations of the occupation measure inside a set [6] and Γ-expansions of level-two rate functions [20, 21], and the joint density of the local times of a continuous-time chain has explicit formulas [23, 24]. We ask a different question, namely where the single most likely composition lies, in a regime where the chain makes only of order $\log N$ switches. Section 1.3 compares our results with this work.

The simplest case has two states, with $q_{12} = q_{21} = 1$ and a uniform start. The chain is then the sequence of directions of the persistent random walk on $\mathbb{Z}$, which repeats its previous step with probability $p = 1 - T/N$, and the composition records its numbers of steps to the right and to the left. The two faces of one state are the endpoints $(N, 0)$ and $(0, N)$, each reached only by the path that never switches, and the face of both states holds the interior compositions. The question is then when the middle composition is as likely as an endpoint. For the two-state chain this question is Problem 131 of Kagey's Open Problem Collection [1], posed for a Galton board. The two are equally likely when T is close to $\log N$. More precisely, the crossing value z_N of T satisfies $z_N + \frac{1}{2}\log z_N = \log N + \frac{1}{2}\log\frac{\pi}{8} + o(1)$ [2].

1.1 Face selection

Put $T = \tau \log N$. How likely the best composition with support F is depends on two effects.

- *The cost of staying.* The chain must stay inside F for all N steps. If the start gives F positive mass and the chain can move between any two states of F without leaving it, this has probability about $e^{-T\lambda_F} = N^{-\tau\lambda_F}$. A larger face is harder to leave, so the chain stays on it more easily.

- *The cost of spreading.* The mass of F is shared among the compositions with support F, and there are about $N^{|F|-1}$ of them, one factor N for each free coordinate. Up to powers of $\log N$, the best single composition gets a share $N^{-(|F|-1)}$.

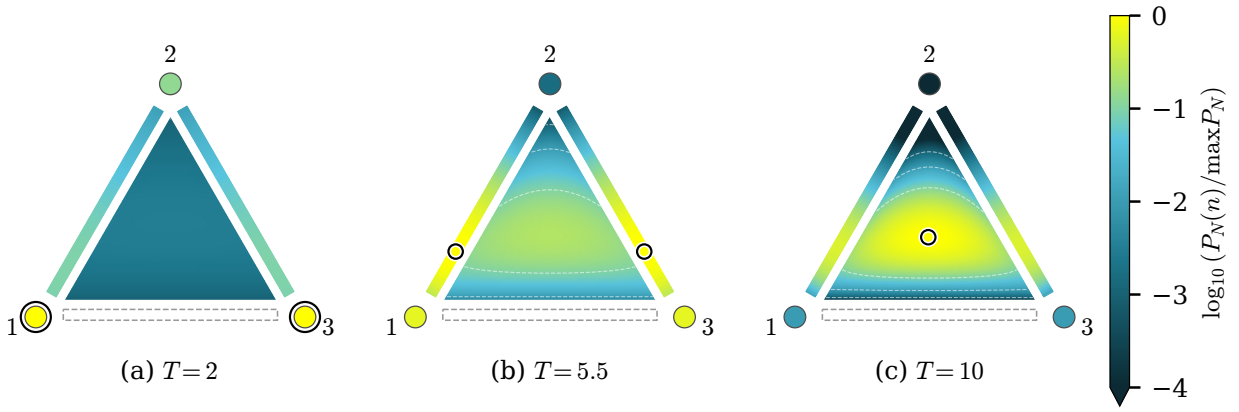
Call a face admissible if some path of positive probability has exactly this support, write M_F for the largest probability of a composition with support F, and put

$$\Phi_\tau(F) = \tau\lambda_F + |F| - 1.$$

Theorem A (first-order face selection)**.** *Fix the rates q_{ij} and the law of X_1, and let $T = \tau \log N$ with $\tau > 0$ fixed. Every admissible face F has $M_F = N^{-\Phi_\tau(F)+o(1)}$. For large N the support of every most likely composition is an admissible face that minimizes Φ_τ among the admissible faces. If the chain can pass between any two states of each such minimizing face without leaving it, then the visit fractions of every most likely composition are within $O(1/\log N)$ of the quasi-ergodic law of its support (Definition 2.1).*

Theorem A is Theorem 3.3(ii) to (iv). In words, a sticky walk prefers a well-separated community, a set of states that is hard to leave for its size. As τ grows staying counts for more, and the mode moves to larger faces. For two states an endpoint has $\Phi_\tau = \tau$ and the interior has $\Phi_\tau = 1$, so the middle takes over at $\tau = 1$, which is the first-order form of the crossing z_N above. Section 3.1 works out four states on a cycle by hand, and Theorem 3.4 reads the winners off the convex hull of the points $(\lambda_F, |F| - 1)$.

Figure 1 shows the rule at work on three states in a row.



(a) $T = 2$ (b) $T = 5.5$ (c) $T = 10$

Figure 1: The exact law of the composition for the path $1 - 2 - 3$ (rate 1 on each edge, uniform start, $N = 90$), on the simplex of visit fractions. The faces are drawn apart. The triangle holds the compositions that use all three states, each band holds the compositions that use two states, and each disk is the composition that spends all N steps in one state. The color is the probability of a composition relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. (a) For small T the mode is at an end of the path. (b) Then it sits on a band, that is, on a pair of neighbors. (c) Then it is inside. The dashed band is the pair $\{1, 3\}$, which the chain cannot use without visiting 2, so it has probability zero. Here λ_F is 1 for an end, $(3 - \sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1 + \sqrt{5})/2$ and at $\tau = (3 + \sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$. At $N = 90$ it moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 4.

The rule also explains a crossing of the planar persistent walk. This walk moves on the square lattice $\mathbb{Z}^2$, one step east, north, west or south at a time. It starts in a uniform direction, and at each step it keeps its direction, turns left or right, or reverses. Turns and reversals are rare. A turn to each side has probability rT/N and a reversal has probability sT/N, where $r, s \geq 0$ are fixed rates with $2r + s > 0$. The corner $(N, 0)$ is reached only by the path that never switches. In a companion paper in preparation [4] we show that for every even $N \geq 4$ there is exactly one value T_N^* of T at which the walk is as likely to end at the corner as at the origin. The corner is more likely below it and the origin above it, and as $N \to \infty$ through even integers

$$\frac{T_N^*}{\log N} \to \min\Big(\frac{2}{2r + s}, \frac{1}{s}\Big). \tag{1}$$

Here $1/s = \infty$ when $s = 0$. The two values in the minimum belong to two kinds of paths to the origin, and face selection explains them. The corner is the face of the single direction east, with exit rate $2r + s$, so it has probability about $N^{-(2r+s)\tau}$. A path to the origin either stays on one axis, a face of two states with exit rate $2r$, or uses all four directions, a face with exit rate 0. On an axis the origin is a single composition, of probability about $N^{-(2r\tau+1)}$. With all four directions a single composition has probability about N^{-3}, but about $N/2$ of them end at the origin, which gives N^{-2}. This suggests that the origin beats the corner when $(2r + s)\tau > \min(2r\tau + 1, 2)$, which is (1). It is only a heuristic, since the origin is not a single composition, and the companion paper [4] proves the limit directly.

1.2 Second order and an application

First order fixes a crossing only up to a factor $1 + o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F, and the start gives F positive mass. For such faces we prove a local limit theorem inside the face (Theorem 4.4), and it gives the following result.

Theorem B (face maxima and crossings)**.** *Fix the rates q_{ij} and the law of X_1.*

(i) Let F satisfy (H)*, with $k = |F|$ states. Uniformly for $1 \le T \le N^{1/3}$,*

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F\, e^{-T\lambda_F}\big(1 + O(1/T + T^2/N)\big),$$

where $K_F > 0$ is the explicit constant of Definition 4.1 (for $k = 1$ it is the start mass of the state).

(ii) Let F and G satisfy (H)*, and suppose that F has $\Delta \ge 1$ more states than G and an exit rate smaller by $\delta > 0$. For large N the equation $M_F = M_G$ has a root T in $[1, N^{1/3}]$, and every such root satisfies*

$$\delta T + \frac{\Delta}{2}\log T = \Delta \log N - \log\frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

Part (i) is Theorem 4.6 and part (ii) is Theorem 4.7. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2}\log\log N + O(1)]$. So the coefficient $\frac{1}{2}$ of $\log\log N$ inside the bracket, which appears in the crossings of the two-state walk [2] and of the complete graph [3], holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility.

The rest of Sections 4 and 5 add the following.

- *Ties.* When several faces tie at first order, Theorem 4.8 describes the window of T in which they compete.
- *The constant.* For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 4.2), and the constants of the two-state walk and the complete graph are short corollaries (Corollaries 4.10 and 4.11). On a face that is not reversible the single count is replaced by two, a count of oriented spanning trees for the flow of the tilted chain and a count of spanning trees for the symmetrized flow (Proposition 4.15).
- *The rate.* On a face that is not reversible the rate of the visit fractions α is the mean jump rate of the chain at α, exits included, minus the mean jump rate of the tilted chain. It is also the rate of leaving the face plus the least value, over balanced flows, of the rate of Bertini, Faggionato and Gabrielli [26], and on a cycle it comes from one equation for the net flow (Proposition 4.16).
- *The next term and the role of* (H). Proposition 4.14 adds the next term, of relative order $1/T$, to the local law, to M_F and to the crossing equation. Without (H) the coefficient of $\log\log N$ can change. On a directed 3-cycle it is 1 at one crossing and 0 at the next (Proposition 4.13).
- *Examples.* Section 5 computes all the constants on the 4-cycle, shows on the complete graph K_3, as in [3, Example 4.9], that the start can decide which face wins, and lists the winners on other graphs.

Section 6 is an application, and nothing before it depends on it. It applies face selection to sticky priors for hidden Markov models. There the exit rate of a set of hidden states is one minus the Perron root of the kernel of the switches restricted to the set, and as the stickiness decreases the most likely count vector, given the kernel and a common exit probability, moves from one state to larger sets (Corollary 6.1 and Proposition 6.2). Among symmetric kernels only the uniform one jumps from one state straight to all of them (Corollary 6.3). Section 1.4 says what each section does.

1.3 Related work

This subsection places the results among earlier work on occupation measures and on exact laws of transition counts. Section 3 introduces a rate I_F for the visit fractions inside a face. It is the Donsker–Varadhan rate of the occupation measure of the chain killed on leaving F. Dupuis and Liu give this rate explicitly for reversible jump processes [5, Section 3.1], and on a finite chain their formula agrees with Proposition 3.1(c). Chen and Deng [7] study large deviations for absorbing Markov chains and use them to derive a variational representation of the principal Dirichlet eigenvalue. Guillin, Nectoux and Wu [6] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. For reversible processes they show that the exit rate of the set is the least value of the Donsker–Varadhan rate over laws on the set (their Cor. 2). In general they prove an inequality between their rate, the Donsker–Varadhan rate and the exit rate, and they conjecture equality (their (2.15)). For a finite chain with Q_F irreducible, the fact that λ_F is the least value of I_F is the variational formula of Donsker and Varadhan [8], which needs no reversibility. Proposition 3.1(a) gives a short proof in our notation. This does not address the conjecture of Guillin, Nectoux and Wu, which concerns their conditional rate. The quasi-ergodic law goes back to Darroch and Seneta [9, 10], the book of Collet, Martínez and San Martín [11] is the standard reference for quasi-stationary and quasi-ergodic laws, and Colonius and Rasmussen [12] treat the reducible case. Dietz and Sethuraman [13] study chains that switch from i to j at time n with probability $G(i,j)/n$, where every $G(i,j) > 0$. Their chains also make about $\log n$ switches, but spread log-uniformly in time, and their occupation fractions have a limit law with full support on the simplex. In [14] they prove large deviations for chains whose transition matrices converge to a reducible matrix.

Exact laws of transition counts go back to Whittle [15], Dawson and Good [16] and Goodman [17]. Polettini [18] and Carugno, Vivo and Coghi [19] derive them from the BEST theorem and the matrix-tree theorem. The expression of Carugno, Vivo and Coghi for the moment generating function is exact at every time, and they use it for large deviations, where every count grows linearly. For two states the composition has the Markov binomial law, and Dekking and Kong [28] find when it has one, two or three modes. Stadje [29, Theorems 1 and 2] gives the generating functions of the joint numbers of visits of a finite chain, and limit laws when one state becomes nearly absorbing and the chain leaves it with probability of order $1/N$ [29, Theorems 3 and 4]. The other rows of his transition matrix stay fixed, so the counts of the other states stay bounded. We group the words by runs instead (Lemma 2.3), which suits a regime with $O(\log N)$ switches. In continuous time, Brydges, van der Hofstad and König [23] give the joint density of the local times of a chain on the event that its range is a given set, and Huang, Kious, Sidoravicius and Tarrès [24] write it with Bessel functions and spanning trees for symmetric rates. Brydges, van der Hofstad and König use their formula for large deviation upper bounds. Theorem 4.4 gives the leading term of the density, with its constant, in our regime. For a fixed geometrically ergodic chain and a bounded real functional, Kontoyiannis and Meyn [30, Theorem 6.3] give exact large

deviation asymptotics of Bahadur–Rao type. Theorem 4.4 is a local form for the whole vector of counts, for a chain that is killed on leaving F and whose switch probabilities shrink with N, uniformly in T and with an explicit constant. Bertini, Gabrielli and Landim [20] and Landim [21] expand the level-two rate functions of finite chains whose rates depend on a parameter. They describe the time scales of such chains but do not ask where the mode of the composition sits. Kim and Landim [25] extend this to the joint rate function of the empirical measure and the flow [26], and they give conditions under which the rate function of the occupation times determines the chain. Proposition 4.16 is the form of the rate of [26] for a killed chain, and Remark 4.17 lists the statements of this kind that are known without killing [21, 25, 27]. The sticky HDP-HMM of Fox, Sudderth, Jordan and Willsky [22] adds a self-transition bias to the hierarchical Dirichlet process hidden Markov model.

We did not find the face-selection question in the literature.

1.4 Plan of the paper

Section 2 fixes the notation and gives the exact law of the composition as a sum over runs. Section 3 works out four states on a cycle by hand and then proves first-order face selection (Theorems 3.3 and 3.4). Section 4 proves the local limit theorem inside a face, the face maxima, the crossing equation and the tie windows. It then recovers the constants of the two-state walk and of the complete graph, shows on a directed 3-cycle what can change without (H), adds the next term (Section 4.6), gives a tree formula for the constant on every irreducible face (Section 4.7), and writes the rate through flows (Section 4.8). Section 5 computes the 4-cycle with all its constants and lists the winners on other graphs. Section 6 applies face selection to sticky priors for hidden Markov models, and Section 7 describes the computer checks and the open questions. Appendix A proves the local law, the uniform upper bound of Lemma 3.2, the first-order bounds for reducible faces and the forest formula of Lemma A.3. Appendix B proves the Laplace lemma that Appendix A uses, and the next term. Appendix C has the proofs for the table of outcomes, and Appendix D the limit of large concentration for the direct jump under the sticky prior.

We quote a few results of [2] and [3]. From [2] we use the run count, the fact that the bin polynomials of Section 2.1 grow toward the middle, the uniform Bessel bound (recalled in (2)) and the expansion of z_N, and from [3] only its expansion of the balanced compositions, for one constant in Proposition 4.14(iv).

2 Setting, notation and the exact law

This section fixes the chain and its notation, says which faces can be the support of a path (Lemma 2.2), and writes the law of the composition as a sum over runs (Lemma 2.3), which every later section uses.

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). (The scalar $q = 1 - p$ of the two-state walk appears only where we recall [2], and the scalar q of Proposition 4.9 is a common exit rate.) The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \le \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \le N : X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i,j \in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i \in F}$. Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \ne j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [34].

Definition 2.1 (faces)**.** Let F be a face. We call F *irreducible* if G_F is strongly connected. In particular every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [9, 10]. The *face maximum* is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ and r_F is positive. So the chance that the continuous-time chain started at a state of F stays in F up to time T, which is a row sum of e^{TQ_F}, lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For irreducible F the quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F, and the best composition on F sits near $N\alpha_F^*$ (Theorem 4.6).

Later sections add their own symbols where they are needed, namely the rate I_F (Section 3) and the tilted chain and the constant K_F (Definition 4.1).

The two-state walk. The persistent random walk on $\mathbb{Z}$ makes its first step left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p and switches with probability $q = 1 - p$. Its directions form the chain of Example 2.5(i). As on a Galton board, we also call a composition a *bin*, and for two states we identify the bin with its number of right steps. For $a, b \ge 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w. Each such word has probability $\frac{1}{2} p^{N-1} (q/p)^{s(w)}$ for the walk with $N = a + b$ steps. So $S_{a,b}(q/p)$ is the probability of bin a divided by the probability $\frac{1}{2} p^{N-1}$ of an endpoint. For $N \ge 2$ let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$, which belongs to a middle bin. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N, and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. By [2], z_N satisfies the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit remainder $R_N = O(1/L^2)$. We also use

the uniform Bessel bound of [2, Theorem 3.3(i)]. Put $\psi(z) = z(I_0(z) + I_1(z))$, with the modified Bessel functions I_0 and I_1. In our notation that bound says that for every $N \geq 2$ and every $z > 0$

$$\Big(1 - \frac{z(z+2)}{2N}\Big)\psi(z) \leq \frac{N}{2}\, S_N\Big(\frac{z}{N}\Big) \leq \psi(z). \tag{2}$$

Conventions. Unless we say otherwise, Q, μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds "uniformly for $1 \leq T \leq N^{1/3}$" holds for all $N \geq N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

This subsection says which faces can be the support of a path. All later statements about face maxima are for these faces.

Lemma 2.2 (admissible faces)**.** *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. We translate paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp}\mu$ and visits every state of F. Conversely, every such walk $v_1, \dots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \geq J$. (The path follows the walk and then stays in its last state, and a stay has positive probability.) So F is admissible exactly when G_F has a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. (If it did, y could be reached from C and C from y, so y would belong to C.) Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. So no edge joins two components, and the condition in (a) forces $r = 1$. □

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

This subsection writes the probability of a composition as a sum over the runs of a path, and gives its continuum form. The upper bound of Lemma 3.2 and the local law of Section 4 both start from these sums.

For $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion)**.** *Let $hq_{\max} \leq 1$ and let n have support F. Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

Proof. We group the paths with composition n by the word of their runs. The states $c_1 \cdots c_M$ of the maximal runs of a path form a word as in $W(m)$, where m_i is the number of runs in state i. The path starts at c_1, which has probability μ_{c_1}. It has $M-1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. So every path with the word c and composition n has the same probability, namely $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, h^{M-1} \prod_i (1-hq_i)^{n_i - m_i}$. The number of these paths is $\prod_i \binom{n_i-1}{m_i-1}$, because the lengths of the m_i runs in state i can be any of the $\binom{n_i-1}{m_i-1}$ ways to write n_i as an ordered sum of m_i positive parts. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

Lemma 2.3 is checked in Lean (Section 7). For $F = \{i\}$ the only path is i^N, so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{3}$$

For $t \in (0,\infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i - 1}/(m_i - 1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. The next lemma says that $e^{-\langle q,t\rangle} H(t)$ is the density of the occupation times of the continuous-time chain on the event that it stays in F. It is the continuum form of Lemma 2.3, in which the binomial coefficients become the powers $t_i^{m_i-1}/(m_i-1)!$. Lemma 4.5 below shows that the discrete law is close to it.

Lemma 2.4 (the continuum density)**.** *Let $(Y_s)_{s \geq 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\, ds$, and let E_F be the event that Y stays in F during $[0,T]$ and visits every state of F. Then $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ has the density $e^{-\langle q,t\rangle} H(t)$ in $(t_2, \ldots, t_k)$ on $\{t \in (0,\infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q,\alpha\rangle} H(T\alpha)$$

in $(\alpha_2, \ldots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

Proof. We split E_F by the sequence of states that Y visits, and for each sequence we integrate out the holding times. Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \ldots, c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$.

The density of the holding times. Fix c, and let s_j be the time that Y spends in its j-th run during $[0,T]$, so that $s_1 + \cdots + s_M = T$. For $j < M$ the j-th run is complete. Its holding time has the density $q_{c_j} e^{-q_{c_j} s_j}$, and it ends with a jump to c_{j+1}, which has probability $q_{c_j c_{j+1}}/q_{c_j}$. The last run only has to last longer than $s_M = T - \sum_{j<M} s_j$, which has probability $e^{-q_{c_M} s_M}$. We multiply these factors. On the set where $s_j > 0$ for every $j \leq M$, the event E_c has the density

$$\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,\mathcal{O}\rangle}$$

in $(s_1, \dots, s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$.

From holding times to occupation times. This density depends on the holding times only through $\mathcal{O}$, so we change variables. For each state $i \neq c_M$ we replace the last holding time in state i by $\mathcal{O}_i$. This change of variables is triangular with unit diagonal, so it preserves Lebesgue measure. The new variables are the $k-1$ occupation times $\mathcal{O}_i$ with $i \neq c_M$, and for each state i the other $m_i - 1$ holding times in state i (for $i = c_M$, those before the last run). Now fix $\mathcal{O} = t$. The $m_i - 1$ free holding times in state i are positive and their sum is less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. (For two states and the word $c = 121$ the only free variable is the first holding time, which ranges over $(0, t_1)$, and the factor is t_1.) Integrating the free holding times out, we find that $\mathbb{P}(E_c,\ \mathcal{O} \in \cdot\,)$ has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_{i \in F} t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \neq c_M$.

Summing over the words. On $\{\sum_i t_i = T\}$ any $k-1$ of the coordinates t_i give the same Lebesgue measure, so we may use $(t_2, \dots, t_k)$ for every c. Summing over the words c with given m gives $W(m)$, and summing over m gives $e^{-\langle q,t\rangle} H(t)$. The substitution $t = T\alpha$ has Jacobian T^{k-1}, which gives f_T, and $T/N = h$ gives the identity for $t = hn$. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (the four models)**.** (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is the persistent walk of [2], with $p = 1 - h$. So $T = N(1-p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints, and Lemma 2.3 is the run count of [2]. (ii) For $Q = J - dI$, with J the all-ones matrix, and uniform μ it is the model of [3], which switches to each other state with probability h. (iii) For $Q = \Pi - I$, with Π stochastic with zero diagonal, $\lambda_F = 1 - \varrho(\Pi_F)$. This is the sticky chain of Section 6. (iv) Let $d = 4$, number the directions east, north, west and south as $0, 1, 2, 3$ modulo 4, and put $q_{j,j\pm1} = r$, $q_{j,j+2} = s$ and μ uniform. Then X_t is the direction of step t of a planar walk that turns left or right with probability rh each and reverses with probability sh. This is the planar walk of [4]. For $s = 0$ and $r = 1$ this is the 4-cycle of Section 3.1. In (i), (ii) and (iv), Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

3 First-order face selection

This section proves the first-order rule of the introduction. The mode sits on a face that minimizes Φ_τ (Theorem 3.3), and the winners can be read off a convex hull (Theorem 3.4).

3.1 A worked example with four states on a cycle

Before the general theorem we work out one example by hand. Take $d = 4$ states on a cycle, with rate 1 between neighbors (indices modulo 4), and the uniform start. So at each step the chain moves to each of its two neighbors with probability $h = T/N$ and stays with probability $1 - 2h$. This is the direction of the planar walk of [4] when it turns but never reverses (Example 2.5(iv) with $r = 1$, $s = 0$). Up to rotation and reflection there are four kinds of faces, namely a vertex, an adjacent pair, an arc of three states and the whole cycle. (A face such as $\{1, 3\}$ cannot be the support of a path, since the chain cannot pass from 1 to 3 without visiting 2 or 4.) Which kind carries the most likely composition?

For each kind we ask two questions. With what probability does the chain stay in the face for all N steps? Among how many compositions is this probability shared? The best composition of the face gets about the first answer divided by the second. In this subsection T is large and T^2/N is small, the sign $\approx$ means equality up to a bounded factor, and we do not give proofs. The proofs are in Theorems 3.3 and 4.6, and Corollary 5.1 has the constants.

face F	λ_F	it stays with probability	number of compositions	best composition M_F	exponent $\Phi_\tau(F)$
vertex	2	e^{-2T}	1	e^{-2T}	2τ
adjacent pair	1	e^{-T}	$N/\sqrt{T}$	$N^{-1}T^{1/2}e^{-T}$	$\tau+1$
arc of three	$2-\sqrt{2}$	$e^{-(2-\sqrt{2})T}$	$(N/\sqrt{T})^2$	$N^{-2}\,T\,e^{-(2-\sqrt{2})T}$	$(2-\sqrt{2})\tau+2$
whole cycle	0	1	$(N/\sqrt{T})^3$	$N^{-3}T^{3/2}$	3

Table 1: The four kinds of faces of the 4-cycle with rate 1 between neighbors and the uniform start, up to bounded factors, for T large and T^2/N small. The last column is the exponent of N when $T=\tau\log N$, up to powers of $\log N$.

A vertex, say $\{1\}$. The chain stays at state 1 only if it never switches, which has probability $(1-2h)^{N-1}\approx e^{-2T}$. So the exit rate is $\lambda_{\{1\}}=2$. The only composition is $(N,0,0,0)$, so nothing is shared, and $M_{\{1\}}=\frac{1}{4}(1-2h)^{N-1}\approx e^{-2T}$.

An adjacent pair, say $\{1,2\}$. The chain leaves the pair by the switches $1\to4$ and $2\to3$, while the switches between 1 and 2 keep it inside. So at each step it leaves with probability h, whichever of the two states it is in, and it stays in the pair with probability $(1-h)^{N-1}\approx e^{-T}$. The exit rate is $\lambda_{\{1,2\}}=1$, half that of a vertex. Given that the chain stays, it switches between 1 and 2 about T times, so it makes about T runs of length about N/T each. The number a of visits to state 1 is the total length of half of these runs. So a is close to $N/2$, with fluctuations of order $\sqrt{T}\cdot N/T=N/\sqrt{T}$. Hence the mass e^{-T} is shared by about $N/\sqrt{T}$ compositions $(a,N-a,0,0)$, and the best one gets

$$M_{\{1,2\}}\approx\frac{e^{-T}}{N/\sqrt{T}}=N^{-1}\,T^{1/2}e^{-T}.$$

For the pair we can check this against an exact formula. A path with support $\{1,2\}$ is a word in the letters 1 and 2. Each stay has probability $1-2h$ and each switch probability h, so a word with j switches has probability $\frac{1}{4}(1-2h)^{N-1}u^j$ with $u=h/(1-2h)$. Summing over the words with a letters 1 gives $P_N(a,N-a,0,0)=M_{\{1\}}S_{a,N-a}(u)$, the bin polynomial of the two-state walk (Section 2.1). It is largest for the balanced compositions, and the uniform Bessel bound of [2], recalled in (2), shows that $S_N(u)$ is $\frac{2}{N}\sqrt{2T/\pi}\,e^T$ times a factor that tends to 1. So $M_{\{1,2\}}$ is $\frac{1}{\sqrt{2\pi}}N^{-1}T^{1/2}e^{-T}$ times such a factor.

An arc of three states, say $\{1,2,3\}$. Only the end states 1 and 3 can leave (both to state 4), each at rate 1, and the middle state 2 cannot. So the chain stays longer if it prefers the middle state, and we need an eigenvalue to find the exit rate. The chain killed on leaving the arc has the matrix $I+hQ_F$ with $-Q_F=2I-A$, where A is the adjacency matrix of the path $1-2-3$. Its largest eigenvalue is $1-(2-\sqrt{2})h$, with eigenvector $(1,\sqrt{2},1)$. So the chain stays in the arc with probability about $e^{-(2-\sqrt{2})T}$, and $\lambda_{\{1,2,3\}}=2-\sqrt{2}=0.586$. The visit fractions now fluctuate in two directions, so the mass is shared by about $(N/\sqrt{T})^2$ compositions, and $M_{\{1,2,3\}}\approx N^{-2}\,T\,e^{-(2-\sqrt{2})T}$.

The whole cycle. The chain never has to leave, so $\lambda_{[4]}=0$ and the mass is 1. It is shared by about $(N/\sqrt{T})^3$ compositions near $(\frac{N}{4},\frac{N}{4},\frac{N}{4},\frac{N}{4})$. So $M_{[4]}\approx N^{-3}T^{3/2}$.

Comparison. Table 1 collects the four cases, up to bounded factors. Now put $T=\tau\log N$. Then $e^{-\lambda T}=N^{-\tau\lambda}$, while the powers of T are only powers of $\log N$. So, up to such powers, $M_F\approx N^{-\Phi_\tau(F)}$ with the exponent $\Phi_\tau(F)=\tau\lambda_F+|F|-1$ in the last column. The mode sits on the face with the smallest exponent. Figure 2(a) plots the four exponents against τ. The vertices

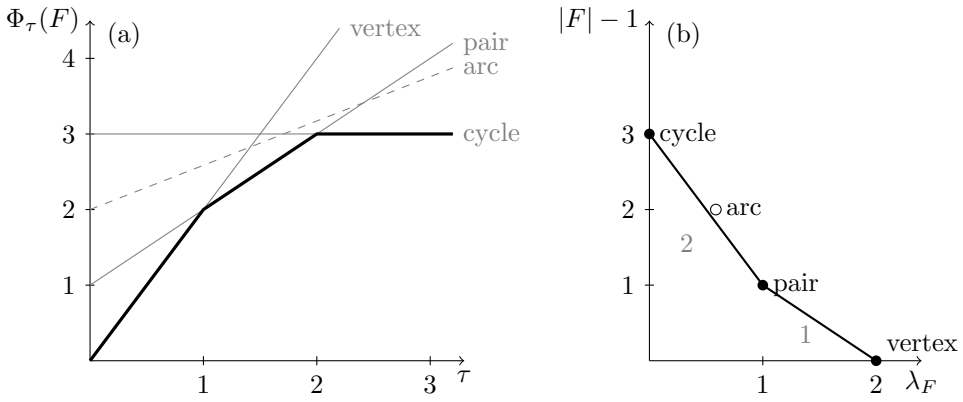


Figure 2: The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 3.4). The arc (open) lies above the boundary.

win for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs of three states never win. They beat the pairs only for $\tau > 1 + \sqrt{2}$, and there the whole cycle beats them, since that needs only $(2 - \sqrt{2})\tau > 1$. Note also that the whole cycle carries almost all the mass for every $\tau > 0$. It carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions.

The general rule. For any admissible face F the same two effects give $M_F \approx N^{-(|F|-1)} e^{-T\lambda_F} = N^{-\Phi_\tau(F)}$ up to powers of $\log N$, as announced in Section 1.1. There is a second way to see the factor $N^{-(|F|-1)}$. A path with support F must enter each of the $|F| - 1$ states of F where it does not start, and each first entry is a switch of probability about T/N. Every later switch also has probability about T/N, but it multiplies the number of ways to place the runs by a factor of order N, so it loses no power of N. Within F the best composition sits near the quasi-ergodic law α_F^*. Here it is $(\frac{1}{2}, \frac{1}{2})$ on a pair, $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ on an arc (the normalized squared Perron vector) and the uniform law on the cycle. Other visit fractions α lose an extra factor $e^{-T(I_F(\alpha) - \lambda_F)}$, with a rate I_F that we now introduce.

The rest of this section proves the rule. Proposition 3.1 studies the rate inside a face, Theorem 3.3 proves the rule for every Q and every start, and Theorem 3.4 reads the winners off a convex hull as in Figure 2(b). Section 5 returns to the 4-cycle with all constants (Corollary 5.1).

3.2 The rate inside a face

This subsection defines the rate I_F of the visit fractions inside a face and proves the properties that Theorems 3.3 and 4.6 use. For $\alpha \in \Delta_F$ let

$$I_F(\alpha) = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i} \Big) = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(q_i - \sum_{j \in F \setminus \{i\}} q_{ij} \frac{u_j}{u_i} \Big),$$

and let $q_i^{\mathrm{out}}(F) = \sum_{j \notin F} q_{ij}$, the rate of leaving F from i. The function I_F is the Donsker–Varadhan rate [8] of the occupation times of the chain killed on leaving F (Section 1.3). The chance that the

chain stays in F with visit fractions near α decays like $e^{-TI_F(\alpha)}$ (Theorem 3.3(i)). Taking $u = \mathbf{1}$ gives $I_F(\alpha) \geq \sum_i \alpha_i q_i^{\text{out}}(F)$, and dropping the terms with q_{ij} gives $I_F(\alpha) \leq \sum_i \alpha_i q_i$. So I_F lies between the rate of leaving F and the rate of leaving the current state, both averaged over α. On the pair $\{1,2\}$ of Section 3.1, part (c) below gives $I_F(\alpha) = 2 - 2\sqrt{\alpha_1\alpha_2}$, which is 2 at a vertex and has its least value 1 at $(\frac{1}{2}, \frac{1}{2})$. We need the properties of I_F without reversibility, and the proofs are short.

Proposition 3.1 (the rate on a face)**.** *Let F be irreducible with $k = |F| \geq 2$.*

(a) The rate satisfies $I_F \geq \lambda_F$ on Δ_F, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate I_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $I_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$I_F(\alpha) = \sum_{\{i,j\}\subseteq F,\ i\neq j} \bigl(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}}\bigr)^2 + \sum_{i\in F} \alpha_i q_i^{\text{out}}(F).$$

(d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $I_F(\alpha) = I_{F'}(\alpha) \geq \lambda_{F'} > \lambda_F$.

Idea of the proof. For $u = r_F$ the quantity $-(Q_F u)_i/u_i$ equals λ_F at every state i. So its average over any α is λ_F, and $I_F \geq \lambda_F$. For the other properties we write $u = e^\psi$. Then the function inside the supremum is concave in ψ and affine in α, and ψ is a maximizer exactly when α is stationary for the chain on F with rates $q_{ij}e^{\psi_j - \psi_i}$. For $u = r_F$ this stationary law is α_F^*, so r_F is optimal there and $I_F(\alpha_F^*) = \lambda_F$. Strict convexity then shows that α_F^* is the only minimizer in the open face, and (d) handles the boundary. Part (d) needs the lower bound on smaller faces, which may have no positive eigenvector. So in the proof a resolvent applied to $\mathbf{1}$ replaces r_F.

Proof. We prove the lower bound in (a) and the value at α_F^* first, and then (b), (d), the rest of (a) and (c), in this order.

Setup. With $u = e^\psi$ we have $I_F(\alpha) = \sup_\psi \Psi_\alpha(\psi)$, where

$$\Psi_\alpha(\psi) = \sum_i \alpha_i q_i - \sum_{i\neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i}$$

and i, j run over F. Each exponential is convex in ψ, so Ψ_α is concave in ψ. It is affine in α, and it does not change when we add a constant to ψ. Let G^ψ be the conservative generator on F (its rows sum to 0) with off-diagonal entries $q_{ij}e^{\psi_j - \psi_i}$. It has the graph G_F, so it is irreducible and has exactly one stationary law. Differentiating in each ψ_i gives $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$. A differentiable concave function attains its maximum exactly at its critical points. So ψ attains $I_F(\alpha)$ if and only if α is the stationary law of G^ψ. We call this the criterion.

The lower bound in (a)*, for every face F'.* We show that $I_{F'} \geq \lambda_{F'}$ on $\Delta_{F'}$ for every face F', irreducible or not. Let $s > \varrho(Q_{F'}) = -\lambda_{F'}$. It is enough to find a positive vector u with $-(Q_{F'}u)_i/u_i > -s$ at every state i. Then $I_{F'}(\alpha) > -s$ for every α, and letting s decrease to $-\lambda_{F'}$ gives the claim. Write $Q_{F'} = B - cI$ with a matrix $B \geq 0$ and a number c. Then $s + c > \varrho(B) \geq 0$, so the series

$$u = (sI - Q_{F'})^{-1}\mathbf{1} = \sum_{\ell\geq 0} (s+c)^{-\ell-1} B^\ell \mathbf{1}$$

converges, and u is positive because the term with $\ell = 0$ is. Since $(sI - Q_{F'})u = \mathbf{1}$, we get $-(Q_{F'}u)_i/u_i = -s + 1/u_i > -s$.

The value at α^*_F. Take $\psi = \log r_F$ and $R = \operatorname{diag}(r_F)$. Then $G^\psi = R^{-1}(Q_F + \lambda_F I)R$, since the off-diagonal entries agree and the rows of the right side sum to 0 by $Q_F r_F = -\lambda_F r_F$. As $\alpha^*_F = l_F \circ r_F$, we get $(\alpha^*_F)^\top G^\psi = l_F^\top (Q_F + \lambda_F I)R = 0$. By the criterion ψ attains the supremum. Its value is λ_F, because $-(Q_F r_F)_i/r_{F,i} = \lambda_F$ at every i. So $I_F(\alpha^*_F) = \lambda_F$.

(b) The function I_F is convex as a supremum of functions that are affine in α. Now let $\alpha \in \Delta^\circ_F$.

The supremum is attained. We normalize $\min_i \psi_i = 0$. Let $\bar\psi = \max_i \psi_i$, and let a be the least positive rate q_{ij} in F. Take a shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar\psi$. It has at most $k-1$ edges and ψ increases by $\bar\psi$ along it, so one of its edges $i \to j$ has $\psi_j - \psi_i \ge \bar\psi/(k-1)$. Keeping only this edge in the second sum of Ψ_α gives

$$\Psi_\alpha(\psi) \le \sum_i \alpha_i q_i - a\,(\min_i \alpha_i)\, e^{\bar\psi/(k-1)}.$$

The right side tends to $-\infty$ as $\bar\psi \to \infty$. So the supremum is the supremum over a bounded set of normalized ψ, and the continuous function Ψ_α attains it. We write $r_\alpha = e^\psi$ for a maximizer. By the criterion α is the stationary law of G^ψ, which is the last claim of (b).

The maximizer is unique up to a constant. Suppose that ψ and $\psi + \delta$ both attain the supremum. By concavity Ψ_α is constant on the segment $\psi + s\delta$, $0 \le s \le 1$. On this segment Ψ_α is a constant minus the sum of the functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, and each of these is convex in s. A sum of convex functions is constant only if each of them is affine, and an exponential e^{x+sy} is affine in s only if $y = 0$. As every α_i is positive, this gives $\delta_j = \delta_i$ on every edge of G_F. Since G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \ne \alpha^1$ lie in Δ°_F, let $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $I_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α,

$$I_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \le (1-s)I_F(\alpha^0) + sI_F(\alpha^1).$$

Equality would mean that ψ attains both $I_F(\alpha^0)$ and $I_F(\alpha^1)$. By the criterion α^0 and α^1 would both be the stationary law of G^ψ, which is impossible. So the inequality is strict.

(d) Let $\alpha \in \Delta_{F'}$. Only the rows $i \in F'$ enter $\sum_i \alpha_i \big(q_i - \sum_{j \in F\setminus\{i\}} q_{ij}u_j/u_i\big)$, and the terms with $j \in F \setminus F'$ are subtracted. Dropping them leaves the function that defines $I_{F'}(\alpha)$, at the vector $(u_i)_{i\in F'}$. So $I_F(\alpha) \le I_{F'}(\alpha)$. If we put $u_j = \varepsilon$ for $j \in F \setminus F'$ and let $\varepsilon \to 0$, the dropped terms tend to 0, so $I_F(\alpha) = I_{F'}(\alpha)$. Then $I_{F'}(\alpha) \ge \lambda_{F'}$ by the lower bound above. Finally $\lambda_{F'} > \lambda_F$, because the Perron root of an irreducible nonnegative matrix, here $Q_F + cI$ for a large number c, is larger than that of each proper principal submatrix [34, Ch. 2].

(a) The lower bound and the value at α^*_F are proved above, so let $\alpha \ne \alpha^*_F$. If α lies on the boundary of Δ_F, then $\alpha \in \Delta_{F'}$ for the proper face $F' = \operatorname{supp}\alpha$, and (d) gives $I_F(\alpha) > \lambda_F$. If $\alpha \in \Delta^\circ_F$ and $I_F(\alpha) = \lambda_F$, then strict convexity gives $I_F(\frac12(\alpha + \alpha^*_F)) < \lambda_F$, against the lower bound.

(c) We bound the function inside the supremum pair by pair. Since $q_i = \sum_{j\in F\setminus\{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, this function is $\sum_i \alpha_i q_i^{\mathrm{out}}(F)$ plus, for each pair $\{i,j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality, $\alpha_i q_{ij} u_j/u_i + \alpha_j q_{ji} u_i/u_j \ge 2\sqrt{\alpha_i q_{ij}\alpha_j q_{ji}}$, so the term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$. This gives the upper bound for $I_F(\alpha)$. For the lower bound first let $\alpha \in \Delta^\circ_F$ and take $u_i = \sqrt{\alpha_i/\nu_i}$. Then $\nu_i q_{ij} = \nu_j q_{ji}$ gives $\alpha_i q_{ij} u_j/u_i = \alpha_j q_{ji} u_i/u_j$, so there is equality in the AM–GM inequality for every pair at once. If some α_i vanish, we take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1 - \varepsilon/u_j)$, which tends to its bound $\alpha_j q_{ji}$. A pair with $\alpha_i = \alpha_j = 0$ has the term 0, and the other pairs are as before. □

In words, the most likely way to stay on F is to spend the fractions α_F^* of the time in its states, and the rate at any other fractions is strictly larger. Part (d) says that fractions on the boundary of Δ_F, which miss some states of F, have the rate of the smaller face that they use, and this rate is larger than λ_F. The lower bound in (a) and the value at α_F^* are checked in Lean, and so is the interior form of (c) when Q_F is symmetric. For symmetric Q_F, $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2. On a face that is not reversible the formula in (c) is only an upper bound, and Proposition 4.16 gives the rate and the gap.

3.3 Face selection

We also need an upper bound that holds on a whole face, including near its boundary, where the local law of Section 4 does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Note that $q - \lambda_F \mathbf{1} \in V_F$, since $A_F - \operatorname{diag}(q - \lambda_F \mathbf{1}) = Q_F + \lambda_F I$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha\rangle$, and I_F is the supremum of these linear rates. Indeed, for $u \in (0,\infty)^F$ the vector θ with $\theta_i = (A_F u)_i / u_i$ lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a positive eigenvector, and $\langle q - \theta, \alpha\rangle$ is then the function of u inside the supremum that defines $I_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector u of $A_F - \operatorname{diag}\theta$.

Lemma 3.2 (a uniform upper bound)**.** *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ, such that for every $n \in \mathbb{Z}_{\geq 0}^F$ with $\operatorname{supp} n = F$ and every h with $h q_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp\bigl(-h\langle n, q - \theta\rangle + C_\theta h^2 |n|\bigr)$.*

The proof is in Appendix A. With $\theta = q - \lambda_F \mathbf{1}$ it gives $P_N(n) \leq C h^{k-1} e^{-T\lambda_F + CT^2/N}$ for every n with support F. As $h^{k-1} = T^{k-1} N^{-(k-1)}$, this is the upper half of the rule $M_F \approx N^{-(k-1)} e^{-T\lambda_F}$ of Section 3.1, up to a power of T. The lemma is stated for a path of any length $|n|$, not only N, so it also applies to the blocks of a path.

The next theorem uses the local law (Theorem 4.4) and the face maxima (Theorem 4.6) of Section 4, which comes later. There is no circularity, since those two theorems are proved there without Theorem 3.3.

Theorem 3.3 (first-order face selection)**.** *Fix Q and μ.*

(i) Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,

$$\log P_N(n) = -(k-1)L - T\, I_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$

$$\log M_F \geq -(k_F - 1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \leq -(k_F - 1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp} \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp} \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 3.4(c)).

In words, (i) says that a composition whose fractions α stay in a compact part of the open face has probability $N^{-(k-1)} e^{-TI_F(\alpha)}$ up to a bounded factor and a power of T. Part (ii) says that the best composition on an admissible face has probability $N^{-(k_F-1)} e^{-T\lambda_F}$ times a power of T with exponent between 0 and $k_F - 1$, up to a bounded factor. With $T = \tau L$ the exponent of N is then $-\Phi_\tau(F)$, and part (iii) says that for large N the mode lies on an admissible face that minimizes it among the admissible faces, whatever the start. By (iv), if every such face is irreducible, the mode sits near the quasi-ergodic law of its face.

Idea of the proof. Part (i) comes from the local law of Section 4, which we prove independently of this theorem. Parts (ii) and (iii) are elementary. A face is visited in blocks, one for each strongly connected component (a single block if the face is irreducible). For the upper bound in (ii) we apply Lemma 3.2 to each block, and for the lower bound we write down explicit paths that spend almost all their time in a component of least exit rate. Part (iii) compares the faces, and (iv) follows from Theorem 4.6.

Proof. (i) For $k = 1$ this is (3), since $I_{\{i\}} = q_i$. For $k \geq 2$, Lemma 4.5 gives $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$ with $T^2/N \leq N^{-1/3}$. We claim that $\log f_T(\alpha) = \frac{k-1}{2} \log T + \log C_F(\alpha) - TI_F(\alpha) + O_{\mathcal{K}}(1)$ (with $C_F(\alpha)$ as in Definition 4.1) for $\alpha \in \mathcal{K}$ and $T \geq 1$. For $T \geq T_1$, with T_1 large, this is Theorem 4.4. For $1 \leq T \leq T_1$ it holds because $f_T(\alpha)$ is positive and continuous on the compact set $[1, T_1] \times \mathcal{K}$ (see the remarks after Theorem 4.4). Finally C_F is positive and continuous on $\mathcal{K}$, so $\log C_F$ is bounded there. So $\log f_T(n/N) = -TI_F(n/N) + O_{\mathcal{K}}(1 + \log T)$, and adding $\log N^{-(k-1)} = -(k-1)L$ gives (i).

(ii) If F is not admissible, $M_F = 0$. Let F be admissible. A path visits each component in one block of consecutive times (proof of Lemma 2.2), and an irreducible face is one component, with $k^* = k_F$. (For irreducible F, Theorem 4.6 later gives the sharper factor $T^{(k_F-1)/2} K_F$, but we do not need it here.) For the upper bound, Lemma 3.2 (or a direct count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1} e^{-h\lambda_C N_C})$, and each of the $r - 1$ switches between blocks gives a factor h. As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F-1} e^{-T\lambda_F})$. For the lower bound the path spends about N/T visits in each component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) We compare the faces by (ii). For large N we have $T_0 \leq T = \tau L \leq N^{1/3}$, so (ii) applies. There $T\lambda_F = \tau L \lambda_F$ and $\log T = O(\log L)$, so both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ and there are finitely many faces, the first claim follows. For the second claim let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \geq m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. So the support of a global mode is not such a face. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which is irreducible and admissible and so satisfies (H) by Lemma 2.2(b). So Theorem 4.6 with $T = \tau L \leq N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

For a reducible face we only have the bounds in (ii), and Proposition 4.13 shows that the power of T can differ from $(k_F - 1)/2$.

3.4 Reading the winners off a convex hull

This subsection turns Theorem 3.3(iii) into a picture. The winners are the points on the lower boundary of a convex hull.

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of S that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv} S$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv} S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv} S$. Figure 2(b) shows it for the 4-cycle.

Theorem 3.4 (geometry of the winners)**.** *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv} S$ *is a path whose vertices lie in S. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \le i \le m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \le i \le m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \ge \lambda_G$ and $|F| \le |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

In words, as τ grows the winners walk up the lower-left boundary of the hull, from small faces with large exit rates to large faces with small exit rates, and they change at the values τ_i, minus the slopes of its edges.

Idea of the proof. The least value $g(\tau)$ of Φ_τ is a minimum of finitely many affine functions of τ, so it is concave and piecewise affine. Its pieces are the vertices v_i and its kinks are the values τ_i. Part (b) adds two inequalities, and part (c) replaces a reducible face by one of its components.

Proof. (a) We study the least value $g(\tau) = \min_{(x,y)\in S}(\tau x + y)$ for $\tau > 0$. Each point (x, y) of S gives the affine function $\tau \mapsto \tau x + y$, whose slope is x, and different points give different functions. So g is a minimum of finitely many affine functions. It is concave and piecewise affine. Let $\tau_1 < \cdots < \tau_m$ be the points of $(0, \infty)$ where its slope changes, and put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Between two kinks. On (τ_i, τ_{i+1}) the function g is affine. The affine function of a point of S is nowhere below g. So if it equals g at one τ inside this interval, it equals g on the whole interval. Hence exactly one point of S attains g there. We call it $v_i = (\lambda^{(i)}, y^{(i)})$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $P_F = v_i$. The slope of g on this interval is $\lambda^{(i)}$. The slope of a concave function decreases, and it changes at each τ_i, so $\lambda^{(0)} > \cdots > \lambda^{(m)}$. By continuity both v_{i-1} and v_i attain $g(\tau_i)$, namely

$$\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}.$$

This is the formula for τ_i in the statement, and since $\tau_i > 0$ it gives $y^{(i)} > y^{(i-1)}$.

At a kink. The points (x, y) of S that attain $g(\tau_i)$ are those on the line $\tau_i x + y = g(\tau_i)$, which passes through v_{i-1} and v_i. Let (x, y) be one of them. For τ slightly below τ_i we have $\tau x + y \ge g(\tau) = \tau \lambda^{(i-1)} + y^{(i-1)}$, with equality at $\tau = \tau_i$. Subtracting gives $(\tau - \tau_i)(x - \lambda^{(i-1)}) \ge 0$, so $x \le \lambda^{(i-1)}$. In the same way, τ slightly above τ_i gives $x \ge \lambda^{(i)}$. So (x, y) lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of S on this edge lies on the line and so attains $g(\tau_i)$.

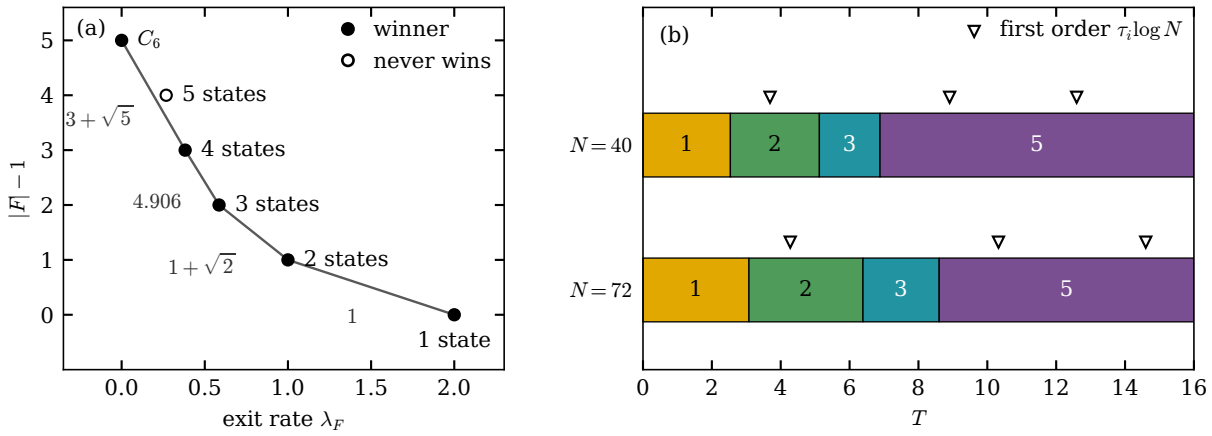


Figure 3: (a) The points $(\lambda_F, |F|-1)$ of the arcs of the 6-cycle C_6 and of C_6 itself, with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 3.4. The winners (filled) are the arcs of one to four states and C_6, and the arc of five states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T, at $N = 40$ and $N = 72$. The mode skips the arc of four states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1$, $1+\sqrt{2}$ and $2+\sqrt{2}$ (Table 2).

The boundary. On $\operatorname{conv} S$ the function $\tau x + y$ is least exactly on the convex hull of the points of S where it is least. By the two steps above this hull is the vertex v_i for $\tau_i < \tau < \tau_{i+1}$ and the edge $[v_{i-1}, v_i]$ for $\tau = \tau_i$. So the lower-left boundary is the path $v_0 \cdots v_m$, and its vertices lie in S.

(b) We add two inequalities. Since F wins at τ and G wins at τ', we have $\Phi_\tau(F) \le \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \le \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \ge 0$, so $\lambda_F \ge \lambda_G$. Then the first inequality gives $|F| - |G| \le \tau(\lambda_G - \lambda_F) \le 0$. If $\lambda_F = \lambda_G$, the two inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$. So if $\mathcal{F}_\tau$ and $\mathcal{F}_{\tau'}$ are disjoint, then $\lambda_F > \lambda_G$, and the first inequality gives $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \le \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

Lean checks part (b), the fact that a face with a proper subface of the same exit rate never wins, and part (a) when the least exit rate of each size decreases strictly with the size. In that case the vertices of the hull are sizes, and Lean checks which size wins between consecutive kinks and that consecutive vertices tie at τ_i. The general path structure in (a) and the ties along a closed edge are proved only here. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community (Table 2). For symmetric Q, λ_F is the least value of $\sum_{i<j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Proposition 3.1(d). Since faces of one size share the term $|F| - 1$, only the faces with the least exit rate of each size can win. Figure 3(a) shows S for the 6-cycle.

4 The second-order theory

The first-order rule locates a crossing only up to a factor $1+o(1)$ in T. To locate it up to $o(1)$ we need the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H), together with the power of T. The plan is as follows. Section 4.1 defines the constant K_F. Section 4.2 proves a local limit theorem inside a face and deduces M_F up to a factor $1+O(1/T+T^2/N)$. Section 4.3 compares two faces, Section 4.4 recovers the crossings of the two-state walk and of the complete graph, and Section 4.5 shows what can change without (H). Section 4.6 adds the next term, of relative order $1/T$, Section 4.7 gives a tree formula for K_F on every irreducible face, and Section 4.8 writes the rate I_F through flows. In this section gradients and Hessians in α are taken in the coordinates $\alpha'=(\alpha_2,\dots,\alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q=(q_i)_{i\in F}$.

4.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $I_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TI_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 4.1 (tilt and constants)**.** Let F be irreducible with $k\ge 2$. For $\alpha\in\Delta_F^\circ$ let r_α attain the supremum defining $I_F(\alpha)$ (Proposition 3.1(b)), and put $\theta(\alpha)=A_Fr_\alpha/r_\alpha$ and $l_\alpha=\alpha/r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha=R^{-1}(A_F-\operatorname{diag}\theta(\alpha))R$ with $R=\operatorname{diag}r_\alpha$. With $D=\operatorname{diag}\alpha$ let

$$Z_\alpha=(\mathbf{1}\alpha^\top-\widehat{Q}_\alpha)^{-1}-\mathbf{1}\alpha^\top,\qquad \Gamma_\alpha=DZ_\alpha+Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha)=\frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det\Sigma_\alpha}},\qquad K_F=C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}}=\mu_i$. Nothing in Definition 4.1 changes when r_α is scaled. Note that $l_\alpha\cdot r_\alpha=1$ and $I_F(\alpha)=\langle q-\theta(\alpha),\alpha\rangle$. Since r_α minimizes $u\mapsto\sum_{i\ne j}\alpha_iq_{ij}u_j/u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i}q_{ij}-l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top\big(A_F-\operatorname{diag}\theta(\alpha)\big)=0. \tag{4}$$

Also $(A_F-\operatorname{diag}\theta(\alpha))r_\alpha=0$ with $r_\alpha>0$, so $\theta(\alpha)\in V_F$ [34]. So $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (4) its stationary law is α. If $(\mathbf{1}\alpha^\top-\widehat{Q}_\alpha)v=0$, multiplying by $\alpha^\top$ gives $\alpha^\top v=0$, so $\widehat{Q}_\alpha v=0$. Hence v is a multiple of $\mathbf{1}$, and $\alpha^\top v=0$ gives $v=0$. Thus Z_α exists, and

$$Z_\alpha\mathbf{1}=0,\qquad \alpha^\top Z_\alpha=0,\qquad -\widehat{Q}_\alpha Z_\alpha=I-\mathbf{1}\alpha^\top. \tag{5}$$

So $\Gamma_\alpha\mathbf{1}=0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det\Sigma_\alpha$. At α_F^* the vector $u=r_F$ gives the value $\lambda_F=I_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*)=q-\lambda_F\mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F. In the local law below, $\mu\cdot r_\alpha$ accounts for the start of the path, $l_\alpha\cdot\mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij}=c_{ji}$ on F let $\operatorname{tree}(c)=\sum_{\mathcal{T}}\prod_{\{i,j\}\in\mathcal{T}}c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 4.2 (the constant)**.** *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $I_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 I_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 I_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})\sqrt{\det \nabla^2 I_F(\alpha)/(2\pi)^{k-1}}$.

(b) Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0,\infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i\alpha_j}\, q_{ij}\sqrt{\nu_i/\nu_j}$. Also $I_F(\alpha) = \sum_i q_i\alpha_i - \sum_{i\neq j} c_{ij}(\alpha)$, and

$$\det \Sigma_\alpha = \frac{2^{k-1}(\prod_i \alpha_i)^2}{\operatorname{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\operatorname{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

In words, the Gaussian factor in C_F is, up to a power of 2π, the square root of the determinant of the Hessian of the rate. For a reversible face this determinant is a weighted count of spanning trees divided by $2^{k-1}(\prod_i \alpha_i)^2$.

Proof. The proof of (a) is in Appendix A and uses Lemma A.1.

(b) We guess the vector r_α and check that the guess is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ (entrywise). As we saw before (4), the derivative of $\sum_{i\neq j} \alpha_i q_{ij} u_j/u_i$ in u_j is $\sum_i l_i q_{ij} - l_j (A_F u)_j/u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j(A_F u)_j}{u_j} = \nu_j \sum_i q_{ji}\sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. So the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.1(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij}\sqrt{\nu_i/\nu_j} = q_{ji}\sqrt{\nu_j/\nu_i}$. So $\widehat{Q}_\alpha$ is reversible with these conductances. Also $\alpha_i\theta_i(\alpha) = l_i(A_F u)_i = \sum_{j\neq i} c_{ij}(\alpha)$, and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$ gives the formula for I_F.

For the determinant we express the quadratic form of Σ_α through the Laplacian of the conductances, namely $\mathcal{L} = -D\widehat{Q}_\alpha$. It is symmetric, its off-diagonal entries are $-c_{ij}(\alpha)$, and $\mathcal{L}\mathbf{1} = 0$. Fix $\eta \in \mathbb{R}^F$ with $\eta_1 = 0$, and put $w = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha\cdot\eta)\alpha$. By (5), $\mathcal{L}w = D(I - \mathbf{1}\alpha^\top)\eta = \zeta$ and $\alpha \cdot w = 0$. So $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top w = 2\zeta^\top w$, where the second step uses $\alpha \cdot w = 0$. Next we remove state 1. Let $v = w - w_1\mathbf{1}$, so that $v_1 = 0$. Since $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we have $\zeta^\top w = \zeta^\top v$ and $\mathcal{L}v = \zeta$. We write primes for the coordinates $2, \dots, k$, $D' = \operatorname{diag}\alpha'$, and $\mathcal{L}_{(1)}$ for $\mathcal{L}$ without row and column 1. As $v_1 = 0$, the rows $2, \dots, k$ of $\mathcal{L}v = \zeta$ say that $\mathcal{L}_{(1)}v' = \zeta'$, and as $\eta_1 = 0$ we have $\zeta' = (D' - \alpha'\alpha'^\top)\eta'$. The matrix-tree theorem [35] gives $\det \mathcal{L}_{(1)} = \operatorname{tree}(c(\alpha))$, which is positive because F is irreducible. So $\mathcal{L}_{(1)}$ is invertible, and

$$\eta'^\top \Sigma_\alpha \eta' = 2\zeta'^\top v' = 2\eta'^\top (D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)\eta'$$

for every $\eta' \in \mathbb{R}^{k-1}$. Both matrices are symmetric, hence $\Sigma_\alpha = 2(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)$. Now we take determinants. The matrix determinant lemma gives $\det(D' - \alpha'\alpha'^\top) = \prod_{i\geq 2}\alpha_i\,(1 - \sum_{i\geq 2}\alpha_i) = \prod_i \alpha_i$. So $\det \Sigma_\alpha = 2^{k-1}(\prod_i \alpha_i)^2/\operatorname{tree}(c(\alpha))$, and putting this into Definition 4.1 gives the formula for $C_F(\alpha)$. □

Part (b) needs a reversible face. Proposition 4.15 at the end of this section gives a formula of the same kind for every irreducible face.

Corollary 4.3 (symmetric faces)**.** *Let F satisfy* (H) *with $k \geq 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j \in F\setminus\{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F, and*

$$K_F = \mu(F)\, k^{(k+1)/2} \sqrt{\operatorname{tree}(q_F)}\, (4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ *is* $\operatorname{tree}(c)$ *for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.*

Proof. The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 4.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)} \operatorname{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. □

4.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition gets the factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of a lattice cell.

Theorem 4.4 (local law)**.** *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,*

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} \big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

In words, the density of the visit fractions on the event that the chain stays in F is a Gaussian factor $T^{(k-1)/2}$ times the large-deviation factor $e^{-TI_F(\alpha)}$ times the constant $C_F(\alpha)$, with a relative error $O(1/T)$ that is uniform away from the boundary of the face.

Idea of the proof. The proof is in Appendix A. The generating function of the sum over words that defines H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point is computed from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α. The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 4.5 (discrete and continuum)**.** *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

In words, the probability of a composition is the continuum density times the volume $N^{-(k-1)}$ of a lattice cell. The proof is in Appendix A. With Theorem 4.4 the lemma replaces the term $O_{\mathcal{K}}(1 + \log T)$ in Theorem 3.3(i) by $\frac{k-1}{2} \log T + \log C_F(n/N) + O_{\mathcal{K}}(1/T + T^2/N)$.

Theorem 4.6 (face maxima)**.** *Let F satisfy (H). If $k \geq 2$, then uniformly for $1 \leq T \leq N^{1/3}$,*

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} \big(1 + O(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}}$ is given by (3). In both cases, uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F - T\lambda_F + O(1/T + T^2/N). \tag{6}$$

In words, the best composition on F sits within $O(N/T)$ of $N\alpha_F^*$, and its probability is the Gaussian peak of the local law at α_F^*. The factor $N^{-(k-1)}$ is the volume of a lattice cell, and $T^{(k-1)/2}$ is the height of a Gaussian density whose spread in each free direction is of order $T^{-1/2}$.

Idea of the proof. Near α_F^* we use the local law and Lemma 4.5, and maximize a Gaussian with a small linear perturbation. Away from α_F^* the rate exceeds λ_F by a fixed amount, and Lemma 3.2, applied with finitely many vectors θ, shows that these compositions are exponentially less likely in T. For bounded T positivity and continuity suffice.

Proof. For $k = 1$ this is (3). Let $k \geq 2$. Put $X = N^{-(k-1)}T^{(k-1)/2}K_F e^{-T\lambda_F}$, the value that the theorem claims for M_F. We compare every $P_N(n)$ with X, and we write $x = \alpha' - \alpha_F^{*\prime}$ for the displacement of α from α_F^*. We fix a ball U around α_F^*, bound the compositions outside U, then those inside U, and at the end we locate the maximizer.

The ball U. By Propositions 3.1(a) and 4.2(a), I_F and $\log C_F$ are real-analytic near α_F^*, and I_F has its minimum λ_F there with a positive definite Hessian. So there are a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ and positive constants c_U, C_U, C_1 and C_3 such that on U

$$c_U|x|^2 \leq I_F - \lambda_F \leq C_U|x|^2, \qquad \bigl|\log(C_F/K_F) - \gamma \cdot x\bigr| \leq C_1|x|^2,$$

where $\gamma = \nabla \log C_F(\alpha_F^*)$, and such that $\log C_F$ is C_3-Lipschitz on U.

Outside U. We show that a composition with $n/N \notin U^\circ$ has $P_N(n) = X \cdot O(T^{(k-1)/2}e^{-2\eta T})$ for some $\eta > 0$. The set $\Delta_F \setminus U^\circ$ is compact and does not contain α_F^*, and I_F is lower semicontinuous. So Proposition 3.1(a) gives $\eta > 0$ with $I_F \geq \lambda_F + 3\eta$ on this set. Recall that $I_F(\beta) = \sup_{u>0}\langle q - A_F u/u, \beta\rangle$ and that $A_F u/u \in V_F$. So every β in the set has a vector $\theta \in V_F$ with $\langle q-\theta, \beta\rangle > \lambda_F+2\eta$, and the same θ works for all points near β. By compactness finitely many vectors $\theta_j \in V_F$ are enough, that is, $\max_j\langle q - \theta_j, \beta\rangle > \lambda_F + 2\eta$ for every $\beta \in \Delta_F \setminus U^\circ$. Now let $\operatorname{supp} n = F$ with $n/N \notin U^\circ$, and take j that attains the maximum at $\beta = n/N$. Then $h\langle n, q - \theta_j\rangle > T(\lambda_F + 2\eta)$, and Lemma 3.2 with $\theta = \theta_j$ gives

$$P_N(n) \leq C(T/N)^{k-1}e^{-T(\lambda_F+2\eta)} = X \cdot O\bigl(T^{(k-1)/2}e^{-2\eta T}\bigr).$$

Inside U. By Theorem 4.4 and Lemma 4.5 with $\mathcal{K} = U$, for $\alpha = n/N \in U$,

$$P_N(n) = X\,\tfrac{C_F(\alpha)}{K_F}\,e^{-T(I_F(\alpha)-\lambda_F)}(1 + E), \qquad |E| \leq C_2(1/T + T^2/N). \tag{7}$$

For the upper bound we use the bounds on U. For $T \geq 2C_1/c_U$ the logarithm of the two middle factors is at most

$$\gamma \cdot x + C_1|x|^2 - c_U T|x|^2 \leq \gamma \cdot x - \tfrac{c_U T}{2}|x|^2 \leq \frac{|\gamma|^2}{2c_U T},$$

where the last step is the maximum over x of a concave quadratic. So $P_N(n) \leq X(1 + O(1/T + T^2/N))$ for $n/N \in U$. The bound outside U is smaller than this for large T, and hence $M_F \leq X(1 + O(1/T + T^2/N))$.

For the lower bound we take a lattice point next to α_F^*. Let n^0 have support F and $|n_i^0 - N\alpha_{F,i}^*| < 1$ for all i. It exists for $N \geq N_0$ since every $\alpha_{F,i}^*$ is positive, and its displacement has $|x| \leq k/N$. Then $C_F(n^0/N) \geq K_F e^{-kC_3/N}$ and $T(I_F(n^0/N) - \lambda_F) \leq C_U k^2 T/N^2$, so (7) gives $P_N(n^0) \geq X(1 - O(1/T + T^2/N))$.

Bounded T. Fix T_2 so large that the two bounds above give $M_F/X \in [\frac{1}{2}, \frac{3}{2}]$ for $T \geq T_2$. For $1 \leq T \leq T_2$ the error term $O(1/T)$ is of order 1, so it is enough to show that $M_F/X \in [c, C]$ with constants $0 < c < C$. For such T, $N^{k-1}X$ lies between two positive constants, and so does f_T on

U, because it is positive and continuous on $[1,T_2]\times U$. So Lemma 4.5 bounds $N^{k-1}P_N(n)$ above for $n/N\in U$ and bounds $N^{k-1}P_N(n^0)$ below, and the bound outside U is $O(N^{-(k-1)})$. This gives the first display for all $1\le T\le N^{1/3}$, and (6) is its logarithm.

The maximizer. Let $\hat n$ be a maximizer and let $\hat x$ be the displacement of $\hat n/N$. For $T\ge T_3$ with T_3 large, the bound outside U is below $X/2\le P_N(n^0)$, so $\hat n/N\in U$, and the error in (7) has $|E|\le\frac12$. We take logarithms in $P_N(\hat n)\ge P_N(n^0)$ and use (7) on the left. There $\log(C_F/K_F)\le C_3|\hat x|$ and $T(I_F-\lambda_F)\ge c_UT|\hat x|^2$, while the right side is at least $\log X-C(1/T+T^2/N)$. So

$$c_UT|\hat x|^2\le C_3|\hat x|+C_4\Big(\frac1T+\frac{T^2}{N}\Big)\le C_3|\hat x|+\frac{2C_4}{T}.$$

The last step holds because $T\le N^{1/3}$ gives $T^2/N\le 1/T$, and this is the step that needs the exponent $\frac13$. Solving the quadratic inequality gives $|\hat x|\le\big(C_3+\sqrt{C_3^2+8c_UC_4}\big)/(2c_UT)=O(1/T)$. For $1\le T<T_3$ the claim holds since $|\hat x|$ is bounded. □

4.3 Crossings and tie windows

We now compare two faces. By Theorem 4.6, the difference $\log M_F-\log M_G$ is an explicit increasing function of T up to an error $O(1/T+T^2/N)$, and its zero is the crossing.

Theorem 4.7 (the crossing equation)**.** *Let F and G satisfy (H), with $\Delta=k_F-k_G\ge1$ and $\delta=\lambda_G-\lambda_F>0$, and let $D_N(T)=\log M_F-\log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L,\frac{2\Delta}{\delta}L]$.

(ii) For large N, every zero $T_N\in[1,N^{1/3}]$ of D_N satisfies

$$\delta T_N+\frac{\Delta}{2}\log T_N=\Delta L-\log\frac{K_F}{K_G}+O\Big(\frac1L\Big). \tag{8}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N=\frac{\Delta}{\delta}\Big[L-\frac12\log L+b_{FG}\Big]+O\Big(\frac{\log L}{L}\Big),\qquad b_{FG}=-\frac12\log\frac{\Delta}{\delta}-\frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

Proof. We subtract (6) for G from (6) for F. Uniformly on $[1,N^{1/3}]$,

$$D_N(T)=-\Delta L+\Xi(T)+\log\frac{K_F}{K_G}+O\Big(\frac1T+\frac{T^2}{N}\Big),\qquad \Xi(T)=\delta T+\frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi'\ge\delta$.

(i) At $T=\frac{\Delta}{2\delta}L$ the display gives $D_N=-\frac{\Delta}{2}L+O(\log L)<0$, and at $T=\frac{2\Delta}{\delta}L$ it gives $D_N=\Delta L+O(\log L)>0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. So D_N is continuous and has a zero in between.

(ii) First we locate a zero $T_N\in[1,N^{1/3}]$ roughly. The error term is $O(1)$ on $[1,N^{1/3}]$, so $\Xi(T_N)=\Delta L-\log(K_F/K_G)+O(1)$. This is larger than $\Xi(\frac{\Delta}{2\delta}L)=\frac{\Delta}{2}L+O(\log L)$, so $T_N>\frac{\Delta}{2\delta}L$ because Ξ is increasing. Also $\delta T_N\le\Xi(T_N)$, so $T_N\le\frac{2\Delta}{\delta}L$. With T_N of order L the error term is $O(1/T_N+T_N^2/N)=O(1/L)$, and $D_N(T_N)=0$ is (8).

(iii) Let T° be the solution of $\Xi(T^\circ)=\Delta L-\log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N)=\Xi(T^\circ)+O(1/L)$, and $\Xi'\ge\delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, note that $T^\circ=\frac{\Delta}{\delta}L+O(\log L)$, so $\log T^\circ=\log L+\log\frac{\Delta}{\delta}+O(\log L/L)$. Putting this into $\delta T^\circ=\Delta L-\frac{\Delta}{2}\log T^\circ-\log(K_F/K_G)$ and dividing by δ gives the explicit form. □

In words, the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$. The crossing balances the three. In the numerical test after Corollary 5.1 the explicit form in (iii) is still off by 0.06 at $N = 10^{14}$, so computations should solve (8). We do not prove that the zero of D_N is unique in general.

At a switch value τ^* of Theorem 3.4 several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph (Corollary 4.11). Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 4.8 (tie windows)**.** *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 3.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F-1}{2}\log\tau^* - \tau^*\lambda_F t$ and $E(t) = \max_{G\in\mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2}\log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \le \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \le N^{-\varepsilon_0} \max_{G\in\mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \le t \le \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \le t \le -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

Consequently the following holds for every $\eta > 0$ and large N. If $|t| \le \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \le \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j runs over the kinks of E, and it changes near each kink. Faces with identical lines (the same k_F, λ_F and K_F) tie at this order and give no crossing.

In words, inside the window the logarithm of every tied face maximum is the same leading term plus the line $\ell_F(t)$, faces that do not tie are smaller by a power of N, and far out in the window the face with the least (for $t > 0$) or largest (for $t < 0$) exit rate wins. In between, the mode changes face only where the highest line changes, namely near the kinks of E.

Idea of the proof. In the window T/L is close to τ^*. For a tied face the terms $-(k_F - 1)L$ and $-T\lambda_F$ of (6) add up to $-m^*L + \tau^*\lambda_F(\frac{1}{2}\log L - t)$, because $\tau^*\lambda_F + k_F - 1 = m^*$. The term $\frac{k_F-1}{2}\log T$ is $\frac{k_F-1}{2}\log L$ plus a constant, up to a small error. So the multiples of $\log L$ add up to $\frac{m^*}{2}\log L$. This and $-m^*L$ are the same for all tied faces, and what is left is the line $\ell_F(t)$. This is (i). A face that does not tie has a larger exponent Φ_{τ^*}, and the first-order bounds already make it smaller by a power of N. This is (ii). For (iii), two tied faces with different exit rates have lines of different slopes, so the gap between their lines grows linearly in $|t|$ and beats every error term once $|t|$ is large.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). We first put T into (6). Write $T = \tau^*L(1 + y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \le \frac{1}{2}$ for $|t| \le L/4$ and large N. Then $\log T = \log\tau^* + \log L + \log(1 + y)$ and $T\lambda_F = \tau^*\lambda_F(L - \frac{1}{2}\log L + t)$, and the error term of (6) is $O(1/L)$. Since $\tau^*\lambda_F + k_F - 1 = m^*$, we get, uniformly for $|t| \le L/4$,

$$\log M_F = -m^*L + \tfrac{m^*}{2}\log L + \ell_F(t) + \tfrac{k_F-1}{2}\log(1 + y) + O(1/L) \qquad (F \in \mathcal{F}^*). \tag{9}$$

(i) For t in a compact set, $y = O(\log L/L)$, so $\log(1+y) = O(\log L/L)$ in (9).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. (If there is no such face, (ii) says nothing, and we put $g = 1$.) Put $\varepsilon_1 = \min\bigl(\tau^*/2,\, g/(4q_{\max}+4)\bigr)$. (Any constant below $g/(2q_{\max})$ would do in the second entry. We use that $0 \le \lambda_F \le q_{\max}$ for every face, since the Perron root of Q_F lies between its largest diagonal entry and 0.) The entry $\tau^*/2$ keeps T of order L. Namely, if $|T/L - \tau^*| \le \varepsilon_1$, then $\frac{1}{2}\tau^* L \le T \le \frac{3}{2}\tau^* L$. So for large N we have $T_0 \le T \le N^{1/3}$, which is the range of Theorem 3.3(ii), and $\log T = O(\log L)$. Take an admissible $F \notin \mathcal{F}^*$ and any $G \in \mathcal{F}^*$. The difference $\Phi_\tau(F) - \Phi_\tau(G)$ is at least g at $\tau = \tau^*$, and its slope in τ is $\lambda_F - \lambda_G$, which is at most $q_{\max}$ in absolute value. So $\Phi_\tau(F) - \Phi_\tau(G) \ge g/2$ for $|\tau - \tau^*| \le \varepsilon_1$. With $\tau = T/L$ the upper bound of Theorem 3.3(ii) for F and the lower bound for G give $\log M_F - \log M_G \le -(\Phi_\tau(F) - \Phi_\tau(G))L + O(\log L) \le -\frac{g}{4}L$ for large N. So (ii) holds with $\varepsilon_0 = g/4$.

(iii) Take $\varepsilon = \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$. Then for $|t| \le \varepsilon L$ and large N we have $|T/L - \tau^*| = \tau^*|y| \le \varepsilon_1$, so (ii) applies and no global mode lies on a face outside $\mathcal{F}^*$. It remains to compare the tied faces. Let $t_0 \ge 1$ be larger than every $|t_j|$, so that E is a single line on $[t_0, \infty)$. Let $F \in \mathcal{F}^*$ have this line, and let $G \in \mathcal{F}^*$ have another line. We show that $M_F > M_G$ for $t_0 \le t \le \varepsilon L$ if t_0 and N are large. Since $\ell_F \ge \ell_G$ on $[t_0, \infty)$, the slopes satisfy $\lambda_F \le \lambda_G$. As both faces are tied,

$$\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \ge 0.$$

There are two cases.

First let $\Delta = 0$. Then the two lines are parallel, and $\ell_F - \ell_G = \log(K_F/K_G) > 0$. The terms with $\log(1+y)$ in (9) are the same for F and G, so $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L) > 0$ for large N.

Now let $\Delta > 0$. Then $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$, and (9) gives

$$\log \frac{M_F}{M_G} = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O\Bigl(\frac{1}{L}\Bigr).$$

For large N and $t \ge 1$ we have $|\log(1+y)| \le 2|y| \le (2t + \log L)/L \le t/2$. So the right side is at least $\frac{3}{4}\Delta t - C$, which is positive for $t \ge t_0$ once t_0 is large.

So for $t_0 \le t \le \varepsilon L$ every global mode lies on a face whose line is E on $[t_0, \infty)$. The slope of ℓ_F is $-\tau^*\lambda_F$, and E follows the line of largest slope on the far right, so such a face has the least λ_F in $\mathcal{F}^*$. The case $t \le -t_0$ is the same with the signs reversed.

The consequence. Fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$, a finite union of closed intervals without kinks. Take $G \in \mathcal{F}^*$. The function $E - \ell_G$ is convex, piecewise linear and nonnegative, and its slope changes only at kinks of E. So its zero set is empty, or a single kink of E, or an interval whose finite endpoints are kinks of E. Hence each interval of S_η lies inside the zero set or is disjoint from it. In the first case $\ell_G = E$ on the interval, and in the second $E - \ell_G \ge g_\eta$ there for some $g_\eta > 0$. Now let $t \in S_\eta$ and let $F \in \mathcal{F}^*$ have $\ell_F(t) = E(t)$. By (i), every $G \in \mathcal{F}^*$ whose line is not E near t has $\log(M_F/M_G) \ge g_\eta - O(\log L/L) > 0$, and by (ii) the faces outside $\mathcal{F}^*$ are smaller still. This is the claim for $t \in S_\eta$. For $t_0 \le |t| \le \varepsilon L$ it is (iii). Since η is arbitrary, the line of the face that carries the mode can change only at $t = t_j + o(1)$. It does change near each kink, because E is a different line on the two sides of a kink. □

4.4 The two-state walk and the complete graph

We use the polynomials $S_{a,b}$ and S_N, the root u_N, the crossing p_N, $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$ of Section 2.1. The first result shows that, at every N, the comparison of a symmetric pair with one of its states is exactly the two-state walk.

Proposition 4.9 (an exact pair identity)**.** *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N-1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u), \qquad u = \frac{ah}{1-qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F, $M_F = M_{\{i\}} S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof. We group the paths by their number of switches. A path with this composition is a word in i, j with m letters i. Each stay has probability $1 - qh$ and each switch ah, so a word with s switches has probability $\mu_i(1-qh)^{N-1-s}(ah)^s = \mu_i(1-qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1-qh)^{N-1}$. Sum over these words and divide. This gives $\sum_w u^{s(w)} = S_{m,N-m}(u)$, which is the identity. By [2, Proposition 4.1] the bin polynomials grow coefficientwise as the bin moves toward the middle, and $S_{m,N-m} = S_{N-m,m}$. So the balanced m give the largest $S_{m,N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). □

Lean checks the identity for a chain with two states. For $d \geq 3$ the paths with this composition never leave F, so the same computation applies, but this case is not in Lean.

Corollary 4.10 (the two-state walk)**.** *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [2] with $T = N(1-p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1, 2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2}\log z_N = L + c_0 + O(1/L)$.*

Proof. Corollary 4.3 with $k = 2$, $\mathrm{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 4.9 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only zero of D_N is z_N, and by Theorem 4.7(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N. Then Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation. □

This is the crossing equation of [2] without its terms $1/(8z_N)$ and R_N. Proposition 4.14 recovers the term $1/(8z_N)$. In its notation this term is $(c_{\{1\}} - c_{\{1,2\}})/z_N$ with $c_{\{1\}} = 0$ and $c_{\{1,2\}} = -\frac{1}{8}$, and (11) with $\Delta = \delta = 1$ gives $z_N + \frac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2})$.

Corollary 4.11 (the complete graph)**.** *Let $Q = J - dI$, where J is the all-ones matrix, and let μ be uniform, the model of [3]. A face of k states has $\lambda_F = d - k$, α_F^* uniform and $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. The crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k = \frac{1}{2}\log(4\pi) - \frac{k+1/2}{k-1}\log k$, the constant of the vertex crossing of [3].*

Proof. Each Q_F is symmetric with row sums $-(d-k)$, and the complete graph on F has k^{k-2} spanning trees by Cayley's formula. So Corollary 4.3 with $\mu(F) = k/d$ gives λ_F, α_F^* and K_F. A vertex has $K = \frac{1}{d}$ and $\lambda = d - 1$, so $b_{FG} = -\frac{1}{k-1}\log(dK_F) = a_k$. The arithmetic that gives b_{FG} from K_F here and in Corollary 4.10 is checked in Lean. At $\tau^* = 1$, Theorem 4.8 describes the tie window, and [3] resolves this window in full. □

4.5 Reducible faces

This subsection shows by an example that the coefficient $\frac{1}{2}$ can fail on a face that is not irreducible.

Remark 4.12 (scope of the $\frac{1}{2}$). The coefficient $\frac{1}{2}$ of $\log\log N$ in Theorems 4.7 and 4.8 is due to the power $T^{(k-1)/2}$ in Theorem 4.6, so it needs only (H). At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^*\delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 4.7. The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the $\frac{1}{2}$ can fail, and in the next example it is 1 and then 0.

Proposition 4.13 (a directed 3-cycle)**.** *Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$.*

(i) The admissible faces are $\{1\}$, $\{1,2\}$ (reducible, $\lambda = 1$) and $\{1,2,3\}$ (irreducible, $\lambda = 0$). Exactly $M_{\{1\}} = (1-bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1,b,b)/(1+2b)$, $\det\nabla^2 I = (1+2b)^4/(3b^2)$ and $K_{\{1,2,3\}} = (1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$. It satisfies $(b-1)T + \log T = L - \log b + O(T^2/N)$, so $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log\frac{b-1}{b}] + O(\log L/L)$, and the coefficient of $\log\log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log\frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0. These roots lie in an interval of length $O(1/L)$.

(iv) Both are changes of the global mode. For large N and $T \le N^{1/3}$, the mode is $(N,0,0)$ for $T < T_N^{(1)}$, it is $(1, N-1, 0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

In words, the face $\{1,2\}$ is reducible, and its best composition $(1, N-1, 0)$ visits state 1 once and then stays in state 2. Its maximum is about $b(T/N)e^{-T}$, with the power T^1 where an irreducible pair would have $T^{1/2}$. This changes the coefficient of $\log\log N$ at both crossings.

Proof. (i) *The faces.* From state 1 the chain can only move along $1 \to 2 \to 3 \to 1$. So a path that starts at 1 has support $\{1\}$, $\{1,2\}$ or $\{1,2,3\}$, and these are the admissible faces. Inside $\{1,2\}$ the chain cannot return from 2 to 1, so this face is reducible with components $\{1\}$ and $\{2\}$, of exit rates b and 1, and $\lambda_{\{1,2\}} = 1$. The only path with support $\{1\}$ stays at 1 for $N-1$ steps, which gives $M_{\{1\}}$. A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$. Raising n_1 by one multiplies this by $(1-bh)/(1-h) < 1$, so the maximum is at $n_1 = 1$ only, and it is $bh(1-h)^{N-2}$.

The full face. Here Q is a generator on $\{1,2,3\}$, so $\lambda = 0$ and $r = \mathbf{1}$. The left vector l is the stationary law. In it the three flows $b\,l_1$, l_2 and l_3 around the cycle are equal, so $l = \alpha^* = (1,b,b)/(1+2b)$ and $(\mu\cdot r)(l\cdot\mathbf{1}) = 1$. By Proposition 4.2(a) it remains to find $\det\nabla^2 I$ at α^*, and then $K_{\{1,2,3\}} = \sqrt{\det\nabla^2 I}/(2\pi)$.

We first find the rate in closed form. By definition

$$I(\alpha) = \sum_i q_i\alpha_i - \inf_{u>0}\Big[b\alpha_1\frac{u_2}{u_1} + \alpha_2\frac{u_3}{u_2} + \alpha_3\frac{u_1}{u_3}\Big].$$

The three ratios are positive numbers with product 1, and every such triple occurs. So by the AM–GM inequality the infimum is $3(b\alpha_1\alpha_2\alpha_3)^{1/3}$, and

$$I(\alpha) = \sum_i q_i\alpha_i - 3b^{1/3}g(\alpha), \qquad g(\alpha) = (\alpha_1\alpha_2\alpha_3)^{1/3}.$$

The first sum is linear, so the Hessian of I is $-3b^{1/3}$ times the Hessian of g. In all three coordinates, $\partial_i\partial_j g = -g\,\Omega_{ij}$ with $\Omega = \frac{1}{3}\operatorname{diag}(w) - \frac{1}{9}ww^\top$ and $w = 1/\alpha$ (entrywise). In the coordinates (α_2, α_3) we have $\alpha_1 = 1 - \alpha_2 - \alpha_3$, so raising α_2 or α_3 moves α in the direction $v_2 = e_2 - e_1$ or $v_3 = e_3 - e_1$. Hence $\nabla^2 I$ is the 2×2 matrix with entries $3b^{1/3}g\, v_i^\top \Omega v_j$.

Now we evaluate at α^*. Put $s = 1 + 2b$ and $y = 1/b$. Then $3b^{1/3}g = 3b/s$ and $w = s(1, y, y)$, and a short computation gives

$$v_2^\top \Omega v_2 = v_3^\top \Omega v_3 = \tfrac{2}{9}s^2(1 + y + y^2), \qquad v_2^\top \Omega v_3 = \tfrac{1}{9}s^2(2 + 2y - y^2).$$

The determinant of this 2×2 matrix is a difference of two squares, namely

$$\frac{s^4}{81}\big[(2 + 2y + 2y^2)^2 - (2 + 2y - y^2)^2\big] = \frac{s^4}{81} \cdot 3y^2(2 + y)^2 = \frac{s^6}{27b^4},$$

since $2 + y = s/b$. Hence $\det \nabla^2 I = (3b/s)^2 s^6/(27b^4) = s^4/(3b^2)$ and $K_{\{1,2,3\}} = s^2/(2\sqrt{3}\,\pi b)$. The exact parts of (i) are checked in Lean.

(ii) We show that $D_1 = \log M_{\{1,2\}} - \log M_{\{1\}}$ increases in h. By (i), $D_1 = \log(bh) + (N - 2)\log(1 - h) - (N - 1)\log(1 - bh)$, so

$$D_1'(h) = \frac{1}{h} - \frac{N - 2}{1 - h} + \frac{b(N - 1)}{1 - bh} > 0 \qquad (bh < 1),$$

because $b > 1$ gives $b/(1 - bh) > 1/(1 - h)$. Also $D_1 \to -\infty$ as $h \to 0$ and $D_1 \to \infty$ as $bh \to 1$. So there is exactly one root (this is checked in Lean). To locate it we expand in $T = Nh$, which gives

$$D_1 = \log b + \log T - L + (b - 1)T + O(T^2/N).$$

This is negative at $T = L/(2(b - 1))$ and positive at $T = L/(b - 1)$ for large N. So the root lies between them, and it satisfies the equation in (ii). Since the root is of order L, that equation gives $\log T = \log L - \log(b - 1) + O(\log L/L)$, and putting this back gives the expansion.

(iii) By (i), $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$. The full face satisfies (H), and Theorem 4.6 with $k = 3$ and $\lambda = 0$ gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/T + T^2/N)$. So on $[L/2, 2L]$

$$\log M_{\{1,2\}} - \log M_{\{1,2,3\}} = L - T + \log b - \log K_{\{1,2,3\}} + O(1/L).$$

At a root the left side is 0, which gives the formula in (iii) with the value of $K_{\{1,2,3\}}$ from (i). Two roots then differ by $O(1/L)$.

(iv) We compare the three face maxima on three ranges of T. Note first that Lemma 3.2 with $\theta = q$ gives $M_{\{1,2,3\}} \le C(T/N)^2$ for $T \le N^{1/3}$.

Let $T < T_N^{(1)}$. Then $M_{\{1\}} > M_{\{1,2\}}$ by (ii). Also $T < L/(b-1)$ by (ii), so $M_{\{1\}} = (1-bh)^{N-1} \ge N^{-b/(b-1)-o(1)}$. This is larger than $C(T/N)^2$ because $b > 2$ gives $b/(b - 1) < 2$. So the mode is $(N, 0, 0)$.

Let $T > T_N^{(1)}$. Then $M_{\{1,2\}} > M_{\{1\}}$ by (ii), and we compare $\{1, 2\}$ with the full face. For $T \le L/2$ the bound on $M_{\{1,2,3\}}$ gives $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} \ge L - T - \log T - C > 0$. On $[L/2, 2L]$ the difference in (iii) is continuous in T, positive at $L/2$ and negative at $2L$. So it is positive below its smallest root and negative above its largest root. For $2L \le T \le N^{1/3}$ the same expansions give a difference of at most $-L + C < 0$. Since the maximum of $M_{\{1,2\}}$ is attained only at $(1, N - 1, 0)$, this proves (iv). □

4.6 The next term

The local law, the face maxima and the crossing equation each have a next term, of relative order $1/T$. For a face F that satisfies (H) with $k \geq 2$ we write $\Sigma_F = \Sigma_{\alpha_F^*}$ and $\gamma_F = \nabla \log C_F(\alpha_F^*)$, in the coordinates α'. The function c and the constants c_F below are not related to the conductances $c_{ij}(\alpha)$ of Proposition 4.2.

Proposition 4.14 (the next term)**.** *Let F satisfy* (H).

(i) Let $k \geq 2$. There is a real-analytic function c on Δ_F°, given by (22), such that for every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} \Big(1 + \frac{c(\alpha)}{T} + O_{\mathcal{K}}\Big(\frac{1}{T^2}\Big)\Big).$$

(ii) Put $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$ if $k \geq 2$, and $c_F = 0$ if $k = 1$. For large N, uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2}\log T + \log K_F - T\lambda_F + \frac{c_F}{T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big). \tag{10}$$

(iii) Let G also satisfy (H)*, with Δ, δ and D_N as in Theorem 4.7. For large N, every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies*

$$T_N\Big(\Delta L - \frac{\Delta}{2}\log T_N - \delta T_N - \log\frac{K_F}{K_G}\Big) = c_F - c_G + O\Big(\frac{1}{L}\Big). \tag{11}$$

(iv) Let $F = \{i, j\}$ with $q_{ij} = q_{ji} > 0$, $q_i = q_j$ and $\mu_i = \mu_j > 0$. Then $c_F = -1/(8q_{ij})$. Let $Q = J - dI$, and let F have $k \geq 2$ states with μ constant and positive on F. Then $c_F = -(k-1)/8$.

In words, the local law has a relative correction $c(\alpha)/T$, where $c(\alpha)$ is the Gaussian mean of a polynomial in the Taylor coefficients of the torus integrand at its peak, divided by the value of the amplitude of the integrand there. The best composition does not sit exactly at $N\alpha_F^*$. It moves by about $N\Sigma_F\gamma_F/T$ toward larger C_F, and this gains $\frac{1}{2}\gamma_F^\top\Sigma_F\gamma_F/T$ in the logarithm. The crossing equation (8) then gains the term $(c_G - c_F)/T_N$ on its right side, with an error $O(L^{-2})$.

Idea of the proof. The proof is in Appendix B. For (i) we carry the Laplace lemma one step further (Lemma B.2). The terms of order $T^{-1/2}$ and $T^{-3/2}$ are odd and integrate to 0, so the error is $O(T^{-2})$. For (ii) the logarithm of the local law near α_F^* is a concave quadratic plus the linear term $\gamma_F \cdot x$, and we complete the square. The maximizer is within $O(1/T)$ of α_F^*, and moving it to the nearest lattice point changes the logarithm only by $O(T/N^2)$. Part (iii) subtracts (10) for the two faces. In (iv) we do not evaluate c from its formula. We compare our expansion with a second expansion of the same probability.

The two values in (iv) have different sources. The value for a symmetric pair is proved here. It uses only Proposition 4.9, the Bessel bound (2) of [2] and the classical expansion of $I_0 + I_1$. These give

$$\log M_{\{i,j\}} = -L + \tfrac{1}{2}\log T + \log K_{\{i,j\}} - T\lambda_{\{i,j\}} - \frac{1}{8q_{ij}T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big), \tag{12}$$

for large N and uniformly for $1 \leq T \leq N^{1/3}$. This expansion does not use the Laplace lemma. The value for the complete graph rests on the expansion of the balanced bins in [3, Theorem 3.1(i)],

which we quote and do not prove again. In both cases the formula (22) gives the same value. For the pair this is a short computation in the proof of Proposition 4.14(iv) in Appendix B.2, and for the full faces of K_3 and K_4 the script `verify_second_order.py` evaluates (22) exactly. For other faces c_F has to be computed from (22), γ_F and Σ_F, as we do for the 4-cycle in Section 5.

4.7 A tree formula for every face

The next proposition extends Proposition 4.2(b) from reversible faces to every irreducible face. It uses two counts of spanning trees in place of one, namely tree as before and the count arb of spanning trees oriented toward a root, which is defined just before Lemma A.3. Its proof rests on Lemma A.3 in Appendix A.

Proposition 4.15 (the constant on a general face)**.** *Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ put*

$$f_{ij}(\alpha) = l_{\alpha,i}\, q_{ij}\, r_{\alpha,j} \qquad (i \neq j \text{ in } F),$$

which does not change when r_α is scaled.

(a) The weights $f(\alpha)$ are the stationary flow of the tilted chain, that is, $f_{ij}(\alpha) = \alpha_i(\widehat{Q}_\alpha)_{ij}$, and $\sum_{j\neq i} f_{ij}(\alpha) = \sum_{j\neq i} f_{ji}(\alpha) = \alpha_i\theta_i(\alpha)$ for every i. The number $\mathrm{arb}_i(f(\alpha))$ does not depend on the root i, and

$$\det \Sigma_\alpha = \frac{(\prod_i \alpha_i)^2\ \mathrm{tree}\big(f(\alpha) + f(\alpha)^\top\big)}{\mathrm{arb}(f(\alpha))^2}$$

and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\mathrm{arb}(f(\alpha))}{\prod_i \alpha_i\, \sqrt{\mathrm{tree}(f(\alpha) + f(\alpha)^\top)}}.$$

(b) Also $\mathrm{arb}(f(\alpha))/\prod_i \alpha_i = \sum_{i\in F} \det\big(\mathrm{diag}\,\theta(\alpha) - A_F\big)_{(i)}$. At $\alpha = \alpha_F^*$ this is $\prod_\rho(\rho - \lambda_F)$, the product over the other $k-1$ eigenvalues ρ of $-Q_F$, counted with multiplicity. So with $l_F \cdot r_F = 1$ and $f_{ij} = l_{F,i}\, q_{ij}\, r_{F,j}$,

$$K_F = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\mathrm{arb}(f)}{\prod_i \alpha^*_{F,i}\, \sqrt{\mathrm{tree}(f + f^\top)}} = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\prod_\rho(\rho - \lambda_F)}{\sqrt{\mathrm{tree}(f + f^\top)}}.$$

(c) In the reversible case of Proposition 4.2(b), $f(\alpha)$ is symmetric and equals $c(\alpha)$, and (a) *reduces to Proposition 4.2(b).*

(d) Let $F = \{1, \dots, k\}$ be a directed cycle, that is, $q_{i,i+1} = a_i > 0$ with indices modulo k, and $q_{ij} = 0$ for all other $i \neq j$ in F. Then for every $\alpha \in \Delta_F^\circ$

$$\det \Sigma_\alpha = \frac{k\,(\prod_i \alpha_i)^2}{\big(\prod_i \alpha_i a_i\big)^{(k-1)/k}}.$$

If moreover $q_i = a_i$ for all i, then $\alpha^*_{F,i} = \varphi^*/a_i$ with $\varphi^* = (\sum_i 1/a_i)^{-1}$, and

$$\det \Sigma_{\alpha_F^*} = \frac{k\,(\varphi^*)^{k+1}}{\prod_i a_i^2}, \qquad K_F = \frac{\mu(F)\,\prod_i a_i}{(2\pi)^{(k-1)/2}\,\sqrt{k}\,(\varphi^*)^{(k+1)/2}}.$$

In words, up to the factor $(2\pi)^{-(k-1)/2}/\prod_i \alpha_i$, the Gaussian factor in C_F is a count of oriented spanning trees for the stationary flow of the tilted chain, divided by the square root of a count of spanning trees for the symmetrized flow. For the directed 3-cycle of Proposition 4.13 part (d) gives $\varphi^* = b/(1+2b)$ and $K_F = (1+2b)^2/(2\sqrt{3}\,\pi b)$ again.

Idea of the proof. Lemma A.3 applies to the tilted generator $\widehat{Q}_\alpha$, whose stationary law is α. Divided by $\prod_i \alpha_i$, the count of oriented trees is the sum of the principal minors of order $k-1$ of $-\widehat{Q}_\alpha$, and these minors do not change under the diagonal conjugation that defines $\widehat{Q}_\alpha$. At α_F^* the sum of these minors is the elementary symmetric polynomial of degree $k-1$ in the eigenvalues of $-Q_F - \lambda_F I$, namely the product of the numbers $\rho - \lambda_F$ over the other eigenvalues ρ of $-Q_F$. (Up to the sign $(-1)^{k-1}$ this is the derivative of the characteristic polynomial of $-Q_F$ at its root λ_F.)

Proof. (a) By Definition 4.1, $(\widehat{Q}_\alpha)_{ij} = q_{ij} r_{\alpha,j}/r_{\alpha,i}$ for $i \neq j$, and $\alpha_i = l_{\alpha,i} r_{\alpha,i}$. So $\alpha_i (\widehat{Q}_\alpha)_{ij} = f_{ij}(\alpha)$. The diagonal entry of $\widehat{Q}_\alpha$ is $-\theta_i(\alpha)$. So $\widehat{Q}_\alpha \mathbf{1} = 0$ gives $\sum_{j\neq i} f_{ij}(\alpha) = \alpha_i \theta_i(\alpha)$, and $\alpha^\top \widehat{Q}_\alpha = 0$ gives $\sum_{j\neq i} f_{ji}(\alpha) = \alpha_i \theta_i(\alpha)$. Now apply Lemma A.3 to $G = \widehat{Q}_\alpha$ and $\pi = \alpha$, for which $\Sigma = \Sigma_\alpha$. Part (iii) is the formula for $\det \Sigma_\alpha$, and Definition 4.1 then gives $C_F(\alpha)$.

(b) We have $-\widehat{Q}_\alpha = R^{-1}(\operatorname{diag}\theta(\alpha) - A_F)R$ with R diagonal, and a conjugation by a diagonal matrix does not change principal minors. So Lemma A.3(i) gives the first claim. At α_F^* we have $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, so $\operatorname{diag}\theta(\alpha_F^*) - A_F = -Q_F - \lambda_F I$. For a $k \times k$ matrix M the sum of the principal minors of order $k-1$ is the elementary symmetric polynomial of degree $k-1$ in the eigenvalues of M. The eigenvalues of $-Q_F - \lambda_F I$ are the numbers $\rho - \lambda_F$, where ρ runs over the eigenvalues of $-Q_F$. Exactly one of them is 0, because the Perron root is a simple eigenvalue. So the symmetric polynomial is the product of the other $k-1$ numbers. Finally $r_{\alpha_F^*}$ and $l_{\alpha_F^*}$ are multiples of r_F and l_F with $l_{\alpha_F^*} \cdot r_{\alpha_F^*} = 1$, and (a) gives the two forms of K_F.

(c) By Proposition 4.2(b), $\alpha_i (\widehat{Q}_\alpha)_{ij} = c_{ij}(\alpha)$, so $f(\alpha) = c(\alpha)$ is symmetric. Then $\operatorname{arb}(f(\alpha)) = \operatorname{tree}(c(\alpha))$, because every spanning tree has exactly one orientation toward a given root. Also $\operatorname{tree}(2c) = 2^{k-1}\operatorname{tree}(c)$, because a spanning tree has $k-1$ edges. So (a) gives $\det \Sigma_\alpha = 2^{k-1}(\prod_i \alpha_i)^2/\operatorname{tree}(c(\alpha))$.

(d) Only the pairs $(i, i+1)$ carry flow, so by (a) the flow $f_{i,i+1}(\alpha)$ out of i equals the flow $f_{i-1,i}(\alpha)$ into i. Hence all $f_{i,i+1}(\alpha)$ are equal to one number φ. Their product is $\prod_i l_{\alpha,i}\, a_i\, r_{\alpha,i+1} = \prod_i \alpha_i a_i$, so $\varphi = (\prod_i \alpha_i a_i)^{1/k}$. The only spanning tree oriented toward a root i with nonzero weight is the cycle without the edge that leaves i. So $\operatorname{arb}(f(\alpha)) = \varphi^{k-1}$. The symmetrized flow has conductance φ on each of the k edges of the undirected cycle. For $k \geq 3$ the spanning trees are the cycle without one edge, so $\operatorname{tree}(f(\alpha) + f(\alpha)^\top) = k\varphi^{k-1}$. For $k = 2$ the two edges of the cycle are the same pair, the conductance is 2φ, and the count is again $k\varphi^{k-1}$. Now (a) gives $\det \Sigma_\alpha = k(\prod_i \alpha_i)^2/\varphi^{k-1}$. If $q_i = a_i$, then Q_F is a generator, so $\lambda_F = 0$, $r_F = \mathbf{1}$ and $l_F = \alpha_F^*$ is the stationary law. Its flow $l_{F,i} a_i$ is constant, which gives $\alpha_{F,i}^* = \varphi^*/a_i$ and $\varphi = \varphi^*$ at α_F^*. Then $\prod_i \alpha_{F,i}^* = (\varphi^*)^k/\prod_i a_i$, and $(\mu \cdot r_F)(l_F \cdot \mathbf{1}) = \mu(F)$. Putting this into the formulas above gives $\det \Sigma_{\alpha_F^*}$ and K_F. □

4.8 The rate through flows

Proposition 3.1(c) gives the rate I_F in closed form on a reversible face. The flow $f(\alpha)$ of Proposition 4.15 gives the rate on every irreducible face, and it shows what a face gains when it is not reversible. A *balanced flow* on F is a family $J = (J_{ij})$ of numbers $J_{ij} \geq 0$, one for each ordered pair $i \neq j$ in F, with $J_{ij} = 0$ when $q_{ij} = 0$ and with $\sum_j J_{ij} = \sum_j J_{ji}$ at every state i. The flow $f(\alpha)$ is balanced (Proposition 4.15(a)). The numbers $\alpha_i q_{ij}$ are the flow of the original chain when

its law is α, and they need not be balanced. For $x \ge 0$ and $y > 0$ let

$$\mathsf{h}(x \,|\, y) = x \log \frac{x}{y} - x + y, \qquad \mathsf{h}(0 \,|\, y) = y.$$

It is the relative entropy per unit time of a Poisson process of rate x with respect to a Poisson process of rate y. It is strictly convex in x, and it vanishes only at $x = y$. Finally let $R_F(\alpha)$ be the right side of the formula in Proposition 3.1(c), which is defined on every face.

Proposition 4.16 (the rate through flows)**.** *Let F be irreducible with $k \ge 2$, let $\alpha \in \Delta_F^\circ$, and let $f = f(\alpha)$.*

(a) The flow f is positive on the edges of G_F, and around every directed cycle of G_F it has the same product as the numbers $\alpha_i q_{ij}$. It is the only balanced flow on F with these two properties. In particular $f_{ij} f_{ji} = \alpha_i \alpha_j q_{ij} q_{ji}$ for all $i \ne j$.

(b) We have

$$I_F(\alpha) = \sum_{i \in F} \alpha_i q_i - \sum_{i \ne j} f_{ij} = \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F) + \min_J \sum_{i \ne j,\ q_{ij} > 0} \mathsf{h}\big(J_{ij} \,|\, \alpha_i q_{ij}\big),$$

where the minimum is over the balanced flows on F, and it is attained only at $J = f$.

(c) The difference $R_F(\alpha) - I_F(\alpha)$ equals $\sum_{\{i,j\}} \big(\sqrt{f_{ij}} - \sqrt{f_{ji}}\big)^2 \ge 0$, the sum over the unordered pairs in F. There is equality if and only if F is reversible, that is, $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$ and some $\nu \in (0, \infty)^F$.

(d) Let $F = \{1, \dots, k\}$ with $k \ge 3$ be a cycle with both directions, that is, $q_{i,i+1} = p_i > 0$ and $q_{i+1,i} = m_i > 0$ with indices modulo k, and $q_{ij} = 0$ for all other $i \ne j$ in F. Put $g_i = \sqrt{\alpha_i \alpha_{i+1} p_i m_i}$. Then

$$I_F(\alpha) = \sum_{i \in F} \alpha_i q_i - \sum_{i=1}^{k} \sqrt{\kappa^2 + 4 g_i^2},$$

where κ is the only real number with

$$\prod_{i=1}^{k} \frac{\kappa + \sqrt{\kappa^2 + 4 g_i^2}}{2} = \prod_{i=1}^{k} \alpha_i p_i.$$

It is the net flow $f_{i,i+1} - f_{i+1,i}$ across every edge, and it has the sign of $\prod_i p_i - \prod_i m_i$.

In words, to hold the fractions α the chain keeps the product of its flow around every cycle and changes the flow until it is balanced. The rate is the drop in the total jump rate, from $\sum_i \alpha_i q_i$ to the total flow of f. It is also the rate of leaving F plus the least cost of a balanced flow. A reversible face balances every pair on its own. A face that is not reversible does better than the formula for reversible faces, because it lets a net flow circulate. On a cycle the size κ of this net flow solves one equation.

Idea of the proof. The tilted chain has rates $q_{ij} r_{\alpha,j} / r_{\alpha,i}$, and the factors $r_{\alpha,j} / r_{\alpha,i}$ cancel around a cycle. A flow with the same cycle products as $\alpha_i q_{ij}$ must have this form for some vector in place of r_α, and balance says that this vector is optimal in the definition of $I_F(\alpha)$. The minimum in (b) is a tangent line bound for h, summed over the edges. Part (c) is the AM–GM inequality once more, and (d) solves a quadratic equation on each edge.

Proof. Let $\psi = \log r_\alpha$, so that $f_{ij} = \alpha_i q_{ij} e^{\psi_j - \psi_i}$, by the definition of f and $\alpha_i = l_{\alpha,i} r_{\alpha,i}$.

(a) An edge of G_F has $q_{ij} > 0$, so $f_{ij} > 0$ there. Around a directed cycle the differences $\psi_j - \psi_i$ sum to 0, so f and the numbers $\alpha_i q_{ij}$ have the same product. A pair i, j with $q_{ij} q_{ji} > 0$ is a cycle of length two, and if $q_{ij} q_{ji} = 0$ both sides of the last claim vanish. Now let J be a balanced flow with the two properties, and put $\ell_{ij} = \log(J_{ij}/(\alpha_i q_{ij}))$ on the edges of G_F. A closed walk in G_F is made of directed cycles, so the sum of ℓ along every closed walk is 0. Take two walks from state 1 to a state i, and follow each of them by the same walk from i back to 1, which exists because F is irreducible. Both closed walks have sum 0, so the two walks from 1 to i have the same sum. Call it χ_i, with $\chi_1 = 0$. A walk to i followed by an edge from i to j is a walk to j, so $\chi_j = \chi_i + \ell_{ij}$, which says that $J_{ij} = \alpha_i q_{ij} e^{\chi_j - \chi_i}$. In the notation of the proof of Proposition 3.1, $J_{ij} = \alpha_i G^\chi_{ij}$, and the balance of J says that α is the stationary law of G^χ. By the criterion there, χ attains $I_F(\alpha)$, and by Proposition 3.1(b) it differs from ψ by a constant. Hence $J = f$.

(b) By Proposition 4.15(a), $\sum_{j \neq i} f_{ij} = \alpha_i \theta_i(\alpha)$, and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$. This is the first formula. For the second we use that, for $x \geq 0$, $y > 0$ and real t,

$$\mathsf{h}(x \,|\, y) \geq xt - y(e^t - 1), \qquad \text{with equality exactly when } x = ye^t.$$

Indeed the difference of the two sides is $\mathsf{h}(x \,|\, ye^t)$. Let J be a balanced flow. We use the bound on each edge with $t = \psi_j - \psi_i$ and add. This gives

$$\sum_{q_{ij} > 0} \mathsf{h}\big(J_{ij} \,|\, \alpha_i q_{ij}\big) \geq \sum_{i \neq j} J_{ij}(\psi_j - \psi_i) + \sum_{i \neq j} \alpha_i q_{ij} - \sum_{i \neq j} f_{ij}.$$

The first sum on the right is $\sum_i \psi_i \big(\sum_j J_{ji} - \sum_j J_{ij} \big) = 0$. The second is $\sum_i \alpha_i (q_i - q_i^{\mathrm{out}}(F))$. So the right side is $I_F(\alpha) - \sum_i \alpha_i q_i^{\mathrm{out}}(F)$ by the first formula. There is equality exactly when $J_{ij} = \alpha_i q_{ij} e^{\psi_j - \psi_i} = f_{ij}$ on every edge.

(c) Since $(\sqrt{x} - \sqrt{y})^2 = x + y - 2\sqrt{xy}$ and $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, we have $R_F(\alpha) = \sum_i \alpha_i q_i - 2 \sum_{\{i,j\}} \sqrt{\alpha_i q_{ij} \alpha_j q_{ji}}$, and the square root is $\sqrt{f_{ij} f_{ji}}$ by (a). With the first formula of (b),

$$R_F(\alpha) - I_F(\alpha) = \sum_{\{i,j\}} \big(f_{ij} + f_{ji} - 2\sqrt{f_{ij} f_{ji}} \big) = \sum_{\{i,j\}} \big(\sqrt{f_{ij}} - \sqrt{f_{ji}} \big)^2.$$

This vanishes exactly when f is symmetric. If F is reversible, then $f = c(\alpha)$ is symmetric by Proposition 4.2(b). Conversely, $f_{ij} = f_{ji}$ reads $\alpha_i q_{ij} r_{\alpha,j} / r_{\alpha,i} = \alpha_j q_{ji} r_{\alpha,i} / r_{\alpha,j}$, which is $\nu_i q_{ij} = \nu_j q_{ji}$ for $\nu_i = \alpha_i / r_{\alpha,i}^2$.

(d) Only the pairs $\{i, i+1\}$ carry flow. The balance at state i says that $f_{i,i+1} - f_{i+1,i} = f_{i-1,i} - f_{i,i-1}$, so the net flow across an edge is one number κ for all edges. By (a), $f_{i,i+1} f_{i+1,i} = g_i^2$. So $f_{i,i+1}$ is the positive root of $x^2 - \kappa x - g_i^2 = 0$, namely $\frac{1}{2}(\kappa + \sqrt{\kappa^2 + 4g_i^2})$, and $f_{i+1,i} = \frac{1}{2}(-\kappa + \sqrt{\kappa^2 + 4g_i^2})$. Their sum is $\sqrt{\kappa^2 + 4g_i^2}$, and (b) gives the formula for $I_F(\alpha)$. As $k \geq 3$, the states $1, 2, \ldots, k$ form a directed cycle, and (a) gives $\prod_i f_{i,i+1} = \prod_i \alpha_i p_i$, which is the equation for κ. Its left side is continuous and strictly increasing in κ, with the limits 0 and ∞, so κ is the only solution. At $\kappa = 0$ the left side is $\prod_i g_i = \prod_i \alpha_i \sqrt{p_i m_i}$, which is less than the right side exactly when $\prod_i m_i < \prod_i p_i$. This gives the sign of κ. □

Remark 4.17 (three special cases, and the sources). (1) If the pairs $\{i, j\}$ with $q_{ij} > 0$ form a tree, then removing a pair cuts F in two parts, and a balanced flow has no net flow between them. So f is symmetric, such a face is reversible, and Proposition 3.1(c) is the whole answer. (2) On a directed cycle with rates $q_{i,i+1} = a_i$, the flow f takes one value φ on all edges, and the

cycle product in (a) gives $\varphi^k = \prod_i \alpha_i a_i$. So $I_F(\alpha) = \sum_i \alpha_i q_i - k(\prod_i \alpha_i a_i)^{1/k}$, as in the proofs of Propositions 4.13 and 4.15(d). Part (d) joins this formula to the reversible one. (3) By the proof of Proposition 3.1(c), $I_F \le R_F$ on all of Δ_F, and $R_F(\alpha) = -x^\top S_F x$ for $x = \sqrt{\alpha}$ (entrywise), where S_F has the diagonal of Q_F and the off-diagonal entries $\sqrt{q_{ij}q_{ji}}$. The least value of R_F on Δ_F is minus the largest eigenvalue of S_F, because S_F has an eigenvector with nonnegative entries for this eigenvalue. The least value of I_F is λ_F. So the largest real eigenvalue of Q_F is at least that of S_F, which is the inequality of Schwenk [33] for Q_F plus a multiple of the identity. Part (c) adds that the inequality is strict unless F is reversible. Indeed, let R_F be least at $\tilde{\alpha}$. If $\tilde{\alpha} = \alpha_F^*$ then $R_F(\tilde{\alpha}) > I_F(\tilde{\alpha})$ by (c), and otherwise $R_F(\tilde{\alpha}) \ge I_F(\tilde{\alpha}) > \lambda_F$.

The cost $\sum \mathsf{h}$ is the rate function of Bertini, Faggionato and Gabrielli [26] for the empirical measure together with the empirical flow of a chain that is not killed. Its least value over the flows is the Donsker–Varadhan rate (see [25, (1.3) to (1.7)]). The second formula in (b) is this statement for the chain killed on leaving F, with a direct proof, and (a) names the minimizer. In fact the killing only adds a linear term. Writing $Q_F u$ out shows that $I_F(\alpha) = \sum_{i\in F} \alpha_i q_i^{\mathrm{out}}(F) + I_F^0(\alpha)$, where I_F^0 is the Donsker–Varadhan rate of the chain on F with the rates q_{ij}, $i, j \in F$, and no killing. So several statements of this section are killed forms of known ones. The uniqueness of the tilt r_α up to scaling is [21, Lemmas A.2 and A.3] (stated as [25, Lemmas 5.5 and 5.6]), the existence and uniqueness of the minimizing flow in (b) is [27, Theorem 1.6], and its form $\alpha_i q_{ij} r_{\alpha,j}/r_{\alpha,i}$ is [25, Lemma 5.7]. The rate of the directed cycle in (2) is [25, Example 5.4] for three states with unit rates, and [25, Proposition 5.13] computes the second derivative of I_F^0.

Example 4.18 (a cycle with a drift)**.** Let F be the whole state space $\{1, \dots, k\}$ with $k \ge 3$, and let $q_{i,i+1} = p$ and $q_{i+1,i} = m$ with $p > m > 0$, and no other rates. Then $q_i = p + m$, the chain is not reversible, and Proposition 4.16(d) gives

$$I_F(\alpha) = p + m - \sum_{i=1}^{k} \sqrt{\kappa^2 + 4pm\,\alpha_i\alpha_{i+1}}, \qquad \prod_{i=1}^{k} \frac{\kappa + \sqrt{\kappa^2 + 4pm\,\alpha_i\alpha_{i+1}}}{2} = p^k \prod_{i=1}^{k} \alpha_i.$$

At the uniform law, $\kappa = (p-m)/k$ solves the equation, and $I_F = 0$, as it must be at the stationary law. The formula for reversible faces gives $R_F = (\sqrt{p} - \sqrt{m})^2 > 0$ there. For $k = 3$, $p = 2$, $m = 1$ and $\alpha = (0.5, 0.3, 0.2)$ we get $\kappa = 0.3059$ and $I_F(\alpha) = 0.1600$, while $R_F(\alpha) = 0.3173$. If k is even and α alternates between two values a and b, then all g_i are equal, $\kappa = (p-m)\sqrt{ab}$, and $I_F(\alpha) = (p+m)(1 - k\sqrt{ab})$. As $m \to 0$ the equation becomes $\kappa^k = p^k \prod_i \alpha_i$, and we recover the directed cycle of Remark 4.17.

5 The 4-cycle and other graphs

In this section Q is symmetric. So the admissible faces are the connected faces with positive start mass, and they satisfy (H) (Lemma 2.2). By Theorem 3.4(c), Theorem 4.8 applies at every tie. We call the faces in $\mathcal{F}_\tau$ the *winners* at τ. We first complete the worked example of Section 3.1 with all its constants, then show on K_3 that the start can decide the winner, and then list the outcomes for other graphs.

Corollary 5.1 (the 4-cycle cascade)**.** *Let Q be the 4-cycle C_4 with $q_{i,i\pm1} = 1$ (indices mod 4), and let μ be uniform.*

(a) The admissible faces are the vertices, the adjacent pairs, the arcs of three states and C_4, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. The winners are the vertices for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and C_4 for $\tau > 2$. The arcs of three states never win.

(b) *The constants are* $K_{\text{vertex}} = \frac{1}{4}$, $K_{\text{pair}} = 1/\sqrt{2\pi}$, $K_{\text{arc3}} = (4+3\sqrt{2})/(4\pi)$ *and* $K_{C_4} = 8/\pi^{3/2}$, *and* $\det \Sigma = 1/512$ *on* C_4.

(c) *For every* $N \geq 2$ *and* $0 < h < \frac{1}{2}$, $M_{\text{pair}} = M_{\text{vertex}} S_N(u)$ *with* $u = h/(1-2h)$. *So the face maxima of a vertex and an adjacent pair cross exactly at* $u = u_N$, *the crossing of [2]. For large* N *this crossing satisfies* $T + \frac{1}{2}\log T = L + c_0 + O(1/L)$.

(d) *For large* N, *every crossing* $T_N \in [1, N^{1/3}]$ *of the face maxima of an adjacent pair and* C_4 *satisfies* $T_N + \log T_N = 2L - \log(8\sqrt{2}/\pi) + O(1/L)$.

(e) *Put* $b = -\frac{1}{2}\log 2 - \frac{1}{2}\log(8\sqrt{2}/\pi)$. *In the tie window at* $\tau = 1$ *the face that carries the global mode changes from the vertices to the adjacent pairs only at* $T = L - \frac{1}{2}\log L + c_0 + o(1)$. *In the tie window at* $\tau = 2$ *it changes from the adjacent pairs to* C_4 *only at* $T = 2(L - \frac{1}{2}\log L + b) + o(1)$.

In words, the face maxima of a vertex and of an adjacent pair are equal exactly when the odds $h/(1-2h)$ equal the odds u_N at the crossing of the two-state walk, and this holds at every N. For large N this is where the mode moves from the vertices to the adjacent pairs. The mode then moves from the pairs to the whole cycle at a crossing whose equation has the coefficient $1 = \frac{\Delta}{2}$ of $\log T$, since the cycle has two more states than a pair.

Proof. (a) A proper face that is not an arc is disconnected, so it is not admissible. For an adjacent pair $-Q_F$ has rows $(2,-1)$ and $(-1,2)$, with least eigenvalue 1. For an arc of three states $-Q_F = 2I - A$, where A is the adjacency matrix of the path on three states, and $r = (1/\sqrt{2}, 1, 1/\sqrt{2})$ is a positive eigenvector with eigenvalue $2-\sqrt{2}$. Also $Q\mathbf{1} = 0$. So Φ_τ equals 2τ, $\tau+1$, $(2-\sqrt{2})\tau+2$ and 3, and comparing the first, second and fourth gives the winners. Since $2-\sqrt{2} > \frac{1}{2}$, the 3-arc has $\Phi_\tau > \frac{1}{2}(\tau+1) + \frac{1}{2}\cdot 3$, which is at least the smaller of the exponents of the pair and of C_4. So it never wins.

(b) A vertex has $K = \mu_i = \frac{1}{4}$. The pair and C_4 have symmetric Q_F with constant row sums, so Corollary 4.3 applies, with $\mu(F) = \frac{1}{2}$ and one spanning tree for the pair, and $\mu(F) = 1$ and four spanning trees for C_4. On C_4 the law α^* is uniform and every edge has conductance $\frac{1}{4}$, so Proposition 4.2(b) gives $\det \Sigma = 2^3 \cdot 4^{-8}/(4 \cdot 4^{-3}) = 1/512$. The 3-arc has unequal row sums, and we use Proposition 4.2(b) with $\nu = \mathbf{1}$. Here $\alpha^* = r \circ r/|r|^2 = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and $r_\alpha = l_\alpha = \sqrt{\alpha^*}$, so $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \frac{1}{4}(1 + 1/\sqrt{2})^2 = (3+2\sqrt{2})/8$. The only spanning tree is the path, with $\text{tree}(c) = c_{12}c_{23} = \frac{1}{8}$, and $\prod_i \alpha_i^* = \frac{1}{32}$. So $K_{\text{arc3}} = \frac{3+2\sqrt{2}}{8} \cdot \frac{32}{2\sqrt{2}} \cdot \frac{1}{4\pi} = \frac{4+3\sqrt{2}}{4\pi}$.

(c) The identity is Proposition 4.9 with $a = 1$ and $q = 2$. So the two face maxima are equal only at $u = u_N$, and by Theorem 4.7(i) this crossing lies in $[L/2, 2L]$ for large N. As $K_{\text{pair}}/K_{\text{vertex}} = 2\sqrt{2/\pi} = e^{-c_0}$, Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation.

(d) This is Theorem 4.7 with $F = C_4$, G an adjacent pair, $\Delta = 2$, $\delta = 1$ and $K_F/K_G = 8\sqrt{2\pi}/\pi^{3/2} = 8\sqrt{2}/\pi$.

(e) At $\tau = 1$ the tied faces are the vertices and the adjacent pairs, and at $\tau = 2$ they are the adjacent pairs and C_4. Faces of one type have the same line. We write $T = \tau(L - \frac{1}{2}\log L + t)$ as in Theorem 4.8, and let E be the upper envelope of the lines in that theorem. By Remark 4.12 the kink of E is at $t = b_{FG}$, which is c_0 at $\tau = 1$ by (c) and b at $\tau = 2$ by (d). So Theorem 4.8 gives (e). □

The finite parts of (a) are checked in Lean. Lean also checks that a closed form for the constants of all arcs, which we do not use, gives the values of K_{pair} and K_{arc3} above.

Proposition 4.14 gives the next term c_F/T of the face maxima, with the explicit coefficient $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$, where c is given by (22). For an adjacent pair, Proposition 4.14(iv) gives $c_{\text{pair}} = -\frac{1}{8}$. For the 3-arc and for C_4 we evaluated the coefficient exactly by computer algebra (the script `verify_second_order.py`), and the values are $c_{\text{arc3}} = \frac{5}{4}\sqrt{2} - 2$ and $c_{C_4} = -\frac{9}{16}$. So these two constants are exact evaluations of the formula of Proposition 4.14, done by computer and not by hand. With the evaluated constant c_{C_4}, Proposition 4.14(iii) gives the next term at the crossings in (d), namely

$$T_N\Big(2L - \log T_N - T_N - \log\frac{8\sqrt{2}}{\pi}\Big) = c_{C_4} - c_{\text{pair}} + O\Big(\frac{1}{L}\Big) = -\frac{7}{16} + O\Big(\frac{1}{L}\Big).$$

Theorem 4.7 also applies to a face that never carries the mode, and this is a good test of the implicit equation (8). Take the 3-arc F against a vertex G, so that $\Delta = 2$, $\delta = \sqrt{2}$ and $\log(K_F/K_G) = \log((4+3\sqrt{2})/\pi) = 0.964591$. At $N = 10^{14}$ a numerical computation of the face maxima, over the symmetric compositions $(a, N-2a, a)$, puts the crossing at $T_N = 42.26334$, where $\Delta L - \frac{\Delta}{2}\log T_N - \delta T_N = 0.959070$. The root of the implicit equation is within 0.004 of T_N, while the explicit form in Theorem 4.7 gives 42.2059. Also $T_N(0.959070 - 0.964591) = -0.2333$, close to the limit $c_{\text{arc3}} - c_{\text{vertex}} = -0.2322$ that Proposition 4.14(iii) gives with the evaluated constant c_{arc3} (a vertex has $c_{\text{vertex}} = 0$).

By Corollary 4.3 the start enters K_F only through $\mu(F)$, and this can change which face wins inside a tie window.

Corollary 5.2 (the start on K_3)**.** *Let $d = 3$ and $Q = J - 3I$, put $T = L - \frac{1}{2}\log L + t$, and let $w = \log\frac{16}{9\sqrt{3}} = 0.026$.*

(a) Let $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$. For every $\eta > 0$ and large N, if $c_0 + \eta \le t \le c_0 + w - \eta$, then every global mode lies on the face $\{1, 2\}$.

(b) Let μ be uniform. For every compact set of t and large N, no global mode lies on a face of two states.

Proof. We compare the lines of Theorem 4.8. All admissible faces tie at $\tau^* = 1$, because $\lambda_F = 3 - k_F$ (Corollary 4.11), and Theorem 4.8 applies since Q is symmetric. By Corollary 4.3 the lines are

$$\log\mu_i - 2t \ \text{ for a vertex } i, \qquad \log\big((\mu_i + \mu_j)\sqrt{2/\pi}\big) - t \ \text{ for an edge } \{i, j\}, \qquad \log K_{[3]} \ \text{ for } [3],$$

with $K_{[3]} = 3^{5/2}/(4\pi)$. A vertex line is the steepest and the line of [3] is flat.

(a) The admissible faces are the vertices 1 and 2, the three edges and [3]. The edge $\{1, 2\}$ has the start mass 1 and the other two edges have $\frac{1}{2}$, so its line is the highest edge line. It lies above the vertex line $-\log 2 - 2t$ exactly when $t > -\log 2 - \frac{1}{2}\log\frac{2}{\pi} = c_0$, and above the line of [3] exactly when $t < \frac{1}{2}\log\frac{2}{\pi} - \log K_{[3]} = c_0 + w$. So the upper envelope E of the lines in Theorem 4.8 is the line of $\{1, 2\}$ on $[c_0, c_0 + w]$, its kinks are c_0 and $c_0 + w$, and no other face has this line. The consequence in Theorem 4.8 gives (a).

(b) Now every vertex has the line $\log\frac{1}{3} - 2t$ and every edge has the line $\log(\frac{2}{3}\sqrt{2/\pi}) - t$. Let t^* be the point where the vertex line meets the line of [3]. The gap between $\max(\log\frac{1}{3} - 2t, \log K_{[3]})$ and the edge line decreases for $t < t^*$ and increases for $t > t^*$, so it is least at t^*, and there it is $\frac{1}{2}\log(3^{7/2}/32) > 0.18$. So the edge line lies below E by at least 0.18 for every t, and by Theorem 4.8(i) the face maximum of an edge is smaller than that of a vertex or of [3], for t in a compact set and large N. □

graph	winners in order, with the value of τ where each one starts	status
K_d	$\lambda_F = d - \lvert F\rvert$. Vertices for $\tau < 1$ and $[d]$ for $\tau > 1$. Every face ties at $\tau = 1$, and [3] resolves the tie at second order.	Lean
symmetric Q	$\lambda_{[d]\setminus\{x\}} \le q_x/(d-1)$, with equality exactly when $q_{yx} = q_x/(d-1)$ for all $y \ne x$. With unit rates on a connected graph, the winners go from vertices straight to $[d]$ exactly when the graph is complete.	proved here
C_d, $d \ge 4$	Arcs of $k = 1, \dots, k_d$ states with $k_d = \lfloor 2d/3 \rfloor$, then C_d. The arc of $k \ge 2$ states wins from $1/(\lambda_{k-1} - \lambda_k)$ with $\lambda_k = 2 - 2\cos\frac{\pi}{k+1}$, and C_d from $\lceil d/3 \rceil/\lambda_{k_d}$. These values are 1, $1+\sqrt{2}$, 4.906 and 8.771 for $k = 2, \dots, 5$, and they grow like $k^3/(2\pi^2)$. Arcs of $k \ge 3$ states win exactly when $d \ge \lceil 3k/2 \rceil$.	Lean, asymptotics by hand
P_d, $d \ge 3$	End-arcs of $1, \dots, \lfloor 2d/3 \rfloor$ states, then P_d. The end-arc of k states has exit rate $2 - 2\cos\frac{\pi}{2k+1}$, and the first switch is at the golden ratio $(1+\sqrt{5})/2$.	Proposition C.3
P_4 at finite N	The end-arc of three states never wins, and its exit rate is only 0.0071 too large to win. Yet with a uniform start it is the exact mode for $T \in (9.381, 12.502)$ at $N = 40$, and on a window at every N tested up to 300.	Proposition C.3, finite N numerical
Θ_ε	Triangles $\{1,2,3\}$ and $\{4,5,6\}$ joined by a link 3–4 of rate $\varepsilon < \varepsilon_1 = (64+\sqrt{46})/60$. The states $1,2,5,6$, then $\{1,2\}$ and $\{5,6\}$ from 1, the triangles from $1 + \varepsilon/3 + O(\varepsilon^2)$, and [6] from $9/\varepsilon + 2 + O(\varepsilon)$.	Proposition C.4, with two exact resultants
paw	A triangle $\{1,2,3\}$ with a leaf 4 at 1. First $\{4\}$, then $\{1,4\}$ from $1+\sqrt{2}$, $\{1,2,3\}$ from $\sqrt{2}+\sqrt{3}$, and all four states from $2+\sqrt{3}$.	Lean
house	The square $1,2,3,4$ with a roof state 5 joined to $1,2$. First $\{3\}, \{4\}, \{5\}$, then $\{3,4\}$ from 1, $\{1,2,5\}$ from $1+\sqrt{2}$, and all five states from $2+\sqrt{2}$.	Lean
K_4 plus a vertex	A state 5 joined to $1,2$ of K_4. First $\{5\}$, then $\{1,2,3,4\}$ from $3(\sqrt{17}+1)/8 = 1.921$, and all five states from $(5+\sqrt{17})/4 = 2.281$.	Lean

Table 2: First-order winners on some graphs with unit rates and a start of full support (the link of Θ_ε has rate ε, and the second row allows any symmetric Q). The last column says where each row is proved (Remark 5.3).

Part (a) recovers [3, Example 4.9], which identifies the global modes exactly, with the same window width w. Here it is an instance of Theorem 4.8. The left end c_0 is the crossing of the two-state walk [2]. Indeed Proposition 4.9 with $a = 1$ and $q = 2$ shows that for every N the face maxima of $\{1,2\}$ and of a vertex cross exactly at $u = u_N$. The comparison of the lines in (a) and the value of w are checked in Lean. With the uniform start the mode jumps from the vertices straight to [3], as [3] shows for every complete graph. The same paper also compares the window in (a) with exact computations at finite N, where it approaches its width w slowly.

Remark 5.3 (the outcomes table). Table 2 lists the outcomes for other graphs. Each row comes from Theorem 3.4 applied to the exit rates of the connected faces. So a row needs two things, namely the exit rates and the points that lie on the lower-left boundary of the hull. The second row has a short proof. By Rayleigh–Ritz with the unit vector f that equals $(d-1)^{-1/2}$ off x and 0 at x, we get $\lambda_{[d]\setminus\{x\}} \le \sum_{y\ne x} q_{xy}/(d-1) = q_x/(d-1)$, with equality exactly when f is an eigenvector of $-Q$ restricted to $[d] \setminus \{x\}$. Its entry at y is $q_{yx}(d-1)^{-1/2}$, so this happens exactly

when q_{yx} does not depend on y, that is, $q_{yx} = q_x/(d-1)$. With unit rates on a connected graph, take x of least degree δ. The vertices with exit rate δ and the face $[d]$ tie at $\tau_0 = (d-1)/\delta$, and there a component C of $[d] \setminus \{x\}$ of least exit rate has $\lambda_C = \lambda_{[d]\setminus\{x\}}$ (the restricted matrix is block diagonal) and $\Phi_{\tau_0}(C) \le \tau_0 \lambda_{[d]\setminus\{x\}} + d - 2 \le d - 1$. Equality needs every q_{yx} to equal $\delta/(d-1)$, which for unit rates means that the graph is complete. So on an incomplete graph the connected face C beats both at τ_0, and on K_d the first row applies. The other rows marked Lean are proved in Lean 4 (v4.33.1 with Mathlib v4.33.1), namely K_d by `complete_vertex_wins`, `complete_full_wins` and `complete_tie`, C_d by `cycle_cascade` and `arcCost_unique_min`, the paw by `paw_phase1` to `paw_phase4`, the house by `house_phase1` to `house_phase4`, and K_4 plus a vertex by `kfv_phase1` to `kfv_phase3`, with the lemmas listed in `MAP.md` (Section 7). On C_d two steps are by hand. A proper face that is not an arc is disconnected, hence not admissible, and the growth of the switch values follows from $2 - 2\cos x = x^2 + O(x^4)$. The rows P_d and Θ_ε are proved in Appendix C, for every $d \ge 3$ and for every $\varepsilon < \varepsilon_1$ (Propositions C.3 and C.4). The proof for Θ_ε uses two resultants that we computed by computer algebra, and it also shows that the bound ε_1 is sharp. In the row P_4 the first sentence is the case $d = 4$ of Proposition C.3, and the rest is an exact computation of the law at the values of N listed. A well-separated community wins for a long range of τ, such as a triangle of Θ_ε from $\tau \approx 1$ to $\tau \approx 9/\varepsilon$. The mode need not grow around its first vertex. It leaves the leaf of the paw, and on the house and on K_4 plus a vertex it jumps to a disjoint community. The second row explains the last one. The state 5 is joined to only two of the states $1, 2, 3, 4$, so by the second row the point of $\{1, 2, 3, 4\}$ lies strictly left of the chord from $\{5\}$ to $[5]$, and the winners cannot jump from $\{5\}$ straight to $[5]$. At feasible N the switch values are approached slowly (Figure 3(b)), and on P_4 even the order of the phases differs.

6 Application: sticky priors for hidden Markov models

This section is an application for readers from statistics. Nothing earlier depends on it, and a reader who is interested only in the general theory can skip to Section 7.

In a hidden Markov model the hidden states $X_1, \dots, X_N$ form a Markov chain on d states. Before any data are seen, the model already says how the N time steps are likely to be shared among the states. This is the law P_N of the count vector $n = (n_1, \dots, n_d)$, where n_i is the number of time steps in state i (the composition of Section 2). A sticky prior makes the chain leave its current state rarely, which is the regime of this paper. We ask which states the most likely count vector uses.

Here is what we use from the earlier sections. Suppose the chain leaves its state with probability $h = \tau \log N/N$ at each step and then picks the next state from a kernel Π. For a set F of states (a face) let ρ_F be the Perron root of Π restricted to F. The chain stays in F for all N steps with probability about $N^{-\tau(1-\rho_F)}$, and about $N^{|F|-1}$ count vectors use exactly the states of F. Theorem 3.3 says that for large N the most likely count vector uses a set F that minimizes $\Phi_\tau(F) = \tau(1-\rho_F) + |F| - 1$, and that its fractions n/N are then close to the law α_F^*, the entrywise product of the left and right Perron vectors of Π on F. Besides this we use the description of a tie between several minimizers (Theorem 4.8) and the exact identity for a symmetric pair of states (Proposition 4.9), once each.

The answer is as follows. As the stickiness decreases, that is, as τ grows, the most likely count vector moves from one state to larger sets of states. Among symmetric kernels only the uniform one jumps from one state straight to all of them.

Fox et al. [22] add a self-transition bias $\kappa \ge 0$ to the HDP hidden Markov model (HDP stands for hierarchical Dirichlet process), so that the hidden chain stays in each state longer. In their

weak-limit approximation with base weights $\bar\beta$ and concentration ω, row j of the transition matrix $(\tilde\pi_{jl})$ is Dirichlet with parameters $\omega\bar\beta_1,\dots,\omega\bar\beta_j+\kappa,\dots,\omega\bar\beta_d$. By neutrality, row j off the diagonal, divided by $1-\tilde\pi_{jj}$, is Dirichlet with parameters $(\omega\bar\beta_l)_{l\neq j}$ and independent of $\tilde\pi_{jj}$. So κ only sets how sticky the chain is, and the kernel of the switches does not depend on it. In this section we fix every exit probability $1-\tilde\pi_{jj}$ at the same value $h=\tau L/N$, the scale of the crossing of the two-state walk, and study the kernel. Under the prior the exit probabilities are random. By the aggregation property of the Dirichlet law, $1-\tilde\pi_{jj}$ has the law $\mathrm{Beta}(\omega(1-\bar\beta_j),\omega\bar\beta_j+\kappa)$, with mean $\omega(1-\bar\beta_j)/(\omega+\kappa)$, so a mean of order L/N needs $\kappa\asymp N/L$. Mixing over this law favors single states. A single state then keeps probability $\mu_j\mathbb{E}[\tilde\pi_{jj}^{N-1}]$, which decays only like a power of L when $\kappa\asymp N/L$, while at a fixed h of order L/N it decays like a power of N. So the mode of the mixture can differ from the mode studied here, and we do not treat it. All results of this section are about the law given Π and h.

So let $\Pi=(\pi_{ij})$ be a stochastic matrix on $[d]$ with zero diagonal, $\tau>0$ and $h=\tau L/N<1$. The hidden chain has transition matrix $(1-h)I+h\Pi=I+hQ$ with $Q=\Pi-I$, so $q_i=1$ and $T=\tau L$ (Example 2.5(iii)). We assume that $\pi_{ij}>0$ for all $i\neq j$, which we call (A+), and that the start μ has full support. Under the prior, (A+) holds almost surely. Let $\rho_F=\varrho(\Pi_F)$ for the principal submatrix Π_F on F. Then the exit rate of F is $\lambda_F=1-\rho_F$, and Q_F and Π_F have the same Perron vectors. A vertex (a single state) has $\rho_F=0$, and $\rho_{[d]}=1$. If $2\le|F|\le d-1$, then Π_F is irreducible by (A+), and $0<\rho_F<1$ since ρ_F is at least the least row sum of Π_F and the Perron root of the irreducible Π exceeds that of each proper principal submatrix [34]. So $\Phi_\tau(F)=\tau(1-\rho_F)+|F|-1$, and we put

$$\tau_-=\min_{|F|\ge2}\frac{|F|-1}{\rho_F},\qquad \tau_+=\max_{F\neq[d]}\frac{d-|F|}{1-\rho_F}.$$

We call a global mode of P_N, that is, a most likely value of the count vector before any data are seen, given Π and h, a *conditional mode*.

Corollary 6.1 (support and location of the conditional mode)**.** *Assume* (A+)*, let μ have full support, and fix $\tau>0$. Let $\mathcal{F}_\tau$ be the set of faces that minimize Φ_τ. There is $N_0(\tau)$ such that for $N\ge N_0(\tau)$ every conditional mode n^* satisfies* $\operatorname{supp} n^*\in\mathcal{F}_\tau$ *and $n^*/N=\alpha^*_F+O(1/L)$ with $F=\operatorname{supp} n^*$. If $\mathcal{F}_\tau=\{F^*\}$, then $n^*/N\to\alpha^*_{F^*}=l\circ r$, where l and r are the left and right Perron vectors of Π_{F^*} with $l\cdot r=1$. No reversibility is needed.*

In words, for large N a conditional mode uses exactly the states of a set that minimizes Φ_τ, and it shares the N time steps among them in the proportions α^*_F.

Proof. By (A+) every face is irreducible and, as μ has full support, admissible. So Theorem 3.3(iii) and (iv) give the claims, with $\alpha^*_F=l\circ r$. □

The threshold $N_0(\tau)$ is not uniform in τ. At $\tau=\tau_-$ the vertices tie with the faces that attain τ_-, and Theorem 4.8(iii) with $\tau^*=\tau_-$ and a fixed large $t=t_0$ shows that for large N the conditional mode is not a single state at $\tau=\tau_-(1-\frac{\log L-2t_0}{2L})<\tau_-$. So $N_0(\tau)\to\infty$ as $\tau\uparrow\tau_-$.

The next proposition says for which τ the conditional mode uses one state, all states, or a set in between.

Proposition 6.2 (thresholds)**.** *Assume* (A+) *and $d\ge3$.*

(a) The minimizers of Φ_τ are exactly the vertices if and only if $\tau<\tau_-$, and no vertex minimizes Φ_τ if $\tau>\tau_-$. The face $[d]$ is the unique minimizer if and only if $\tau>\tau_+$, and $[d]$ does not minimize Φ_τ if $\tau<\tau_+$.

(b) The thresholds satisfy $1 < \tau_- \le d-1 \le \tau_+$.

(c) The equality $\tau_- = d-1$ *holds if and only if* $\tau_+ = d-1$, *and both hold if and only if*

$$\rho_F \le \frac{|F|-1}{d-1} \qquad \text{for every face } F. \tag{13}$$

If (13) fails, then $\tau_- < d-1 < \tau_+$, *and for* $\tau \in (\tau_-, \tau_+)$ *every minimizer* F *of* Φ_τ *has* $2 \le |F| \le d-1$.

Consequently, let μ *have full support, fix* τ *and let* $N \ge N_0(\tau)$ *as in Corollary 6.1. If* $\tau < \tau_-$, *the prior modes are the* Ne_i *with* $\mu_i = \max_l \mu_l$. *If* $\tau > \tau_+$, *every conditional mode has full support. If* $\tau_- < \tau < \tau_+$, *every prior mode has a support* F *with* $2 \le |F| \le d-1$, *and* $F = F^*$ *if* F^* *is the unique minimizer of* Φ_τ.

Proof. We compare every face with a vertex and with $[d]$. The function $\Phi_\tau(F) = \tau(1-\rho_F) + |F| - 1$ is affine in τ, so each comparison changes sign at one value of τ, and τ_- and τ_+ are the extreme ones.

(a) A vertex has $\Phi_\tau = \tau$. For $|F| \ge 2$ we have $\rho_F > 0$ and

$$\Phi_\tau(F) - \tau = |F| - 1 - \tau\rho_F = \rho_F\Big(\frac{|F|-1}{\rho_F} - \tau\Big).$$

So F is worse than a vertex exactly when $\tau < (|F|-1)/\rho_F$. This holds for every F with $|F| \ge 2$ exactly when $\tau < \tau_-$, and if $\tau > \tau_-$ some F is better than every vertex. In the same way $\Phi_\tau([d]) = d-1$, and for $F \ne [d]$ we have $1 - \rho_F > 0$ and

$$\Phi_\tau(F) - (d-1) = \tau(1-\rho_F) - (d - |F|) = (1-\rho_F)\Big(\tau - \frac{d-|F|}{1-\rho_F}\Big).$$

So F is worse than $[d]$ exactly when $\tau > (d-|F|)/(1-\rho_F)$, which gives the two claims on $[d]$.

(b) In the minimum that defines τ_-, a face with $2 \le |F| \le d-1$ has $\rho_F < 1$ and so $(|F|-1)/\rho_F > 1$, and $[d]$ gives the value $d-1 \ge 2$. So $\tau_- > 1$ and $\tau_- \le d-1$. In the maximum that defines τ_+, a vertex gives the value $d-1$, so $\tau_+ \ge d-1$.

(c) By (b), $\tau_- = d-1$ exactly when $(|F|-1)/\rho_F \ge d-1$ for all $|F| \ge 2$, and $\tau_+ = d-1$ exactly when $(d-|F|)/(1-\rho_F) \le d-1$ for all $F \ne [d]$. After clearing the denominators both conditions say $(d-1)\rho_F \le |F| - 1$. This inequality always holds for the vertices and for $[d]$, so each condition is (13). If (13) fails, then neither holds, so $\tau_- < d-1 < \tau_+$ by (b), and (a) excludes the vertices and $[d]$ on (τ_-, τ_+).

For the consequence, Corollary 6.1 says that the support of a prior mode minimizes Φ_τ, and (a) and (c) say which faces can do so. If $\tau < \tau_-$ the conditional modes are vertex compositions, and $P_N(Ne_i) = \mu_i(1-h)^{N-1}$ is largest for the largest μ_i. Parts (a) to (c) are checked in Lean. □

We say that Π has a *direct jump* when (13) holds. Then at first order the conditional mode goes from one state straight to all d states at $\tau = d-1$. For the uniform kernel $\pi_{ij} = 1/(d-1)$ the matrix Π_F has constant row sums, so $\rho_F = (|F|-1)/(d-1)$. Hence (13) says that every set of states is at least as easy to leave as under the uniform kernel. That kernel has $Q = (J - dI)/(d-1)$, and all its faces tie at $\tau = d-1$. This is the degeneracy of [3] (Corollary 4.11). The next corollary shows that a direct jump is special.

Corollary 6.3 (only the uniform kernel)**.** *Let $d \geq 3$ and assume* (A+). (a) *If Π is symmetric, then Π has a direct jump if and only if $\pi_{ij} = 1/(d-1)$ for all $i \neq j$.* (b) *If $d = 3$, then Π has a direct jump if and only if $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$ for all three pairs. In particular every cyclic kernel $\pi_{i,i+1} = a$, $\pi_{i,i-1} = 1 - a$ (indices mod 3) has a direct jump.*

Proof. We test (13) on the pairs. The matrix $\Pi_{\{i,j\}}$ has zero diagonal and off-diagonal entries π_{ij} and π_{ji}, so $\rho_{\{i,j\}} = \sqrt{\pi_{ij}\pi_{ji}}$.

(a) Here $\rho_{\{i,j\}} = \pi_{ij}$, so for a pair (13) reads $\pi_{ij} \leq 1/(d-1)$. The d rows sum to 1 and have $d(d-1)$ off-diagonal entries, so the average of the π_{ij} is $1/(d-1)$. If none of them exceeds the average, they are all equal to it. So a direct jump forces the uniform kernel. Conversely the uniform kernel has $\rho_F = (|F|-1)/(d-1)$ for every face, as we saw above, so it satisfies (13) with equality.

(b) For $d = 3$ the faces are the vertices, the three pairs and [3]. For the vertices and [3], (13) reads $0 \leq 0$ and $1 \leq 1$. For a pair it reads $\sqrt{\pi_{ij}\pi_{ji}} \leq \frac{1}{2}$. The cyclic kernel has $\pi_{ij}\pi_{ji} = a(1-a) \leq \frac{1}{4}$ for every pair. Both parts are checked in Lean. □

So for a symmetric kernel that is not uniform there is a range of τ in which the conditional mode uses more than one state but not all of them (Proposition 6.2(c)). Without symmetry a direct jump is possible, and for three states we can say how likely it is under the prior.

Proposition 6.4 (the direct jump under the prior)**.** *Let $d = 3$ and let the base weights be uniform. In the weak-limit sticky HDP-HMM the rows of Π are then independent, and the first off-diagonal entry of each row has the law* Beta$(\omega/3, \omega/3)$. *Write $\mathbb{P}_\omega$ for this law of Π and D for the event that Π has a direct jump. Then* (a) $\mathbb{P}_\omega(D) \to \frac{1}{4}$ *as $\omega \to 0$, and* (b) $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ *as $\omega \to \infty$.*

Proof. Let B_1, B_2, B_3 be independent with law Beta$(\omega/3, \omega/3)$, and put $\pi_{12} = B_1$, $\pi_{21} = B_2$ and $\pi_{31} = B_3$, so that the other off-diagonal entries are $1 - B_j$. By Corollary 6.3(b), D is the event that B_1B_2, $(1-B_1)B_3$ and $(1-B_2)(1-B_3)$ are all at most $\frac{1}{4}$.

(a) As $\omega \to 0$ each B_j tends in law to a fair coin on $\{0,1\}$. So the law of (B_1, B_2, B_3) tends to the uniform law on the eight points of $\{0,1\}^3$. At these points the three products take only the values 0 and 1, so none of the points lies on the boundary of D (where some product equals $\frac{1}{4}$). Hence $\mathbb{P}_\omega(D)$ tends to the limit probability of D, which is the fraction of the eight points at which all three products are 0. Only $(1,0,1)$ and $(0,1,0)$ have this property (checked in Lean), so the limit is $\frac{2}{8} = \frac{1}{4}$. Part (b) is proved in Appendix D. □

In words, a direct jump has probability close to $\frac{1}{4}$ when the concentration is small and of order $1/\omega$ when it is large. For (b), for large ω the set D is a thin tube around the cyclic kernels, with cross-section of area about $\frac{9}{8}t^4$ at the logit coordinate t along the cyclic kernels (Appendix D), which gives the order $1/\omega$. So the uniform tie is not typical. At $\omega = 1$, Monte Carlo with 10^6 draws gives $\mathbb{P}_\omega(D) = 0.115$. So at $\omega = 1$ about 88 percent of the kernels drawn from the prior have a range of τ in which, at first order, the conditional mode sits on a pair (Proposition 6.2(c)).

The comparison of a single state with a symmetric pair is exact at every N.

Corollary 6.5 (an exact link to the two-state walk)**.** *Let $i \neq j$ satisfy $\pi_{ij} = \pi_{ji} = \pi_0 > 0$ and $\mu_i = \mu_j > 0$. We do not assume* (A+). *Then for every $N \geq 2$ the balanced compositions are the most likely ones with support $\{i,j\}$, and they are more likely than Ne_i if and only if $\tau > \tau^{\mathrm{pair}}(N) := Nu_N/((\pi_0 + u_N)L)$. If also $\mu_i = \mu_j = \max_l \mu_l$, then for every τ with $\tau^{\mathrm{pair}}(N) < \tau < N/L$ no conditional mode is a single state. Finally,*

$$\tau^{\mathrm{pair}}(N) = \frac{1}{\pi_0}\Big[1 - \frac{\frac{1}{2}\log L - c_0}{L} + O\Big(\frac{\log L}{L^2}\Big)\Big],$$

which tends to the first-order value $1/\rho_{\{i,j\}} = 1/\pi_0$.

In words, a single state against a symmetric pair is the two-state walk, so the value of τ at which the pair overtakes the state is known at every N and not only in the limit.

Proof. We apply Proposition 4.9. Every exit rate is 1, so its hypotheses hold with $a = \pi_0$ and $q = 1$, and its odds are $u = \pi_0 h/(1-h)$. It says that the balanced compositions are the most likely ones on $\{i, j\}$ and that they beat Ne_i exactly when $u > u_N$. Solving for h, this is $h > u_N/(\pi_0 + u_N)$, and with $h = \tau L/N$ it is $\tau > \tau^{\mathrm{pair}}(N)$.

For the conditional modes, $P_N(Ne_l) = \mu_l(1-h)^{N-1} \le P_N(Ne_i)$ for every l. So when the pair beats Ne_i it beats every single state. The condition $\tau < N/L$ is $h < 1$, and the range of τ is not empty as $u_N/(\pi_0 + u_N) < 1$.

Finally $Nu_N = z_N/p_N$ and $u_N = O(L/N)$, so $\tau^{\mathrm{pair}}(N) = \frac{z_N}{\pi_0 L}(1 + O(L/N))$. By Corollary 4.10, $z_N + \frac{1}{2}\log z_N = L + c_0 + O(1/L)$ and z_N is of order L, so $\log z_N = \log L + O(\log L/L)$ and $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$. Dividing by $\pi_0 L$ gives the claim. □

So far we looked at the single most likely count vector. The last result gives the total probability that the chain uses only the states of a set F.

Proposition 6.6 (face masses)**.** *Assume* (A+) *and* $0 < h < 1$. *For every face* F *with* $|F| \ge 2$ *and* $\Lambda_F = 1 - h\lambda_F$,

$$\frac{\min r_F}{\max r_F}\mu(F)\Lambda_F^{N-1} \le \mathbb{P}(\operatorname{supp} n \subseteq F) \le \frac{\max r_F}{\min r_F}\mu(F)\Lambda_F^{N-1},$$

and for fixed τ, $\Lambda_F^{N-1} = N^{-\tau\lambda_F} e^{O(L^2/N)}$. *Also* $\mathbb{P}(\text{one state}) = (1-h)^{N-1} \sim N^{-\tau}$ *and* $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$ *for fixed* τ.

In words, the chain stays inside F with probability $\mu(F)N^{-\tau\lambda_F}$ up to a bounded factor, and the probability that it uses every state tends to 1.

Proof. We write the probability with the transition matrix restricted to F and compare the vector $\mathbf{1}$ with the Perron vector. Let $U_F = (1-h)I + h\Pi_F \ge 0$. The chain stays in F for all N steps with probability $\mathbb{P}(\operatorname{supp} n \subseteq F) = \mu_F^\top U_F^{N-1}\mathbf{1}$, and $U_F r_F = \Lambda_F r_F$. As $r_F/\max r_F \le \mathbf{1} \le r_F/\min r_F$ entrywise and U_F^{N-1} has nonnegative entries, we may replace $\mathbf{1}$ by these two multiples of r_F. This gives the bounds. For fixed τ, $(N-1)\log(1-h\lambda_F) = -\tau\lambda_F L + O(L^2/N)$. A vertex composition is a single path that never switches, which gives $\mathbb{P}(\text{one state})$. Every $F \ne [d]$ has $\lambda_F > 0$, so each of the finitely many proper faces has probability tending to 0, and $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$. The finite bounds are checked in Lean. □

Remark 6.7 (conditional mode versus mass). Let $d \ge 3$. For fixed $\tau < \tau_+$ and large N every conditional mode has a proper support, while $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$. For $\tau < \tau_-$ the conditional mode uses one state, an event of probability about $N^{-\tau}$. A face of size k holds about N^{k-1} compositions, so it can carry most of the mass while each of them is less likely than a vertex. Table 3 shows this. At $\tau = 0.8$ the conditional mode is a single state, yet $\mathbb{P}(\text{one state}) = 0.0122$. At $\tau = 8$ first order puts the conditional mode on $\{1, 2\}$, but at $N = 240$ it has full support. So the conditional mode can be a poor summary of a sticky prior, and face selection tells us on which states it sits.

τ	conditional mode	one state	$\{1,2\}$	$[3]$
0.5	one state	0.0643	0.465	0.379
0.8	one state	0.0122	0.423	0.531
3	$(120, 120, 0)$	4.3×10^{-8}	0.129	0.871
8	$(111, 111, 18)$	1.2×10^{-21}	0.0081	0.992

Table 3: The kernel Π with rows $(0, 0.9, 0.1)$, $(0.9, 0, 0.1)$, $(0.5, 0.5, 0)$, the uniform start and $N = 240$, where $\tau_- = 10/9$ and $\tau_+ = 10$. The conditional mode (from exact dynamic programming, with a tie of three compositions at $\tau = 0.5$ and 0.8) and the probabilities of the supports.

7 Verification and open questions

The theorems, propositions, lemmas and corollaries of this paper are proved in the text or in the appendices, with two kinds of outside input. We quote results of [2, 3] where we say so (for example the uniform Bessel bound (2), and the expansion behind the value for the complete graph in Proposition 4.14(iv)). And the rows of Table 2 marked Lean are proved in Lean and not in the text. No proof rests on a floating-point computation. Exact computer algebra enters in two places. First, Proposition C.4, which proves the row Θ_ε of Table 2, rests on two exact resultants of characteristic polynomials, computed in sympy by `verify_examples.py` (Section C.3). Second, the values of c_F for the 3-arc and for C_4 after Corollary 5.1 are exact evaluations, by `verify_second_order.py`, of the formula for c_F in Proposition 4.14(ii). The proposition is proved, and only these two values are computed. The script `verify_forest.py` lists the spanning trees one by one. It checks Lemma A.3 in exact rational arithmetic for 119 generators with two to six states. It checks Proposition 4.15(a) to (c) in floating point against Definition 4.1, and part (d) for the directed 3-cycle. Each check is an assertion in the script, and the floating-point checks use a tolerance. The script `verify_rate_flows.py` checks Proposition 4.16 in floating point on 480 random faces with two to five states and on random cycles, and it gives the numbers of Example 4.18.

The other numbers in the paper are of three kinds. Some are closed forms. Some are computations of the law at a finite N, namely Figure 1, the finite-N row of P_4 in Table 2, Figure 3(b), the crossing of the 3-arc at $N = 10^{14}$ in Section 5 and Table 3. These are done in two independent ways where possible (for example by the run expansion and by a torus integral, or by a dynamic program and an exact formula). They illustrate the theorems, and no statement is deduced from them. Finally, the value 0.115 in Section 6 is a Monte Carlo estimate.

Many finite and algebraic steps, and the rows of Table 2 marked Lean, are proved in Lean 4 in the folder `kagey-problems/problem131/lean/PaperC`, and its file `MAP.md` matches each statement of the paper to its Lean theorems. The asymptotic results are not in Lean.

The file `SUPPLEMENT.md` in the folder `kagey-problems/problem131` lists each check, the two methods behind it and the predictions that we wrote down before most of the computations. The data and code availability statement at the end of the paper says where the files are and how to rerun the checks.

Open questions. We list them in the order of the paper.

The local law at the boundary. Theorem 4.4 holds on compact subsets of the open face, and a local law up to the boundary is open. The same formula cannot hold there. Take a star with center d and $\ell = d - 1 \geq 3$ leaves, and a start of full support. Then $P_N(n) = 0$ whenever $\operatorname{supp} n = [d]$ and $n_d < \ell - 1$, since a path visits the center between any two runs at the leaves, although n/N

lies in the open face. This zero set is checked in Lean.

Reducible faces. For a reducible face, Theorem 3.3(ii) places the power of T in M_F between $k_F - k^*$ and $k_F - 1$, and heuristically it is $(k^* - 1)/2 + k_F - k^*$. We do not prove this.

Uniqueness and constants. Theorem 4.7 puts all crossings of two faces in an interval of length $O(1/L)$, but uniqueness is proved only in special cases, such as the two-state walk [2], the complete graph [3], the pairs of Proposition 4.9 and the planar walk [4]. Proposition 4.15 gives a tree formula for K_F on every irreducible face, but we have no closed form for c_F. At finite N the order of the phases can differ from first order when a face lies close to the hull, as on the path P_4 (Table 2).

Linear functions of the composition. For the planar walk the position after N steps is a linear function of the composition, and the origin is a line of compositions. The heuristic of Section 1.1 applies face selection to it. For the planar walk its conclusion is the subject of the companion paper [4], in preparation, but we have no such rule in general.

Sticky priors. We do not treat the mixture over the exit probabilities under the prior of [22] (Section 6), or the probability of a direct jump for $d \geq 4$.

Acknowledgments. I thank Peter Kagey for posing the problem.

A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 4.2(a), Lemma 3.2, Theorem 4.4, Lemma 4.5 and Theorem 3.3(ii). Its last section proves Lemma A.3, which is used for Proposition 4.15. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [34]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M, and every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$.

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 4.1, with (G, π) in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (5). Put $D = \operatorname{diag} \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root)**.** *Let G, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \operatorname{diag} \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \tag{14}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ, which is Γ without the row and column of state 1, is positive definite.

Proof. We differentiate the eigenvalue equation twice at $\eta = 0$. Since $G + \operatorname{diag}\eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. We write subscripts for derivatives at $\eta = 0$. Differentiating $(G + \operatorname{diag}\eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, since $\pi^\top G = 0$. Then $v_j = Ze_j$, because by (5) both vectors solve $Gx = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i, gives $Gv_{ij} + \operatorname{diag}(e_i)v_j + \operatorname{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$. We multiply by $\pi^\top$ again. The terms with G, κ_i and κ_j vanish, and we get $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (14) let $\bar\eta = \eta - (\pi \cdot \eta)\mathbf{1}$. By (5), $-Gu = \bar\eta$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2\sum_i \pi_i \eta_i u_i = 2\sum_i \pi_i \bar\eta_i u_i = -2\sum_i \pi_i u_i (Gu)_i$. Now expand the squares in (14). By $G\mathbf{1} = 0$ and $\pi^\top G = 0$ the terms with u_i^2 and with u_j^2 each sum to $-\sum_i \pi_i G_{ii} u_i^2$, and the cross terms sum to $-2\sum_i \pi_i u_i (Gu)_i + 2\sum_i \pi_i G_{ii} u_i^2$. So both sides of (14) agree. Also $\Gamma\mathbf{1} = 0$ by (5). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of G, so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $\bar\eta = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ. □

Idea of the proof of Proposition 4.2(a). The gradient of $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag}\vartheta)$ is the stationary law of the chain tilted by ϑ, and its Hessian is the matrix Γ of that chain (Lemma A.1). So $\nabla\Lambda$ maps the tilt to the fractions α. We show that ∇I_F is the map back, so the two Hessians are inverse to each other.

Proof of Proposition 4.2(a). Step 1: the derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag}\vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$ (the shift property). It is a simple eigenvalue with positive right and left eigenvectors $\hat r, \hat l$, normalized by $\sum_i \hat r_i = 1$ and $\hat l \cdot \hat r = 1$, and by the implicit function theorem Λ, $\hat r$ and $\hat l$ are real-analytic in ϑ. Put $\pi = \hat l \circ \hat r$ and $G = \hat R^{-1}(Q_F + \operatorname{diag}\vartheta - \Lambda(\vartheta)I)\hat R$ with $\hat R = \operatorname{diag}\hat r$. Then $G\mathbf{1} = 0$ and $\pi^\top G = 0$ by the two eigenvector equations, so G is an irreducible generator with stationary law π. Since $\hat R$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \operatorname{diag}\eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from (G, π).

Step 2: the gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3: the tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 4.1 and (4). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4: the Hessian of I_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $I_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla I_F = \beta$ remains. So $\nabla^2 I_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 4.1. □

A.2 The torus representation

We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\geq 2}$, $b_1 = (q_{i1})_{i\geq 2}$ and $A' = (q_{ij})_{i,j\geq 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z') b_1$ and

$$b(z') = \big(\mu_1 + \mu'^\top X(z') b_1\big)\big(1 + a_1^\top X(z')\mathbf{1}\big).$$

Lemma A.2 (torus representation)**.** *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \geq s_i$ for every $i \geq 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \leq g(s')$ and $|b(z')| \leq b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Big(t_1 g(z') + \sum_{i\geq 2} t_i z_i\Big)\, b(z') \prod_{i\geq 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

In words, $H(t)$ is a Cauchy integral over $k-1$ circles, one for each state of F except state 1. The circle of state 1 has already been integrated out, and it leaves the factor $e^{t_1 g(z')}$.

Idea of the proof. By Cauchy's formula the factor $\prod_i t_i^{m_i - 1}/(m_i - 1)!$ of a word in $H(t)$ is the integral of $e^{\langle t, z\rangle} \prod_i z_i^{1-m_i}$ over k circles. The sum of $\prod_i z_i^{-m_i}$ over all words, with their weights, is a geometric series, namely $\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1}$ with $\mathcal{Z} = \operatorname{diag}(z)$. The variable z_1 enters this resolvent only through the simple pole $1/(z_1 - g(z'))$, so the integral over the first circle is a residue.

Proof. Step 1: the series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. So the series $X(z') = \sum_{j\geq 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \geq s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2: a radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1} A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon\mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \geq 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3: k circles. Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) = \varrho(S^{-1} A_F) < 1$ (the two matrices are similar), expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}} \prod_{i\in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi}\int_0^{2\pi} e^{t_i z_i} z_i^{1-m}\, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \geq 1$ and 0 if $m \leq 0$. We multiply the series by $e^{\langle t, z\rangle} \prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z\rangle}\, \mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} \prod_{i\in F} z_i \, d\varphi. \tag{15}$$

Step 4: the first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \le g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z')\mathbf{1}.$$

We integrate over φ_1 in (15) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi}\int_0^{2\pi} e^{t_1 z_1} z_1 \, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula. □

All coefficients are nonnegative, so the integrand has its largest modulus at $\varphi = 0$. This gives Lemma 3.2, and a Laplace expansion around $\varphi = 0$ gives Theorem 4.4.

A.3 Proof of Lemma 3.2

We bound the sum of Lemma 2.3 term by term by a continuum sum H_σ with slightly larger rates, and H_σ by the largest modulus of the integrand of Lemma A.2.

Step 1: the radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \ge 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \ge 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1, which reads $a_1^\top u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{16}$$

Step 2: a continuum sum. Fix n with support F and h with $hq_{\max} \le \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{hq_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \le n_i$ for every i are nonzero, and in them $h^{m_i - 1}\binom{n_i - 1}{m_i - 1} \le t_i^{m_i - 1}/(m_i - 1)!$ and $(1 - hq_i)^{n_i - m_i} \le e^{-hq_i(n_i - m_i)} \le e^{-q_i t_i}\sigma^{m_i}$. Note that $h^{M-1} = h^{k-1}\prod_i h^{m_i - 1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F (a word with M letters has $M - 1$ switches). So

$$P_{|n|}(n) \le \sigma\, h^{k-1} e^{-\langle q, t\rangle} H_\sigma(t), \tag{17}$$

where H_σ is H for σA_F.

Step 3: the bound for H_σ. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus with radii $\sigma\theta'$ we have $|g_\sigma| \le \sigma g(\theta') = \sigma\theta_1$ by (16), $|b_\sigma| \le b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t, \theta\rangle} b(\theta')\sigma^{k-1}\prod_{i\ge 2}\theta_i$, and for every $\sigma > 0$

$$H_\sigma(t) \le \sigma^{k-1} b(\theta') \prod_{i \ge 2} \theta_i \, e^{\sigma\langle t, \theta\rangle}. \tag{18}$$

With (17) this gives $P_{|n|}(n) \le \sigma^k b(\theta')\prod_{i\ge 2}\theta_i\, h^{k-1}\exp(-h\langle n, q - \theta\rangle + (\sigma - 1)h\langle n, \theta\rangle)$. Finally $\sigma \le e^{1/2}$ and $\sigma - 1 \le 2hq_{\max}$, so the lemma holds with $C_\theta = \max\bigl(e^{k/2} b(\theta')\prod_{i\ge 2}\theta_i,\ 2q_{\max}\max_i \theta_i\bigr)$. The bound after the lemma is the case $\theta = q - \lambda_F \mathbf{1}$. □

A.4 Proof of Theorem 4.4

For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \operatorname{diag}(\theta')$. By (16) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q,\alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha,\varphi)} \chi(\alpha,\varphi)\, d\varphi, \tag{19}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha,\varphi) = \alpha_1 g(z') + \sum_{i\geq 2} \alpha_i z_i$ and $\chi(\alpha,\varphi) = b(z') \prod_{i\geq 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k-1$.

Hypothesis (iii). By (16), $\Psi(\alpha,0) = \langle \alpha, \theta\rangle$ is real. By Lemma A.2, $|g(z')| \leq g(\theta') = \theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi) \leq \Psi(\alpha,0) - \sum_{i\geq 2} \alpha_i\theta_i(1-\cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'), z') = 0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \operatorname{diag} z)$. With $R = \operatorname{diag} r$ we have $A_F - \operatorname{diag}(\theta - \eta) = R(\widehat{Q}_\alpha + \operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta - \eta) = \varrho(\widehat{Q}_\alpha + \operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta) = -\alpha$ and $\nabla^2\lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \operatorname{diag} z') < 0$, so $(1, X(z')b_1) \geq 0$ by the first fact. This vector lies in the kernel of $A_F - \operatorname{diag}(g(z'), z')$ (rows $i \geq 2$ by the definition of X, and row 1 by the definition of g). By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'), z') = 0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'), z') = 0$ in z_j gives $\lambda_1 g_j + \lambda_j = 0$, and differentiating again in z_i gives $\lambda_1 g_{ij} + \lambda_{11} g_i g_j + \lambda_{1j} g_i + \lambda_{i1} g_j + \lambda_{ij} = 0$. Put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \geq 2$. Then the first equation is $g_j = -\alpha_j/\alpha_1$, so $w_j = e_j + g_j e_1$, and the last four terms of the second equation are $w_i^\top \Gamma_\alpha w_j$. So

$$g_j = -\frac{\alpha_j}{\alpha_1}, \qquad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top \Gamma_\alpha V$, where V has columns $w_2, \ldots, w_k$. Since $\partial/\partial\varphi_j = \mathrm{i} z_j\, \partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi \Psi(\alpha, 0) = 0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla_\varphi^2 \Psi(\alpha,0) = \Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B = I + \mathbf{1}\alpha'^\top/\alpha_1$. For $y \in \mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy + (\alpha'^\top y/\alpha_1)\mathbf{1} = (0, By)$. Since $\Gamma_\alpha \mathbf{1} = 0$, we may replace Vy by $(0, By)$ in $y^\top V^\top \Gamma_\alpha V y$, which gives $\mathcal{H} = B^\top \Sigma_\alpha B$. The matrix determinant lemma gives $\det B = 1 + \alpha'^\top \mathbf{1}/\alpha_1 = 1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det \mathcal{H} = \det \Sigma_\alpha/\alpha_1^2$.

Regularity. By Proposition 4.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0 \in \mathcal{K}$. By (16) and continuity there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \operatorname{diag}(\theta'(\alpha_0) - \delta\mathbf{1})) < 0$. By Lemma A.2, X, g and b are holomorphic where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \geq 2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\} \times \mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ-derivatives are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$.

The constant. The matrix $\mathcal{J} = \Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det \mathcal{J} = \prod_{i\geq 2} \theta_i^2 \det \Sigma_\alpha/\alpha_1^2$. So the lemma gives, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi} \chi\, d\varphi = e^{T\langle\alpha,\theta\rangle} \Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}} \frac{\alpha_1}{\prod_{i\geq 2}\theta_i \sqrt{\det \Sigma_\alpha}} \Big(\chi(\alpha,0) + O\Big(\frac{1}{T}\Big)\Big).$$

Here $\chi(\alpha,0) = b(\theta')\prod_{i\geq 2}\theta_i$. As in the derivation of (16), $r' = r_1 X(\theta')b_1$. In the same way the columns $j \geq 2$ of $l^\top(A_F - \operatorname{diag}\theta) = 0$, which is (4), give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu \cdot r = r_1(\mu_1 + \mu'^\top X(\theta')b_1)$ and $l \cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta')\mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{\alpha_1}.$$

As $r_\alpha, l_\alpha > 0$ and $\mu(F) > 0$ by (H), $\chi(\alpha,0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha,0)(1 + O(1/T))$. Putting this into (19) and using $I_F(\alpha) = \langle q - \theta, \alpha\rangle$ gives $f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-TI_F(\alpha)}(1 + O(1/T))$, uniformly on $\mathcal{K}$. □

A.5 Proof of Lemma 4.5

Idea of the proof. The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them term by term. The two terms for m differ by a factor κ_m that is close to 1 when the number of runs M is small compared with $\sqrt{N}$, and under the continuum weights a path has of order T runs.

Step 1: the ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i\in F} e^{-q_i t_i} \frac{t_i^{m_i - 1}}{(m_i - 1)!}, \qquad m \in \mathbb{Z}_{\geq 1}^F.$$

Since $h^{m_i-1}\binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!}\prod_{1\leq j<m_i}(1 - j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i\in F}\Big[\prod_{1\leq j<m_i}\Big(1 - \frac{j}{n_i}\Big)\Big](1 - hq_i)^{n_i - m_i} e^{hq_i n_i}.$$

Let $w_m = v_m / \sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m / \sum_m v_m = \mathbb{E}_w \kappa_m$.

Step 2: bounds for κ_m. Let $n_{\min} = \min_i n_i$. We claim that for every m

$$\Big(1 - \frac{M^2}{2n_{\min}}\Big) e^{-q_{\max}^2 T^2/N} \leq \kappa_m \leq e^{hq_{\max} M}. \tag{20}$$

First let $m_i \leq n_i$ for every i. The factors $1 - j/n_i$ are at most 1, and $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)}$, so $\kappa_m \leq e^{hq_{\max} M}$. For the lower bound, $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1 - hq_i)^{n_i - m_i} e^{hq_i n_i} \geq e^{-n_i h^2 q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2 T^2/N}$. The factors $1 - j/n_i$ lie in $[0,1]$, so their product is at least 1 minus the sum of the j/n_i, and hence at least $1 - \sum_i m_i^2/(2n_i) \geq 1 - M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then $\kappa_m = 0$, while $M \geq n_{\min} + 1$ gives $M^2/(2n_{\min}) \geq 2$, so the left side of (20) is negative.

By (20) it remains to bound $\mathbb{E}_w e^{hq_{\max} M}$ and $\mathbb{E}_w M^2$ from above.

Step 3: the number of runs under w. Since W for σA_F is $\sigma^{M-1} W(m)$, we have $\mathbb{E}_w \sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0) = 0$, as the logarithm of a moment generating function, and $\Upsilon(1) \geq 0$ since $M \geq 1$. So $\Upsilon(c) \leq c\,\Upsilon(1)$ for $0 \leq c \leq 1$. We show that $\Upsilon(1) \leq C_{\mathcal{K}} T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (18) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1} b(\theta') \prod_{i\geq 2} \theta_i\, e^{eT\langle \alpha, \theta\rangle}$.

For the lower bound note that $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TI_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $I_F(\alpha) = \langle q-\theta, \alpha\rangle$. By Theorem 4.4 this is at least $\frac{1}{2}T^{-(k-1)/2}C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1, T_1] \times \mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \leq C_{\mathcal{K}}T$.

Step 4: the two averages. Let $x = hq_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \leq e^{x(1+C_{\mathcal{K}}T)} = 1 + O(T^2/N)$, because $x \leq q_{\max}T/N$ and $T^2/N \leq 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\Upsilon(1)-y}$ for $y \geq \Upsilon(1)$ and the bound 1 for $y < \Upsilon(1)$. This gives

$$\mathbb{E}_w(M-1)^2 = \int_0^\infty 2y\,\mathbb{P}_w(M-1 \geq y)\,dy \leq \Upsilon(1)^2 + \int_{\Upsilon(1)}^\infty 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2 + 2\Upsilon(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_{\mathcal{K}}N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}}\min_i \alpha_i > 0$. So by (20), $\mathbb{E}_w\kappa_m$ lies between $e^{-q_{\max}^2T^2/N}(1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}}N))$ and $\mathbb{E}_w e^{hq_{\max}M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$. □

A.6 Proof of Theorem 3.3(ii)

Let F be admissible, with components $C_1, \ldots, C_r$. If F is irreducible, then $r = 1$, and the argument below uses only Lemma 3.2 and an explicit family of paths, not the local law.

Idea of the proof. A path with support F visits the components one after the other, in r blocks, and each passage to the next block is one switch. For the upper bound we apply Lemma 3.2 to every block. For the lower bound we choose the block lengths ourselves. A component of least exit rate gets almost all the time, and every other component C gets only about N/T steps, the time scale of a single switch. Such a short block has probability bounded below, and it has only about $(N/T)^{|C|-1}$ compositions instead of $N^{|C|-1}$.

Let a be the least positive rate q_{ij}. Since $T \leq N^{1/3}$, we have $h \leq N^{-2/3}$ and $T^2/N \leq 1$, and we take N_0 so large that h is as small as we need. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j-th block lies in the component D_j and starts at y_j, and whose switch out of block j is $x_j \to y_{j+1}$ for $j < r$. So block j ends at x_j for $j < r$, and the pattern does not fix the end of the last block. Let $\operatorname{supp} n = F$. Block j has length $N_j = \sum_{i\in D_j} n_i$. By the Markov property the paths with this pattern and composition n have total probability $\mu_{y_1}\prod_{j<r} hq_{x_jy_{j+1}}\prod_{j\leq r}\varpi_j$. Here ϖ_j is the probability that the chain started at y_j stays in D_j for N_j visits, with the composition that n prescribes on D_j, and for $j < r$ also ends at x_j. We drop the condition on the end. Then ϖ_j is at most the probability that $X_1, \ldots, X_{N_j}$ have that composition for the chain started at y_j. If $|D_j| \geq 2$, Lemma 3.2 with start y_j and $\theta = q - \lambda_{D_j}\mathbf{1}$ gives $\varpi_j \leq \kappa_1 h^{|D_j|-1}e^{-h\lambda_{D_j}N_j+\kappa_1h^2N_j}$, where κ_1 is the largest of these finitely many constants. If $D_j = \{i\}$, then $\lambda_{D_j} = q_i$ and $\varpi_j = (1-hq_i)^{N_j-1} \leq e^{1/2-h\lambda_{D_j}N_j}$. Now we multiply over j. The powers of h add up to $\sum_j(|D_j|-1)+r-1 = k_F-1$. In the exponent, $\lambda_{D_j} \geq \lambda_F$ and $\sum_j N_j = N$ give $h\sum_j\lambda_{D_j}N_j \geq T\lambda_F$ and $\kappa_1h^2\sum_j N_j = \kappa_1T^2/N \leq \kappa_1$. Summing over the patterns gives $P_N(n) \leq \kappa_2(T/N)^{k_F-1}e^{-T\lambda_F}$.

Lower bound. We write down a family of paths with support F, bound its probability from below and count its compositions. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_jy_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T\rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A. Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \le |C|^2$. Consider the paths $X_1, \dots, X_m$ that make exactly these J switches, in this order, at any J of the $m-1$ transition times. There are $\binom{m-1}{J}$ of them, and each has probability at least $(ha)^J (1 - hq_{\max})^m$. For large N we have $m > |C|^2 \ge J$. Then $\binom{m-1}{J} h^J \ge (hm - Jh)^J / J! \ge (1 - Jh)^J / J!$ and $(1 - hq_{\max})^m \ge e^{-2hq_{\max} m} \ge e^{-4q_{\max}}$, since $1 \le hm \le 2$. So their total probability is at least $(1 - Jh)^J a^J e^{-4q_{\max}} / J!$, and $(1 - Jh)^J \ge 2^{-J}$ for small h.

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \ge 2m$, the event of Claim A, with C_{j^*} in place of C and ℓ visits in place of m, has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \ge 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*}. Since $B^s w = (1 - h\lambda_F)^s w$, the probability of staying in C_{j^*} for s steps from z is $\sum_{z'} (B^s)_{zz'} \ge (B^s w)_z / \max w \ge (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y, then $s = \ell - 2m + 1$ steps inside C_{j^*} to any state z', then Claim A for the last m visits, from z' to x. By the Markov property this gives at least $c_A^2 (\min w / \max w)(1 - h\lambda_F)^\ell$. Also $\lambda_F \le q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \le \frac{1}{2}$ and $\log(1 - h\lambda_F) \ge -h\lambda_F - h^2 \lambda_F^2$.

The family of paths. Give block $j \ne j^*$ the length m and block j^* the length $\ell = N - (r-1)m$, which is at least $2m$ for $T \ge T_0 = r + 2$ and $N \ge N_0$. Consider the paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j+1$ by the switch $x_j \to y_{j+1}$. They all have support F. By the Markov property, and since $h\ell \le T$ and $h^2 \ell \le T^2 / N \le 1$, their total probability is at least $\mu_{y_1} (ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. The composition of such a path is fixed by its restrictions to the blocks. On the block in C_j the restriction is a vector of $|C_j|$ positive integers whose sum is m for $j \ne j^*$ and ℓ for $j = j^*$. A vector of positive integers with a given sum is fixed by all but one of its entries, and each entry is at most the sum. Also $\ell \le N$. So these paths have at most $(N+1)^{k^* - 1} \prod_{j \ne j^*} (m+1)^{|C_j| - 1}$ compositions, and $m + 1 \le 3N/T$. Since $\sum_{j \ne j^*} (|C_j| - 1) = k_F - k^* - (r-1)$ and $h = T/N$, dividing the total probability by the number of compositions gives $M_F \ge c\, N^{-(k_F - 1)} T^{k_F - k^*} e^{-T\lambda_F}$. □

A.7 A forest formula for the covariance

This lemma is used for Proposition 4.15. It writes $\det \Sigma$ through spanning trees, with no reversibility.

For weights $f_{ij} \ge 0$ on the ordered pairs $i \ne j$ of F and a state $i \in F$ let

$$\mathrm{arb}_i(f) = \sum_{\mathcal{T}} \prod_{a \ne i} f_{a\, p_{\mathcal{T}}(a)},$$

the sum over the spanning trees $\mathcal{T}$ of the complete graph on F, where $p_{\mathcal{T}}(a)$ is the neighbor of a on the path from a to i in $\mathcal{T}$. So $\mathrm{arb}_i(f)$ is the weighted count of the spanning trees oriented toward the root i. If f is symmetric, then $\mathrm{arb}_i(f) = \mathrm{tree}(f)$ for every i.

Lemma A.3 (a forest formula for the covariance)**.** *Let G be an irreducible generator on F with stationary law π, where $k = |F| \ge 2$, and let D, Z and Γ be as at the start of this appendix, and Σ as in Lemma A.1 (namely Γ without the row and column of state 1). Put $f_{ij} = \pi_i G_{ij}$ for $i \ne j$, the stationary flow from i to j.*

(i) The number $\mathrm{arb}_i(f)$ *does not depend on i. We write* $\mathrm{arb}(f)$ *for it. Also* $\det(-G)_{(i)} =$

$\pi_i \operatorname{arb}(f)/\prod_j \pi_j$ *for every* i*, where* (i) *means that row and column* i *are deleted, and*

$$\det(\mathbf{1}\pi^\top - G) = \sum_{i \in F} \det(-G)_{(i)} = \frac{\operatorname{arb}(f)}{\prod_j \pi_j}.$$

(ii) Let $S = -(DG + G^\top D)$*. Then* S *is the Laplacian of the conductances* $f_{ij} + f_{ji}$*, and* $\Gamma = Z^\top S Z$*.*

(iii) With $\operatorname{tree}(f + f^\top)$ *the weighted count of spanning trees for the conductances* $f_{ij} + f_{ji}$*,*

$$\det \Sigma = \frac{\operatorname{tree}(f + f^\top)}{\det(\mathbf{1}\pi^\top - G)^2} = \frac{(\prod_j \pi_j)^2 \operatorname{tree}(f + f^\top)}{\operatorname{arb}(f)^2}.$$

In words, the determinant of the covariance of the occupation fractions is the square of the product of the stationary probabilities, times a count of undirected spanning trees for the symmetrized stationary flow, divided by the square of a count of directed spanning trees for the flow itself. No reversibility is needed. For two states with rates a and b the flow is $ab/(a+b)$ in each direction, the two counts are $2ab/(a+b)$ and $ab/(a+b)$, and $\det \Sigma = 2ab/(a+b)^3$, which is $\frac{1}{4}$ for $a = b = 1$.

Idea of the proof. The matrix $\Gamma = DZ + Z^\top D$ factors as $Z^\top S Z$, where S is twice the symmetric part of the Laplacian $-DG$ of the stationary flow. Each of the three factors has rank $k-1$, so the determinant of the product is 0 and tells us nothing. The adjugate is still multiplicative, and the adjugate of a matrix of rank $k-1$ is a rank one matrix that we know up to one scalar from the two kernels. For S the scalar is a count of spanning trees by the matrix-tree theorem. For Z the scalar is $1/\det(\mathbf{1}\pi^\top - G)$, and the directed matrix-tree theorem turns this determinant into a count of oriented spanning trees.

Proof. We use three facts about the adjugate $\operatorname{adj} M$ of a $k \times k$ matrix M. First, $\operatorname{adj}(AB) = \operatorname{adj}(B)\operatorname{adj}(A)$ for all square matrices A and B, by the Cauchy–Binet formula. Second, $\det(M + uv^\top) = \det M + v^\top \operatorname{adj}(M)\, u$ for all vectors u and v. Third, suppose that M has rank $k-1$ and that $Mu = 0$ and $v^\top M = 0$ with u and v nonzero. Then $\operatorname{adj} M = m\, uv^\top$ for some number m. Indeed $M \operatorname{adj} M = \operatorname{adj}(M) M = (\det M) I = 0$, so every column of $\operatorname{adj} M$ lies in the kernel $\mathbb{R}u$ of M, and every row lies in the left kernel $\mathbb{R}v^\top$. A matrix with these two properties is a multiple of $uv^\top$.

(i) Let $\mathcal{L} = -DG$. Its off-diagonal entries are $-f_{ij}$, its row sums vanish because $G\mathbf{1} = 0$, and its column sums vanish because $\pi^\top G = 0$. The kernel of G is $\mathbb{R}\mathbf{1}$ because G is irreducible, so $\mathcal{L}$ has rank $k-1$. By the third fact with $u = v = \mathbf{1}$, all entries of $\operatorname{adj} \mathcal{L}$ are equal. By the matrix-tree theorem for directed graphs [31, 32], the diagonal entry $\det \mathcal{L}_{(i)}$ of $\operatorname{adj} \mathcal{L}$ is $\operatorname{arb}_i(f)$. So $\operatorname{arb}_i(f)$ does not depend on i. Deleting row and column i from $-G = D^{-1}\mathcal{L}$ gives $\det(-G)_{(i)} = \operatorname{arb}(f)/\prod_{j \neq i} \pi_j = \pi_i \operatorname{arb}(f)/\prod_j \pi_j$, and the sum over i is $\operatorname{arb}(f)/\prod_j \pi_j$ because $\sum_i \pi_i = 1$. Next, $-G$ has rank $k-1$, right kernel $\mathbb{R}\mathbf{1}$ and left kernel $\mathbb{R}\pi^\top$. So $\operatorname{adj}(-G) = g\, \mathbf{1}\pi^\top$ for some number g by the third fact. The trace of $\mathbf{1}\pi^\top$ is 1, so g is the trace of $\operatorname{adj}(-G)$, which is $\sum_i \det(-G)_{(i)}$. Since $\det(-G) = 0$, the second fact gives

$$\det(\mathbf{1}\pi^\top - G) = \pi^\top \operatorname{adj}(-G)\, \mathbf{1} = g\, (\pi^\top \mathbf{1})(\pi^\top \mathbf{1}) = g.$$

(ii) The matrix S is symmetric. For $i \neq j$ its entry is $-(\pi_i G_{ij} + \pi_j G_{ji}) = -(f_{ij} + f_{ji})$, and $S\mathbf{1} = -DG\mathbf{1} - G^\top \pi = 0$. So S is the Laplacian of the conductances $f_{ij} + f_{ji}$. By (5),

$-GZ = I - \mathbf{1}\pi^\top$ and $\pi^\top Z = 0$. Hence

$$Z^\top(-DG)Z = Z^\top D(I - \mathbf{1}\pi^\top) = Z^\top D - (\pi^\top Z)^\top \pi^\top = Z^\top D,$$

where we used $D\mathbf{1} = \pi$. The transpose of this identity is $Z^\top(-G^\top D)Z = DZ$. Adding the two gives $Z^\top S Z = Z^\top D + DZ = \Gamma$.

(iii) Put $w = \det(\mathbf{1}\pi^\top - G)$, which is not 0 because Z exists. By the matrix-tree theorem [35] every diagonal entry of $\operatorname{adj} S$ is $\operatorname{tree}(f + f^\top)$. This number is positive because the graph of G is strongly connected. So S has rank $k-1$, and the third fact with $u = v = \mathbf{1}$ gives $\operatorname{adj} S = \operatorname{tree}(f + f^\top)\,\mathbf{1}\mathbf{1}^\top$. Now we turn to Z. By (5), $Z\mathbf{1} = 0$ and $\pi^\top Z = 0$. The matrix $Z + \mathbf{1}\pi^\top = (\mathbf{1}\pi^\top - G)^{-1}$ is invertible, so Z has rank $k-1$, and $\operatorname{adj} Z = z\,\mathbf{1}\pi^\top$ for some number z by the third fact. Since $\det Z = 0$, the second fact gives

$$\frac{1}{w} = \det(Z + \mathbf{1}\pi^\top) = \pi^\top \operatorname{adj}(Z)\,\mathbf{1} = z.$$

Also $\operatorname{adj}(Z^\top) = (\operatorname{adj} Z)^\top = z\,\pi\mathbf{1}^\top$. By (ii) and the first fact,

$$\operatorname{adj}\Gamma = \operatorname{adj}(Z)\operatorname{adj}(S)\operatorname{adj}(Z^\top) = z^2\,\operatorname{tree}(f + f^\top)\,\mathbf{1}(\pi^\top\mathbf{1})(\mathbf{1}^\top\pi)\mathbf{1}^\top = \frac{\operatorname{tree}(f + f^\top)}{w^2}\,\mathbf{1}\mathbf{1}^\top.$$

The entry of $\operatorname{adj}\Gamma$ at state 1 is $\det\Sigma$. This is the first formula in (iii), and the second follows from (i). □

B A Laplace lemma with a parameter, and the next term

This appendix proves the Laplace lemma that Appendix A uses for Theorem 4.4. It then carries the lemma one step further (Lemma B.2) and proves Proposition 4.14. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \operatorname{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m.

Lemma B.1 (Laplace integrals with a parameter)**.** *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha, 0)$ is real and $\nabla_\varphi \Psi(\alpha, 0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re}\Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha, \varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

In words, the integral is the value of the integrand at $\varphi = 0$ times the volume of a Gaussian peak, which is $(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. The error is uniform in the parameter α.

Idea of the proof. By (iii) and compactness the part of the torus away from $\varphi = 0$ is exponentially small. Near 0 we substitute $\varphi = x/\sqrt{T}$ and expand. The integrand becomes a Gaussian weight times $a(\alpha, 0)$, plus a term of order $T^{-1/2}$ that is odd in x and integrates to 0, plus a remainder of order $1/T$. All constants depend on α only through bounds that hold on the whole of $\mathcal{K}$.

Proof. By compactness $B_\Psi = \max_{|\beta|\le 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta|\le 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $\sigma_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \le (mB_\Psi)^m$. Below, C depends only on m, $\sigma_{\min}$, B_Ψ, B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. The set of (α,φ) with $\|\varphi\| \ge \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)} \le CT^{-1-m/2} e^{T\Psi(\alpha,0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \le \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \le \varepsilon$, and it preserves measure. Let $R(\alpha,\varphi) = \Psi(\alpha,\varphi) - \Psi(\alpha,0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6}\Psi_{ijk}\varphi^i\varphi^j\varphi^k + \omega_0$ with $|\omega_0| \le C\|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha,0)$. It also gives $|R| \le \frac{m^3}{6} B_\Psi \|\varphi\|^3 \le \frac{1}{4}\sigma_{\min}|\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \le \varepsilon|\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha,0)} \int_{\|x\| \le \varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J} x}\, e^{w(x)} a(\alpha, x/\sqrt{T})\, dx,$$

and $|w(x)| \le \frac{1}{4}\sigma_{\min}|x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$, $a_0 = a(\alpha,0)$ and $a_i = \partial_{\varphi_i} a(\alpha,0)$. Then $w = T^{-1/2}P + \omega_1$ with $|\omega_1| \le CT^{-1}|x|^4$, and $|w| \le CT^{-1/2}|x|^3$. Since $|e^w - 1 - w| \le \frac{1}{2}|w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2}P| \le CT^{-1}(|x|^4 + |x|^6)e^{\sigma_{\min}|x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \le CT^{-1/2}|x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2}a_i x^i| \le CT^{-1}|x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\big(a_0 P(x) + a_i x^i\big) + \omega(x), \qquad |\omega(x)| \le CT^{-1}(1+|x|)^6 e^{\sigma_{\min}|x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \ge \sigma_{\min}|x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1+|x|)^6 e^{-\sigma_{\min}|x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon\sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \le CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2}x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha,0)} T^{-m/2}\big[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})\big]$. Since $\det \mathcal{J} \le (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha,0)}(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. □

B.1 The next term of the Laplace lemma

With two more derivatives the same proof gives the term of order $1/T$ and an error $O(T^{-2})$.

Lemma B.2 (the next term)**.** *Let $\mathcal{K}$, m, Ψ and a satisfy the hypotheses of Lemma B.1. Assume also that the φ-derivatives of Ψ up to order 6 and of a up to order 4 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Write $a_0 = a(\alpha,0)$, write $\mathcal{J}^{ij}$ for the entries of $\mathcal{J}(\alpha)^{-1}$, and write Ψ_{ijk}, Ψ_{ijkl}, a_i and a_{ij} for the φ-derivatives at $(\alpha,0)$. Put*

$$\begin{aligned} c_1(\alpha) = {} & \tfrac{1}{2}a_{ij}\mathcal{J}^{ij} + \tfrac{1}{2}a_i \Psi_{jkl}\mathcal{J}^{ij}\mathcal{J}^{kl} \\ & + a_0\Big[\tfrac{1}{8}\Psi_{ijkl}\mathcal{J}^{ij}\mathcal{J}^{kl} + \tfrac{1}{12}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{il}\mathcal{J}^{jr}\mathcal{J}^{ks} + \tfrac{1}{8}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{ij}\mathcal{J}^{kl}\mathcal{J}^{rs}\Big]. \end{aligned} \tag{21}$$

Then c_1 is continuous on $\mathcal{K}$, and uniformly for $\alpha \in \mathcal{K}$ and $T \ge 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha,\varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a_0 + \frac{c_1(\alpha)}{T} + O\Big(\frac{1}{T^2}\Big)\Big).$$

In words, c_1 is the mean of an explicit polynomial of degree 6 under the Gaussian law with covariance $\mathcal{J}^{-1}$. As a formal check of the coefficients in (21), take the integrand $e^{T(e^{\mathrm{i}\varphi}-1-\mathrm{i}\varphi)}$ with $m=1$, $a=1$ and integer T. Its integral over $[-\pi,\pi]$ is $2\pi e^{-T}T^T/T!$, and Stirling's formula gives $(2\pi/T)^{1/2}(1-\frac{1}{12T}+O(T^{-2}))$. Here $\mathcal{J}=1$, $\Psi_{111}=-\mathrm{i}$ and $\Psi_{1111}=1$, so (21) gives $c_1=\frac{1}{8}-\frac{1}{12}-\frac{1}{8}=-\frac{1}{12}$ as well. The check is only formal, because $\Psi=e^{\mathrm{i}\varphi}-1-\mathrm{i}\varphi$ is not 2π-periodic. So it is not a function on the torus, and the lemma does not cover it as stated. Only $e^{T\Psi}$ is periodic, for integer T.

Idea of the proof. After the substitution $\varphi=x/\sqrt{T}$ the integrand is a Gaussian weight times a power series in $T^{-1/2}$ whose coefficients are polynomials in x. The coefficients of the odd powers of $T^{-1/2}$ are odd polynomials and integrate to 0. The coefficient of T^{-1} has the Gaussian mean c_1, by Wick's formula.

Proof. We follow the proof of Lemma B.1 with its notation, and expand further. Now B_Ψ and B_a are the maxima over $|\beta|\le 6$ and $|\beta|\le 4$. The localization and the scaling are as before. The integral over $\|\varphi\|\ge\varepsilon$ is at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)}\le CT^{-2-m/2}e^{T\Psi(\alpha,0)}$.

Taylor expansion. Put $s=T^{-1/2}$, so $0<s\le 1$. For $j=3,4,5$ let $P_j(x)$ be the part of degree j of the Taylor polynomial of $\Psi(\alpha,\cdot)$ at 0. So $P_3=P$ and $P_4(x)=\frac{1}{24}\Psi_{ijkl}x^ix^jx^kx^l$. For $j=1,2,3$ let $A_j(x)$ be the part of degree j of the Taylor polynomial of $a(\alpha,\cdot)$, so $A_1(x)=a_ix^i$ and $A_2(x)=\frac{1}{2}a_{ij}x^ix^j$. Taylor's theorem to order 6 for Ψ and to order 4 for a gives, on the cube $\|x\|\le\varepsilon\sqrt{T}$,

$$w=w_0+\omega_1,\qquad w_0=sP_3+s^2P_4+s^3P_5,\qquad a(\alpha,sx)=a_0+sA_1+s^2A_2+s^3A_3+\omega_2,$$

with $|\omega_1|\le Cs^4|x|^6$ and $|\omega_2|\le Cs^4|x|^4$. Let $\mathsf{p}(v)=1+v+\frac{1}{2}v^2+\frac{1}{6}v^3$. We compare e^w with $\mathsf{p}(w)$, then with $\mathsf{p}(w_0)$, and then with

$$Y=1+sP_3+s^2\big(P_4+\tfrac{1}{2}P_3^2\big)+s^3\big(P_5+P_3P_4+\tfrac{1}{6}P_3^3\big),$$

which collects the terms of $\mathsf{p}(w_0)$ of degree at most 3 in s. First, $|e^v-\mathsf{p}(v)|\le\frac{1}{24}|v|^4e^{|v|}$, and we know that $|w|\le Cs|x|^3$ and $|w|\le\frac{1}{4}\sigma_{\min}|x|^2$. So $|e^w-\mathsf{p}(w)|\le Cs^4|x|^{12}e^{\sigma_{\min}|x|^2/4}$. Second, $\mathsf{p}(w)-\mathsf{p}(w_0)=\omega_1\big[1+\frac{1}{2}(w+w_0)+\frac{1}{6}(w^2+ww_0+w_0^2)\big]$, and $|w|$ and $|w_0|$ are at most $C(1+|x|)^5$, by $|w|\le Cs|x|^3$ and $s\le 1$. So $|\mathsf{p}(w)-\mathsf{p}(w_0)|\le Cs^4(1+|x|)^{16}$. Third, $\mathsf{p}(w_0)-Y$ is a sum of terms s^j with $4\le j\le 9$ times polynomials in x of degree at most 15 with bounded coefficients, so $|\mathsf{p}(w_0)-Y|\le Cs^4(1+|x|)^{15}$. Together,

$$|e^w-Y|\le Cs^4(1+|x|)^{16}e^{\sigma_{\min}|x|^2/4}.$$

Now write $e^wa=Y(a_0+sA_1+s^2A_2+s^3A_3)+Y\omega_2+(e^w-Y)a$. Here $|Y|\le C(1+|x|)^9$ and $|a|\le B_a$. In the first product the terms of degree at least 4 in s are at most $Cs^4(1+|x|)^{12}$. Hence

$$e^wa(\alpha,sx)=a_0+s\,\mathcal{Q}_1(x)+s^2\mathcal{Q}_2(x)+s^3\mathcal{Q}_3(x)+\omega(x),\qquad |\omega(x)|\le CT^{-2}(1+|x|)^{16}e^{\sigma_{\min}|x|^2/4},$$

where

$$\mathcal{Q}_1=a_0P_3+A_1,\qquad \mathcal{Q}_2=a_0\big(P_4+\tfrac{1}{2}P_3^2\big)+A_1P_3+A_2,$$
$$\mathcal{Q}_3=a_0\big(P_5+P_3P_4+\tfrac{1}{6}P_3^3\big)+A_1\big(P_4+\tfrac{1}{2}P_3^2\big)+A_2P_3+A_3.$$

Every term of $\mathcal{Q}_1$ and of $\mathcal{Q}_3$ has odd degree, so $\mathcal{Q}_1$ and $\mathcal{Q}_3$ are odd functions.

Integration. As before, ω contributes at most CT^{-2} to the integral over the cube, since $(1+|x|)^{16}e^{-\sigma_{\min}|x|^2/4}$ is integrable. Extending the integrals of the four polynomial terms to $\mathbb{R}^m$

changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \le CT^{-2}$. On $\mathbb{R}^m$ the odd terms $\mathcal{Q}_1$ and $\mathcal{Q}_3$ integrate to 0 against $e^{-\frac{1}{2}x^\top \mathcal{J} x}$. This is why the error is $O(T^{-2})$ and not $O(T^{-3/2})$. So the integral over the cube is

$$e^{T\Psi(\alpha,0)}T^{-m/2}\Big[(2\pi)^{m/2}\det \mathcal{J}^{-1/2}\Big(a_0 + \frac{\mathbb{E}\mathcal{Q}_2(x)}{T}\Big) + O(T^{-2})\Big],$$

where x is a centered Gaussian vector with $\mathbb{E}[x^i x^j] = \mathcal{J}^{ij}$. As in the proof of Lemma B.1, the errors are $O(T^{-2})$ relative to the main factor, because $\det \mathcal{J} \le (mB_\Psi)^m$.

It remains to show that $\mathbb{E}\mathcal{Q}_2 = c_1$. By Wick's formula a Gaussian moment of order 4 or 6 is the sum, over the three or fifteen ways to pair the indices, of the products of the covariances of the pairs. The derivatives are symmetric in their indices. So $\mathbb{E}[A_2] = \frac{1}{2}a_{ij}\mathcal{J}^{ij}$, $\mathbb{E}[A_1 P_3] = \frac{1}{6}\cdot 3\, a_i \Psi_{jkl}\mathcal{J}^{ij}\mathcal{J}^{kl}$ and $\mathbb{E}[P_4] = \frac{1}{24}\cdot 3\,\Psi_{ijkl}\mathcal{J}^{ij}\mathcal{J}^{kl}$. In $\mathbb{E}[P_3^2] = \frac{1}{36}\Psi_{ijk}\Psi_{lrs}\mathbb{E}[x^i x^j x^k x^l x^r x^s]$, six pairings join each index of the first factor to an index of the second, and each gives $\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{il}\mathcal{J}^{jr}\mathcal{J}^{ks}$. The other nine pair two indices inside each factor and join the remaining two, and each gives $\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{ij}\mathcal{J}^{kl}\mathcal{J}^{rs}$. So $\frac{1}{2}\mathbb{E}[P_3^2]$ gives the last two terms of (21). Finally c_1 is a polynomial in a_0, in the derivatives at $\varphi = 0$ and in the entries of $\mathcal{J}^{-1}$, which are continuous on $\mathcal{K}$. □

B.2 Proof of Proposition 4.14

Throughout, F satisfies (H). If $F = \{i\}$, then (3) gives $\log M_F = \log \mu_i - q_i T + O(T^2/N)$, which is (10) with $c_F = 0$. So let $k \ge 2$ until the proof of (iii).

Part (i). We use the proof of Theorem 4.4 in Appendix A. It shows that Ψ and χ in (19) satisfy the hypotheses of Lemma B.1 with $m = k-1$, and that their φ-derivatives of every order are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$. So Lemma B.2 applies with $a = \chi$. Let $c_1(\alpha)$ be the function (21) for this Ψ and $a = \chi$, with $\mathcal{J}(\alpha) = -\nabla_\varphi^2\Psi(\alpha,0)$, and put

$$c(\alpha) = \frac{c_1(\alpha)}{\chi(\alpha,0)}, \qquad \chi(\alpha,0) = b(\theta'(\alpha))\prod_{i\ge 2}\theta_i(\alpha) > 0. \tag{22}$$

Then the bracket $\chi(\alpha,0)+O(1/T)$ in the last step of that proof becomes $\chi(\alpha,0)+c_1(\alpha)/T+O(T^{-2})$. Since $\chi(\alpha,0)$ is bounded below on $\mathcal{K}$, this is $\chi(\alpha,0)(1+c(\alpha)/T+O(T^{-2}))$, and the rest of that proof gives the expansion in (i). By the regularity step of that proof, Ψ and χ are real-analytic in (α,φ) near $\{\alpha_0\}\times\mathbb{T}^{k-1}$ for every $\alpha_0 \in \Delta_F^\circ$. So their φ-derivatives at $\varphi = 0$ are real-analytic in α, and so are the entries of $\mathcal{J}(\alpha)^{-1}$ and c. Finally (i) gives $c(\alpha) = \lim_{T\to\infty} T\big[f_T(\alpha)T^{-(k-1)/2}C_F(\alpha)^{-1}e^{TI_F(\alpha)} - 1\big]$. So c is real, and it does not depend on which state of F we number 1.

Part (ii), *the plan.* We write $\log P_N(n)$ as a fixed term plus a function G_T of $\alpha = n/N$, and we maximize G_T. Far from α_F^*, at distance more than a large multiple of $1/T$, the quadratic growth of I_F makes G_T smaller than $(c_F - 1)/T$. At distance at most that multiple of $1/T$, the function G_T is a constant plus a linear term minus a positive definite quadratic form, up to an error $O(T^{-2})$. Completing the square shows that this quadratic model has the maximum c_F/T, and that it takes it exactly at the point $\alpha' = \alpha_F^{*\prime} + \Sigma_F\gamma_F/T$. So $G_T \le c_F/T + O(T^{-2})$, which gives the upper bound. A lattice point next to that point gives the lower bound.

Part (ii), *the constants.* Write $\Sigma = \Sigma_F$, $\gamma = \gamma_F$, $c^* = c(\alpha_F^*)$, $X = N^{-(k-1)}T^{(k-1)/2}K_F e^{-T\lambda_F}$ and $x = \alpha' - \alpha_F^{*\prime}$. By Propositions 3.1(a) and 4.2(a) and by (i), the functions I_F, $\log C_F$ and c are real-analytic near α_F^*, and I_F has its minimum λ_F at α_F^* with Hessian Σ^{-1}. So there are a closed ball $U = \{|x| \le \rho_0\} \subset \Delta_F^\circ$ and constants $c_U > 0$ and L_1 such that on U

$$\Big|\log\frac{C_F(\alpha)}{K_F} - \gamma\cdot x\Big| \le L_1|x|^2, \qquad \Big|I_F(\alpha) - \lambda_F - \tfrac{1}{2}x^\top\Sigma^{-1}x\Big| \le L_1|x|^3,$$
$$|c(\alpha) - c^*| \le L_1|x|, \qquad |c(\alpha)| \le L_1, \qquad I_F(\alpha) - \lambda_F \ge c_U|x|^2.$$

Put $L_2 = |\gamma| + L_1\rho_0$, so that $|\log(C_F(\alpha)/K_F)| \le L_2|x|$ on U. Fix ρ_1 so large that

$$\rho_1 \ge \frac{L_2}{c_U}, \qquad \rho_1 \ge |\Sigma\gamma| + k, \qquad c_U\rho_1^2 - L_2\rho_1 - L_1 \ge |c_F| + 1.$$

For $\alpha \in U$ let

$$G_T(\alpha) = \log\frac{C_F(\alpha)}{K_F} - T\big(I_F(\alpha) - \lambda_F\big) + \frac{c(\alpha)}{T}.$$

By (i) with $\mathcal{K} = U$ and Lemma 4.5, there are T_0 and N_0 such that for $N \ge N_0$, $T_0 \le T \le N^{1/3}$ and $\operatorname{supp} n = F$ with $\alpha = n/N \in U$,

$$\log\frac{P_N(n)}{X} = G_T(\alpha) + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big). \tag{23}$$

Here we used $\log(1 + c/T + O(T^{-2})) = c/T + O(T^{-2})$, which holds because $|c| \le L_1$ on U. We take $T_0 \ge \rho_1/\rho_0$, and we increase T_0 and N_0 finitely many times below.

Part (ii), *two bounds for* G_T. First let $\alpha \in U$ with $|x| \ge \rho_1/T$. Then $G_T(\alpha) \le L_2|x| - c_UT|x|^2 + L_1/T$. The function $v \mapsto L_2v - c_UTv^2$ decreases for $v \ge L_2/(2c_UT)$, and ρ_1/T lies in this range. So

$$G_T(\alpha) \le \frac{L_2\rho_1 - c_U\rho_1^2 + L_1}{T} \le \frac{c_F - 1}{T} \qquad (\alpha \in U,\ |x| \ge \rho_1/T). \tag{24}$$

Next let $|x| \le \rho_1/T$, and put $y = x - \Sigma\gamma/T$. Then $\alpha \in U$, and the bounds on U give $G_T(\alpha) = \gamma\cdot x - \frac{T}{2}x^\top\Sigma^{-1}x + c^*/T + \epsilon$ with $|\epsilon| \le L_1|x|^2 + L_1T|x|^3 + L_1|x|/T \le L_1(\rho_1^2 + \rho_1^3 + \rho_1)T^{-2}$. Completing the square gives $\gamma\cdot x - \frac{T}{2}x^\top\Sigma^{-1}x = \frac{1}{2T}\gamma^\top\Sigma\gamma - \frac{T}{2}y^\top\Sigma^{-1}y$. So

$$G_T(\alpha) = \frac{c_F}{T} - \frac{T}{2}y^\top\Sigma^{-1}y + O\Big(\frac{1}{T^2}\Big) \qquad (|x| \le \rho_1/T), \tag{25}$$

uniformly in x. Since Σ^{-1} is positive definite, (24) and (25) give $G_T \le c_F/T + O(T^{-2})$ on U.

Part (ii), *the upper bound.* The part of the proof of Theorem 4.6 that treats compositions outside U uses only that U is a closed ball around α_F^* inside Δ_F°. So there is $\eta > 0$ with $P_N(n) \le X\cdot O(T^{(k-1)/2}e^{-2\eta T})$ for $\operatorname{supp} n = F$ and $n/N \notin U$. After increasing T_0 this is at most $Xe^{-\eta T} \le Xe^{(c_F-1)/T}$. For $n/N \in U$ we use (23) and the bound on G_T. Hence $\log(M_F/X) \le c_F/T + O(T^{-2} + T^2/N)$.

Part (ii), *the lower bound.* Let α^0 be the point with $x = \Sigma\gamma/T$. Let n_i^0 be an integer nearest to $N\alpha_i^0$ for $i \ge 2$, and $n_1^0 = N - \sum_{i\ge2} n_i^0$. Then $|n_i^0 - N\alpha_i^0| \le k$ for every i. Since $\alpha^0 \to \alpha_F^*$ as $T \to \infty$ and every $\alpha_{F,i}^*$ is positive, n^0 has support F after increasing T_0 and N_0. For $\alpha = n^0/N$ we have $|y| \le k/N$ and $|x| \le |\Sigma\gamma|/T + k/N \le \rho_1/T$, because $T \le N$. So (25) gives $G_T(n^0/N) \ge c_F/T - O(T/N^2 + T^{-2})$, and (23) gives $\log(M_F/X) \ge \log(P_N(n^0)/X) \ge c_F/T - O(T^{-2} + T^2/N)$. Note that a lattice point near α^0 loses only $O(T/N^2)$, because G_T is flat to first order at its maximizer.

The two bounds give (10) for $N \ge N_0$ and $T_0 \le T \le N^{1/3}$. For $1 \le T \le T_0$ it follows from (6), since c_F/T and $1/T$ are $O(T^{-2})$ there, with a constant that depends on T_0. This proves (ii).

Part (iii). Now F and G are as in Theorem 4.7, and $k_G = 1$ is allowed. Let $T_N \in [1, N^{1/3}]$ be a zero of D_N. The proof of Theorem 4.7(ii) shows that $\frac{\Delta}{2\delta}L < T_N \le \frac{2\Delta}{\delta}L$ for large N. Subtracting (10) for G from (10) for F at $T = T_N$ gives

$$0 = -\Delta L + \frac{\Delta}{2}\log T_N + \delta T_N + \log\frac{K_F}{K_G} + \frac{c_F - c_G}{T_N} + O\Big(\frac{1}{T_N^2} + \frac{T_N^2}{N}\Big).$$

We multiply by T_N. The error becomes $O(1/T_N + T_N^3/N) = O(1/L)$, which is (11).

Part (iv)*, a symmetric pair.* Let $F = \{i, j\}$ with $q_{ij} = q_{ji} = a > 0$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$. Then F satisfies (H). We first prove (12), without the Laplace lemma. Let $1 \le T \le N^{1/3}$ and let N be so large that $qh \le \frac{1}{2}$ and $I + hQ$ is stochastic. By Proposition 4.9, $M_F = \mu_i(1-qh)^{N-1} S_N(u)$ with $u = ah/(1-qh)$. Put $z = Nu = aT/(1-qh)$. Then $aT \le z \le 2aT$ and $0 \le z - aT \le 2aqT^2/N$. By (2),

$$\log S_N(u) = \log \frac{2}{N} + \log \psi(z) + O\Big(\frac{T^2}{N}\Big),$$

since $z(z+2)/(2N) = O(T^2/N)$ and this is at most $\frac{1}{2}$ for large N. By [36, Eq. 10.40.1], as $z \to \infty$, $I_0(z) = e^z(2\pi z)^{-1/2}(1 + \frac{1}{8z} + O(z^{-2}))$ and $I_1(z) = e^z(2\pi z)^{-1/2}(1 - \frac{3}{8z} + O(z^{-2}))$. So $I_0(z) + I_1(z) = 2e^z(2\pi z)^{-1/2}(1 - \frac{1}{8z} + O(z^{-2}))$ and

$$\log \psi(z) = z + \tfrac{1}{2}\log z + \tfrac{1}{2}\log\frac{2}{\pi} - \frac{1}{8z} + \rho(z), \qquad |\rho(z)| \le C_a z^{-2} \quad \text{for } z \ge a.$$

The bound on ρ holds for all $z \ge a$, and not only for large z, because $\rho(z) = O(z^{-2})$ as $z \to \infty$ and ρ is continuous on $[a, \infty)$. The function $z + \frac{1}{2}\log z - \frac{1}{8z}$ has a bounded derivative on $[a, \infty)$. So replacing z by aT in it changes it by $O(T^2/N)$, and

$$\log \psi(z) = aT + \tfrac{1}{2}\log T + \tfrac{1}{2}\log\frac{2a}{\pi} - \frac{1}{8aT} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big).$$

Also $(N-1)\log(1-qh) = -qT + O(T^2/N)$. Adding the three parts,

$$\log M_F = -L + \tfrac{1}{2}\log T + \log\Big(2\mu_i\sqrt{\frac{2a}{\pi}}\Big) - (q-a)T - \frac{1}{8aT} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big).$$

By Corollary 4.3 with $k = 2$, $\mu(F) = 2\mu_i$ and $\mathrm{tree}(q_F) = a$, we have $\lambda_F = q - a$ and $K_F = 2\mu_i \cdot 2^{3/2}\sqrt{a}\,(4\pi)^{-1/2} = 2\mu_i\sqrt{2a/\pi}$. This is (12). Now we compare it with (10) at $T = L$. Both expand $\log M_F$, so their difference $(c_F + \frac{1}{8a})/L$ is $O(L^{-2} + L^2/N) = O(L^{-2})$. Letting $N \to \infty$ gives $c_F = -1/(8a)$.

As a check we compute the same value from (22). Number the states so that i is state 1. Then $A' = 0$, so in the notation of Appendix A we have $g(z_2) = a^2/z_2$ and $b(z_2) = \mu_i(1 + a/z_2)^2$. At $\alpha = \alpha_F^* = (\frac{1}{2}, \frac{1}{2})$ the radius is $\theta_2 = a$, so $\Psi = a\cos\varphi$ and $\chi = \mu_i a(1 + e^{-\mathrm{i}\varphi})^2 e^{\mathrm{i}\varphi} = 2\mu_i a(1 + \cos\varphi)$. Hence $\mathcal{J} = a$, $\Psi_{111} = 0$, $\Psi_{1111} = a$, $\chi(\alpha_F^*, 0) = 4\mu_i a$, $\chi_1 = 0$ and $\chi_{11} = -2\mu_i a$, and (21) gives $c_1 = -\mu_i + \frac{1}{2}\mu_i = -\frac{1}{2}\mu_i$ and $c(\alpha_F^*) = -1/(8a)$. Also $\gamma_F = 0$, because C_F is unchanged when the two states are swapped (Proposition 4.2(b) with $\nu = \mathbf{1}$). So the two values agree.

Part (iv)*, the complete graph.* Let $Q = J - dI$, let F have $k \ge 2$ states and let $\mu_i = \mu_0 > 0$ for $i \in F$. Then F satisfies (H), and Corollary 4.3 with $\mu(F) = k\mu_0$ and $\mathrm{tree}(q_F) = k^{k-2}$ gives $\lambda_F = d - k$, α_F^* uniform on F and $K_F = \mu_0 k^{k+1/2}(4\pi)^{-(k-1)/2}$.

We first show that $\gamma_F = 0$, so that $c_F = c(\alpha_F^*)$. By Proposition 4.2(b) with $\nu = \mathbf{1}$ we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$, and then

$$C_F(\alpha) = \frac{\mu_0\big(\sum_{i\in F}\sqrt{\alpha_i}\big)^2}{(4\pi)^{(k-1)/2}}\,\frac{\sqrt{\mathcal{T}(\alpha)}}{\prod_{i\in F}\alpha_i},$$

where $\mathcal{T}(\alpha)$ is the tree sum of Proposition 4.2(b) for the conductances $\sqrt{\alpha_i\alpha_j}$, $i, j \in F$. A permutation of F permutes the spanning trees of the complete graph on F. So C_F is a symmetric function of $(\alpha_i)_{i\in F}$. Take $j \ge 2$ and swap the states 1 and j. This maps the point $\alpha_F^* + v(e_j - e_1)$ to $\alpha_F^* - v(e_j - e_1)$, so $v \mapsto \log C_F(\alpha_F^* + v(e_j - e_1))$ is even and its derivative at $v = 0$ vanishes. In the coordinates α' this derivative is the j-th entry of γ_F.

Now we compare with [3]. That paper treats the uniform start. It writes r for our h, so its $T = Nr$ is ours, and $p = 1 - (d-1)h$. Its V_N is p^{N-1}/d, and its $C_{N,k}$ is the probability, for the uniform start, of a balanced bin with k positive coordinates. Let n be such a bin with support F, so that $|n_i - N/k| < 1$ for $i \in F$. A path inside F has probability μ_0 times a product of transition probabilities, so for our start $P_N(n) = d\mu_0 C_{N,k}$. By [3, Theorem 3.1(i)], for large N and uniformly for $\frac{1}{2}L \le T \le 2L$,

$$\log \frac{C_{N,k}}{V_N} = (k-1)\Big[T - L + \tfrac{1}{2}\log T - a_k\Big] - \frac{k-1}{8T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big),$$

with a_k as in Corollary 4.11. Also $\log V_N = -\log d - (d-1)T + O(T^2/N)$, and $\log \mu_0 - (k-1)a_k = \log K_F$. So

$$\log P_N(n) = -(k-1)L + \frac{k-1}{2}\log T + \log K_F - (d-k)T - \frac{k-1}{8T} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big). \tag{26}$$

On the other hand $\alpha = n/N$ has $|x| \le k/N$, so (23) applies to n. The bounds on U give $G_T(n/N) = c(\alpha_F^*)/T + O(1/N)$, since $\log(C_F/K_F) = O(|x|)$, $T(I_F - \lambda_F) = O(T|x|^2)$ and $c(\alpha) - c(\alpha_F^*) = O(|x|)$. Hence $\log(P_N(n)/X) = c(\alpha_F^*)/T + O(T^{-2} + T^2/N)$. Comparing with (26) at $T = L$, which lies in $[T_0, N^{1/3}]$ and in $[\frac{1}{2}L, 2L]$ for large N, gives $(c(\alpha_F^*) + \frac{k-1}{8})/L = O(L^{-2})$. So $c_F = c(\alpha_F^*) = -(k-1)/8$. This value rests on the cited theorem of [3]. □

C Proofs for the table of outcomes

This appendix proves the rows P_d and Θ_ε of Table 2. In both rows Q is symmetric and the start has full support. So the admissible faces are the connected faces (Lemma 2.2), and the winners at τ are the connected faces that minimize $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ (Theorem 3.4). We use three facts about a connected face F. First, $-Q_F$ is symmetric, λ_F is its least eigenvalue, and an eigenvector of $-Q_F$ with positive entries belongs to λ_F (the second fact at the start of Appendix A). Second, $-Q_F$ is the Laplacian of the graph on F, which is positive semidefinite, plus the diagonal matrix of the rates out of F. So λ_F is the least value of

$$\sum_{\{i,j\}\subseteq F} q_{ij}(f_i - f_j)^2 + \sum_{i\in F} q_i^{\text{out}}(F) f_i^2 \tag{27}$$

over unit vectors $f \in \mathbb{R}^F$. Third, if $F' \subsetneq F$ is nonempty, then $\lambda_{F'} > \lambda_F$ (Proposition 3.1(d)).

Each proof has two steps, namely the exit rates and then the lower-left boundary of the hull. For the second step we use the following lemma. It says that a chain of points with increasing switch values is its own lower-left boundary.

Lemma C.1 (a chain of lines)**.** *Let $x_0 > x_1 > \cdots > x_m \ge 0$ and $y_0 < y_1 < \cdots < y_m$, and put $\ell_i(\tau) = \tau x_i + y_i$ and $t_i = (y_i - y_{i-1})/(x_{i-1} - x_i)$ for $1 \le i \le m$. Put $t_0 = 0$ and $t_{m+1} = \infty$, and suppose that $t_1 < t_2 < \cdots < t_m$. Then for $1 \le j \le m$ and every τ*

$$\ell_j(\tau) - \ell_{j-1}(\tau) = (x_{j-1} - x_j)(t_j - \tau). \tag{28}$$

Let $0 \le i \le m$ and let τ be finite with $t_i \le \tau \le t_{i+1}$. Then $\ell_0(\tau) \ge \cdots \ge \ell_i(\tau) \le \cdots \le \ell_m(\tau)$. If also $t_i < \tau < t_{i+1}$, then all these inequalities are strict, so $\ell_i(\tau) < \ell_j(\tau)$ for every $j \ne i$.

Proof. The identity (28) is the definition of t_j. If $j \le i$, then $t_j \le t_i \le \tau$, so $\ell_j(\tau) \le \ell_{j-1}(\tau)$. If $j > i$, then $\tau \le t_{i+1} \le t_j$, so $\ell_j(\tau) \ge \ell_{j-1}(\tau)$. If $t_i < \tau < t_{i+1}$, these inequalities are strict. □

C.1 The path

Let P_d be the path with states $1,\dots,d$ and $q_{i,i+1}=q_{i+1,i}=1$ for $1\le i<d$. An *end-arc* of $k<d$ states is one of the two faces $\{1,\dots,k\}$ and $\{d-k+1,\dots,d\}$. For $k\ge 1$ put

$$\lambda'_k = 2-2\cos\frac{\pi}{2k+1}, \qquad \tau'_k=\frac{1}{\lambda'_{k-1}-\lambda'_k}\ (k\ge 2), \qquad D'_k = k+\lambda'_k\tau'_k\ (k\ge 2).$$

Lemma C.2 (the numbers D'_k)**.** *(a) The sequence λ'_k is positive, strictly decreasing and strictly convex in $k\ge 1$. Hence $0<\tau'_2<\tau'_3<\cdots$. Also $\tau'_2=(1+\sqrt5)/2$.*

(b) For every $k\ge 2$, $\frac32 k-\frac12 < D'_k < \frac32 k$.

Proof. (a) Let $g(x)=2-2\cos(\pi/x)$, so that $\lambda'_k=g(2k+1)$. For $x\ge 2$,

$$g'(x)=-\frac{2\pi\sin(\pi/x)}{x^2}<0, \qquad g''(x)=2\pi\Big(\frac{\pi\cos(\pi/x)}{x^4}+\frac{2\sin(\pi/x)}{x^3}\Big)>0.$$

So $\lambda'_{k-1}-\lambda'_k>\lambda'_k-\lambda'_{k+1}>0$ for $k\ge 2$, and the τ'_k increase. Since $\lambda'_1=1$ and $\lambda'_2=2-2\cos\frac{\pi}{5}=(3-\sqrt5)/2$, we get $\tau'_2=2/(\sqrt5-1)=(1+\sqrt5)/2$.

(b) We turn the claim into two bounds on the ratio of two consecutive exit rates, and we prove each bound by comparing $\sin x$ with x. Put $a=\pi/(2(2k-1))$ and $b=\pi/(2(2k+1))$, so that $0<b<a\le\pi/6$. Since $2-2\cos 2x=4\sin^2 x$, we have $\lambda'_{k-1}=4\sin^2 a$ and $\lambda'_k=4\sin^2 b$. So the ratio $\rho=\lambda'_k/\lambda'_{k-1}$ equals $(\sin b/\sin a)^2$. By definition $D'_k-k=\lambda'_k/(\lambda'_{k-1}-\lambda'_k)=\rho/(1-\rho)$, which increases with ρ and takes the values $\frac{k-1}{2}$ and $\frac{k}{2}$ at $\rho=\frac{k-1}{k+1}$ and $\rho=\frac{k}{k+2}$. So the claim is

$$\frac{k-1}{k+1}<\rho<\frac{k}{k+2}.$$

If we replace each sine by its argument, then ρ becomes $(b/a)^2=(2k-1)^2/(2k+1)^2$. This number lies between the two bounds and very close to the lower one. So the lower bound needs only $\sin b/\sin a>b/a$, and the upper bound needs the next term of the sine.

The lower bound. The function $\sin x/x$ decreases on $(0,\pi]$, so $\sin b/\sin a > b/a = (2k-1)/(2k+1)$. Then $\rho>(2k-1)^2/(2k+1)^2>(k-1)/(k+1)$, where the last step holds because $(2k-1)^2(k+1)-(k-1)(2k+1)^2=2>0$.

The upper bound. Write $s(x)=\sin x/x$ and $y=(8k-2)/((k+2)(2k-1)^2)$. Then

$$\rho=\Big(\frac{2k-1}{2k+1}\Big)^2\Big(\frac{s(b)}{s(a)}\Big)^2 \qquad\text{and}\qquad \frac{k}{k+2}=\Big(\frac{2k-1}{2k+1}\Big)^2(1+y),$$

the second because $k(2k+1)^2-(k+2)(2k-1)^2=8k-2$. So we must show that $s(b)/s(a)<\sqrt{1+y}$. We reduce this in three steps to an inequality between two explicit functions of k.

Step 1. We remove the square root. We have $0<y<1$, because $(k+2)(2k-1)^2\ge 12(2k-1)>8k-2$, and $\sqrt{1+y}\ge 1+\frac{y}{3}$ for $0\le y\le 3$. So it is enough to show that $s(b)/s(a)<1+\frac{y}{3}$, that is

$$s(b)-s(a)<\tfrac{y}{3}\,s(a).$$

Step 2. We bound both sides with $1-\frac{x^2}{6}\le s(x)\le 1-\frac{x^2}{6}+\frac{x^4}{120}$, which holds for $x>0$. On the right, $a\le\pi/6$ gives $s(a)\ge 1-\frac{\pi^2}{216}>0.954$, so $\frac{y}{3}s(a)>0.318\,y$. On the left,

$$s(b)-s(a)\le\frac{a^2-b^2}{6}+\frac{b^4}{120}=\frac{\pi^2 k}{3(4k^2-1)^2}+\frac{\pi^4}{1920(2k+1)^4}.$$

So it is enough to show that this is less than $0.318\,y$.

Step 3. We multiply both sides by $(2k-1)^2$, and on the left we use $k/(2k+1)^2 < 1/(4k)$ and $(2k-1)^2/(2k+1)^4 < 1/(4k^2)$. It is then enough to show that

$$\frac{\pi^2}{12k} + \frac{\pi^4}{7680k^2} < 0.318\,\frac{8k-2}{k+2}.$$

The left side decreases in k, and the right side, which is $0.318(8-\frac{18}{k+2})$, increases. At $k=2$ the left side is less than 0.415 and the right side is more than 1.11. So the inequality holds for every $k \geq 2$. □

Proposition C.3 (the path)**.** *Let $d \geq 3$, let Q be the path P_d with unit rates, and let the start have full support. Put $K = \lfloor 2d/3 \rfloor$ and $\tau_{\text{full}} = \lceil d/3 \rceil / \lambda'_K$.*

(a) The connected faces are the sets of consecutive states. An end-arc of $k < d$ states has exit rate λ'_k. A set of k consecutive states that contains neither 1 nor d has exit rate $2-2\cos\frac{\pi}{k+1} > \lambda'_k$.

(b) We have $2 \leq K \leq d-1$ and $\tau'_2 < \cdots < \tau'_K < \tau_{\text{full}}$.

(c) The winners are the states 1 and d for $0 < \tau < \tau'_2$, the two end-arcs of k states for $\tau'_k < \tau < \tau'_{k+1}$ and $2 \leq k \leq K-1$, the two end-arcs of K states for $\tau'_K < \tau < \tau_{\text{full}}$, and P_d alone for $\tau > \tau_{\text{full}}$. No other face wins for any $\tau > 0$.

In words, the mode grows from an end of the path, one state at a time, until it covers $K = \lfloor 2d/3 \rfloor$ states (about two thirds of the path), and then it jumps to the whole path. The first step is at the golden ratio.

Idea of the proof. Growing an end-arc by one state lowers its exit rate by less and less, since λ'_k is convex. So the end-arcs alone win in the order of their size, and the arc of k states starts to win at τ'_k. The whole path enters when its exponent $d-1$ falls below the least exponent of the arcs. At τ'_k that least exponent is $D'_k - 1$, so the arcs of k states win before the whole path exactly when $D'_k < d$. Lemma C.2(b) puts D'_k just below $\frac{3}{2}k$, and this gives $k \leq 2d/3$.

Proof. (a) A connected face of a path is a set of consecutive states. Let F have $k < d$ states. If $F = \{1, \ldots, k\}$, then $-Q_F$ is tridiagonal with off-diagonal entries -1 and diagonal $(1, 2, \ldots, 2)$. Put $\vartheta = \pi/(2k+1)$ and $r_j = \cos((2j-1)\vartheta/2)$ for $1 \leq j \leq k$. Then $r_j > 0$, since $(2j-1)\vartheta/2 < \pi/2$. With $r_0 = r_1$ and $r_{k+1} = \cos(\pi/2) = 0$ we have $r_{j-1} + r_{j+1} = 2\cos\vartheta\, r_j$ for $1 \leq j \leq k$, and this says that $-Q_F r = (2 - 2\cos\vartheta)r$. So $\lambda_F = \lambda'_k$, and the same holds for $\{d-k+1, \ldots, d\}$ by the reflection $i \mapsto d+1-i$. If F contains neither 1 nor d, then $-Q_F = 2I - A$ with A the adjacency matrix of a path on k states. Now $r_j = \sin(j\pi/(k+1))$ is a positive eigenvector with eigenvalue $2 - 2\cos\frac{\pi}{k+1}$, and this is larger than λ'_k because $\frac{\pi}{k+1} > \frac{\pi}{2k+1}$.

(b) Since $d \geq 3$ we have $2 \leq K \leq d-1$. The τ'_k increase by Lemma C.2(a). Note that $d - K = \lceil d/3 \rceil$, so $\tau_{\text{full}} = (d-K)/\lambda'_K$. By Lemma C.2(b), $D'_K < \frac{3}{2}K \leq d$, that is $\lambda'_K \tau'_K < d - K$. So $\tau'_K < \tau_{\text{full}}$.

(c) Write $\phi_k(\tau) = \tau\lambda'_k + k - 1$ for the exponent of an end-arc of k states, $1 \leq k \leq d-1$. First we apply Lemma C.1 to the points $(\lambda'_k, k-1)$ for $1 \leq k \leq K$ followed by $(0, d-1)$, the point of P_d. Its switch values are $\tau'_2, \ldots, \tau'_K$ and $(d-K)/\lambda'_K = \tau_{\text{full}}$, which increase by (b). So on each of the intervals in (c) the face named there has a smaller exponent than the other end-arcs of at most K states and than P_d, and by (a) also than every set of at most K consecutive states that is not an end-arc.

It remains to show that a face F of k consecutive states with $K < k < d$ never wins. By (a) its exponent is at least $\phi_k(\tau)$. We show that $\phi_k(\tau) > \min(\phi_K(\tau), d-1)$ for every $\tau > 0$. Now we apply Lemma C.1 to the points $(\lambda'_j, j-1)$ for $1 \le j \le d-1$, with switch values $\tau'_2 < \cdots < \tau'_{d-1}$. If $\tau < \tau'_{K+1}$, then $\tau < \tau'_j$ for $K < j \le k$, and (28) gives $\phi_k(\tau) > \phi_{k-1}(\tau) > \cdots > \phi_K(\tau)$. If $\tau \ge \tau'_{K+1}$, take j with $K+1 \le j \le d-1$ and $\tau'_j \le \tau$, and with $\tau \le \tau'_{j+1}$ if $j < d-1$. Then

$$\phi_k(\tau) \ge \phi_j(\tau) \ge \phi_j(\tau'_j) = D'_j - 1 > \tfrac{3}{2}j - \tfrac{3}{2} \ge \tfrac{3}{2}K \ge d-1.$$

The first step is Lemma C.1, the second holds since ϕ_j increases, the third is the definition of D'_j, the fourth is Lemma C.2(b), the fifth is $j \ge K+1$, and the last holds because $3K \ge 2d-2$. So F never wins. The winners at τ are the admissible faces with the least exponent, and this gives (c). □

At a switch value the faces on both sides tie. On P_4 we have $K = 2$ and $D'_3 = 4.077 > 4$, so the end-arc of three states never wins. Its point lies only $\lambda'_3 - \frac{1}{2}\lambda'_2 = 0.0071$ to the right of the edge from the end-arcs of two states to P_4. The distance is small because the lower bound in Lemma C.2(b) is nearly attained.

C.2 Two triangles

Let $\varepsilon > 0$. The graph Θ_ε has states $1, \dots, 6$, unit rates on the triangles $A = \{1,2,3\}$ and $B = \{4,5,6\}$, and one more link between 3 and 4 of rate ε. We call $1, 2, 5, 6$ the inner states and $\{1,2\}$, $\{5,6\}$ the inner edges. The graph has the symmetries $1 \leftrightarrow 2$, $5 \leftrightarrow 6$ and the mirror map $i \mapsto 7-i$. Put

$$\begin{aligned} p_A(x) &= x^2 - (3+\varepsilon)x + \varepsilon, \\ p_4(x) &= x^3 - (5+2\varepsilon)x^2 + (6+6\varepsilon)x - 2\varepsilon, \\ p_5(x) &= x^3 - (4+2\varepsilon)x^2 + (3+4\varepsilon)x - \varepsilon, \end{aligned}$$

and $\varepsilon_0 = (64 - \sqrt{46})/60 = 0.95363$, $\varepsilon_1 = (64 + \sqrt{46})/60 = 1.17971$, the roots of $120\varepsilon^2 - 256\varepsilon + 135$.

Proposition C.4 (two triangles joined by a link)**.** *Let the start have full support.*

(a) The two triangles have the same exit rate $\lambda_A = \lambda_B = \frac{1}{2}\big(3 + \varepsilon - \sqrt{\varepsilon^2 + 2\varepsilon + 9}\big)$, which is the smaller root of p_A, and $0 < \lambda_A < 1$.

(b) Among the connected faces of one, two and three states the least exit rates are 2, 1 and λ_A. They are attained exactly by the inner states, by the inner edges and by A and B. Put $\lambda_4 = \lambda_{\{1,2,3,4\}}$ and $\lambda_5 = \lambda_{\{2,\dots,6\}}$. Every connected face of four states has exit rate at least λ_4, and every connected face of five states has exit rate λ_5. Here λ_4 and λ_5 are the least roots of p_4 and p_5, and $\lambda_5 < \lambda_4 < \lambda_A$.

(c) For every $\varepsilon > 0$, $\lambda_5 > \frac{1}{2}\lambda_4$, and $\lambda_4 > \frac{2}{3}\lambda_A$ if and only if $\varepsilon < \varepsilon_1$.

(d) Let $\varepsilon < \varepsilon_1$, and put

$$\tau_A = \frac{1}{1-\lambda_A} = \frac{1 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9}}{4}, \qquad \tau^* = \frac{3}{\lambda_A} = \frac{3\big(3 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9}\big)}{2\varepsilon}.$$

Then $1 < \tau_A < \tau^*$. The winners are the inner states for $0 < \tau < 1$, the inner edges for $1 < \tau < \tau_A$, the triangles A and B for $\tau_A < \tau < \tau^*$, and [6] alone for $\tau > \tau^*$. No other face wins for any $\tau > 0$. As $\varepsilon \to 0$, $\tau_A = 1 + \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\tau^* = \frac{9}{\varepsilon} + 2 + O(\varepsilon)$.

(e) *Let $\varepsilon > \varepsilon_1$. Then at $\tau = 3/\lambda_A$, where the triangles and* [6] *have the same exponent, the face* $\{1,2,3,4\}$ *has a smaller exponent. So the list in (d) fails.*

In words, for a weak link each triangle is a community that the chain rarely leaves. The mode sits on a triangle from $\tau \approx 1$ until $\tau \approx 9/\varepsilon$, and then it spreads over all six states at once. Once the link is stronger than ε_1 this list fails, because a triangle together with the far end of the link beats both the triangles and [6] at the value of τ where they tie.

Idea of the proof. The exit rates of small faces are explicit, and those of the faces with four and five states are roots of cubics. The points of the inner states, the inner edges, the triangles and [6] form a chain as in Lemma C.1. The points of the faces with four and five states must lie strictly to the right of the edge from a triangle to [6], and this is what (c) says. We prove (c) by continuity in ε. A change of sign needs a common root of two polynomials, hence a zero of a resultant, and the resultants are explicit polynomials in ε.

Proof. (a) On the vectors (a,a,b) the matrix $-Q_A$ acts as $(a,b) \mapsto (a-b,\ -2a+(2+\varepsilon)b)$, with characteristic polynomial p_A, and $(1,-1,0)$ is an eigenvector with eigenvalue 3. As $p_A(0) = \varepsilon > 0$ and $p_A(1) = -2 < 0$, the smaller root of p_A lies in $(0,1)$ and the other root is larger than 1. So the smaller root is λ_A. The mirror map gives $\lambda_B = \lambda_A$.

(b) By (27), $\lambda_F \geq 1$ whenever every state of F has rate at least 1 out of F. We go through the connected faces by size. The states 3 and 4 have exit rate $2+\varepsilon$, and the inner states have 2. The connected pairs are the inner edges, with $\lambda = 1$, the pair $\{3,4\}$, with $\lambda = 2$, and $\{1,3\}$, $\{2,3\}$ and their mirror images, with $\lambda = \frac{1}{2}(4+\varepsilon-\sqrt{\varepsilon^2+4}) > 1$. The connected faces of three states are A, B and the sets $\{a,3,4\}$ and $\{3,4,b\}$ with $a \in \{1,2\}$ and $b \in \{5,6\}$, because a connected face that meets both triangles contains 3 and 4. In each of these four sets every state has rate at least 1 out of the set, so $\lambda \geq 1 > \lambda_A$. A connected face of four states meets both triangles, so it contains 3 and 4. So it is $\{1,2,3,4\}$, its mirror image $\{3,4,5,6\}$, or a set $\{a,3,4,b\}$, in which every state has rate 1 out of the set, so that $\lambda \geq 1$. A connected face of five states is the complement of an inner state (the complements of 3 and of 4 are disconnected), and these four faces are images of $\{2,\dots,6\}$ under the symmetries. By the third fact above, $\lambda_5 < \lambda_{\{3,4,5,6\}} = \lambda_4 < \lambda_A < 1$, since $\{2,\dots,6\} \supsetneq \{3,4,5,6\}$ and $\{1,2,3,4\} \supsetneq A$.

For the polynomials, note that $(1,-1,0,0)$ is an eigenvector of $-Q_F$ for $F = \{1,2,3,4\}$ with eigenvalue 3. On the vectors (a,a,b,c) the matrix acts as $(a,b,c) \mapsto (a-b,\ -2a+(2+\varepsilon)b-\varepsilon c,\ -\varepsilon b+(2+\varepsilon)c)$, with characteristic polynomial $(x-1)(x-2)(x-2-2\varepsilon)-2(x-2-\varepsilon) = p_4(x)$. So the eigenvalues of $-Q_F$ are 3 and the roots of p_4. As $\lambda_4 < 1$, it is the least root of p_4. Now let $F = \{2,\dots,6\}$. The vector that is 1 at 5, -1 at 6 and 0 elsewhere is an eigenvector of $-Q_F$ with eigenvalue 3. On the vectors with the value a at 2, b at 3, c at 4 and e at both 5 and 6, the matrix acts as

$$(a,b,c,e) \mapsto \big(2a-b,\ -a+(2+\varepsilon)b-\varepsilon c,\ -\varepsilon b+(2+\varepsilon)c-2e,\ -c+e\big).$$

Expanding the determinant along the first row gives the characteristic polynomial

$$(x-2)\big[(x-1)\big((x-2-\varepsilon)^2-\varepsilon^2\big)-2(x-2-\varepsilon)\big]-\big[(x-2-\varepsilon)(x-1)-2\big] = (x-3)\,p_5(x).$$

So the eigenvalues of $-Q_F$ are 3 (twice) and the roots of p_5. As $\lambda_5 < 1$, it is the least root of p_5. (Both characteristic polynomials are also checked by computer algebra, see Section C.3.)

(c) Put $D_1 = \lambda_5 - \frac{1}{2}\lambda_4$ and $D_2 = \lambda_4 - \frac{2}{3}\lambda_A$. Both are continuous in $\varepsilon > 0$, because the least eigenvalue of a symmetric matrix is continuous in its entries. If $D_1 = 0$, then $x = \lambda_5$ is a common root of $p_5(x)$ and $p_4(2x)$. If $D_2 = 0$, then $x = \lambda_4$ is a common root of $p_4(x)$ and $p_A(\frac{3}{2}x)$. A

common root makes the resultant vanish, and an exact computation gives

$$\begin{aligned}\operatorname{Res}_x\big(p_5(x), p_4(2x)\big) &= -8\varepsilon^3(16\varepsilon^2 + 12\varepsilon + 9),\\ \operatorname{Res}_x\big(p_A(\tfrac{3}{2}x), p_4(x)\big) &= -\tfrac{1}{16}\varepsilon^2(120\varepsilon^2 - 256\varepsilon + 135).\end{aligned}$$

So D_1 has no zero in $\varepsilon > 0$, and D_2 can vanish only at ε_0 and at ε_1. Hence D_1 has one sign on $(0,\infty)$, and D_2 has one sign on each of the intervals $(0,\varepsilon_0)$, $(\varepsilon_0,\varepsilon_1)$ and (ε_1,∞). A zero of the resultant does not force D_2 to vanish, because the common root need not be the least root of p_4. So we find the sign of D_2 for small ε, for large ε and at ε_0.

First let ε be small. By (a) and (b), $0 < \lambda_5 < \lambda_4 < \lambda_A < \varepsilon$. Here $\lambda_5 > 0$ because $\lambda_5 \geq 0$ by (27) and $p_5(0) = -\varepsilon \neq 0$, and $\lambda_A < \varepsilon$ because $p_A(\varepsilon) = -2\varepsilon < 0$. At $\varepsilon = 0$ the cubics are $x(x-2)(x-3)$ and $x(x-1)(x-3)$, with the simple root 0. So the implicit function theorem gives one root $\frac{\varepsilon}{3} + O(\varepsilon^2)$ of each cubic near 0, and no other root near 0. So $\lambda_4 = \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\lambda_5 = \frac{\varepsilon}{3} + O(\varepsilon^2)$, and (a) gives $\lambda_A = \frac{\varepsilon}{3} + O(\varepsilon^2)$. Hence $D_1 = \frac{\varepsilon}{6} + O(\varepsilon^2)$ and $D_2 = \frac{\varepsilon}{9} + O(\varepsilon^2)$ are positive for small ε. So $D_1 > 0$ for every $\varepsilon > 0$, and $D_2 > 0$ on $(0,\varepsilon_0)$.

Next, $D_2 < 0$ for large ε. Indeed $\lambda_A \to 1$ as $\varepsilon \to \infty$ by (a). On the other hand take $f = (a,a,c,c)$ in (27) for $\{1,2,3,4\}$. The value is $2(a-c)^2 + 2c^2$ and $|f|^2 = 2a^2 + 2c^2$, so $\lambda_4 \leq \min_{a,c}((a-c)^2 + c^2)/(a^2 + c^2) = (3 - \sqrt{5})/2 < \frac{1}{2}$ for every ε.

Now we show that $D_2 > 0$ at ε_0. Put $x_0 = \frac{7}{40}$. Then

$$p_4(x_0) = \frac{57743}{64000} - \frac{809}{800}\varepsilon, \qquad p_A(\tfrac{3}{2}x_0) = \frac{59}{80}\varepsilon - \frac{4599}{6400}$$

are both negative for $0.8923 < \varepsilon < 0.9743$, and this interval contains ε_0. For $0 \leq x \leq \frac{1}{2}$ we have $p_4'(x) = 3x^2 - 2(5+2\varepsilon)x + 6 + 6\varepsilon \geq 1 + 4\varepsilon > 0$. So p_4 increases on $[0, x_0]$ and is negative there, and its least root λ_4 is larger than x_0. Also $p_A(\frac{3}{2}x_0) < 0$ puts $\frac{3}{2}x_0$ between the roots of p_A, so $\lambda_A < \frac{3}{2}x_0$. Hence $\lambda_4 > x_0 > \frac{2}{3}\lambda_A$ at ε_0.

So $D_2 > 0$ on $(0,\varepsilon_0]$. It has no zero in $(\varepsilon_0,\varepsilon_1)$, so $D_2 > 0$ on $(0,\varepsilon_1)$. It has no zero in (ε_1,∞) and is negative for large ε, so $D_2 < 0$ on (ε_1,∞). By continuity $D_2 = 0$ at ε_1. This proves (c).

(d) As $p_A(\frac{3}{4}) = \frac{\varepsilon}{4} - \frac{27}{16} < 0$ for $\varepsilon < \frac{27}{4}$, we have $\lambda_A < \frac{3}{4}$. So $1 - \lambda_A > 0$ and $\lambda_A < 3(1-\lambda_A)$, that is $1 < \tau_A < \tau^*$. Apply Lemma C.1 to the points $(2,0)$, $(1,1)$, $(\lambda_A, 2)$ and $(0,5)$ of the inner states, the inner edges, the triangles and [6]. The switch values are 1, τ_A and τ^*. So on each interval in (d) the faces named there have a smaller exponent than the other three kinds, and by (b) than every other connected face of at most three states. Let F be a connected face of four or five states. By (b) and (c), for every $\tau > 0$

$$\begin{aligned}\Phi_\tau(F) &\geq \tau\lambda_4 + 3 > \tfrac{2}{3}(\tau\lambda_A + 2) + \tfrac{1}{3}\cdot 5 && \text{if } |F| = 4,\\ \Phi_\tau(F) &= \tau\lambda_5 + 4 > \tfrac{1}{3}(\tau\lambda_A + 2) + \tfrac{2}{3}\cdot 5 && \text{if } |F| = 5,\end{aligned}$$

where the second line uses $\lambda_5 > \frac{1}{2}\lambda_4 > \frac{1}{3}\lambda_A$. Each right side is at least the smaller of the exponents $\tau\lambda_A + 2$ of a triangle and 5 of [6]. So F never wins, and this gives the winners. The closed forms follow from (a) by rationalizing, since $(\varepsilon^2 + 2\varepsilon + 9) - (1+\varepsilon)^2 = 8$ and $(3+\varepsilon)^2 - (\varepsilon^2 + 2\varepsilon + 9) = 4\varepsilon$. The expansions follow from $\sqrt{\varepsilon^2 + 2\varepsilon + 9} = 3 + \frac{\varepsilon}{3} + O(\varepsilon^2)$.

(e) At $\tau = 3/\lambda_A$ the triangles and [6] have the exponent 5, and $\{1,2,3,4\}$ has $3\lambda_4/\lambda_A + 3 < 5$ by (c). □

At $\varepsilon = 1$, for example, $\tau_A = (1+\sqrt{3})/2$ and $\tau^* = 6 + 3\sqrt{3}$.

C.3 What was checked by computation

The proofs above use computer algebra in one place only, namely the two resultants in the proof of Proposition C.4(c). The script `verify_examples.py` computes them in sympy. It also expands the characteristic polynomials of $-Q_F$ for $F = A$, $\{1,2,3,4\}$ and $\{2,\dots,6\}$ and finds $(x-3)p_A$, $(x-3)p_4$ and $(x-3)^2p_5$, and it recomputes $p_4(\frac{7}{40})$ and $p_A(\frac{21}{80})$ in rational arithmetic. The rest of its checks for these two rows are in floating point, and no proof rests on them. For $3 \le d \le 9$ it computes the exit rates of all connected faces of P_d, compares those of the end-arcs with λ'_k, and finds the sizes of the winners as in Proposition C.3(c). With 50 digits it confirms $\frac{3}{2}k - \frac{1}{2} < D'_k < \frac{3}{2}k$ for $2 \le k \le 2000$. For $\varepsilon = 0.3$, 1 and 1.1 it computes the exit rates of all 63 faces of Θ_ε and finds the least exit rates, the winners and the switch values of Proposition C.4. For $\varepsilon = 1.5$, 3, 4.4, 4.6 and 8 it finds that the faces $\{1,2,3,4\}$ and $\{3,4,5,6\}$ win on an interval of τ, which agrees with Proposition C.4(e).

D The large-concentration limit of the direct jump

We prove Proposition 6.4(b), with B_1, B_2, B_3 independent Beta(a,a) as in part (a) and $a = \omega/3 \ge 1$.

Idea of the proof. We pass to the logits ℓ_j of the B_j. In these coordinates the three conditions that define D say that three linear forms M_1, M_2, M_3 of ℓ, whose sum is 0, are bounded above by quantities of second order in ℓ. For large a the law of ℓ concentrates near 0 on the scale $a^{-1/2}$. There the bounds are small, so on D all three forms are nearly 0. The points where they vanish form the line of the cyclic kernels, so D is a thin tube around this line. Let t be the coordinate along the line. At the point t of the line each of the three bounds is about $\frac{1}{2}t^2$, so the section of D at t is close to the triangle $M_1, M_2, M_3 \le \frac{1}{2}t^2$, which has area $\frac{9}{8}t^4$. So we expect $\mathbb{P}_\omega(D) \approx a^{3/2}\int t^4 e^{-cat^2}dt \approx 1/a$, and a Gaussian integral gives the constant.

Before the proof we set up three things, namely the logits and the form of D in them, two bounds on the function that appears, and coordinates along and across the cyclic kernels.

Logits. Put $\ell_j = \log(B_j/(1-B_j))$, $g(y) = 2\log\cosh(y/2)$ and $g_j = g(\ell_j)$. Since $B_j = e^{\ell_j/2}/(2\cosh(\ell_j/2))$ and $1 - B_j = e^{-\ell_j/2}/(2\cosh(\ell_j/2))$, we have

$$\log(2B_j) = \tfrac{1}{2}(\ell_j - g_j), \qquad \log\bigl(2(1-B_j)\bigr) = -\tfrac{1}{2}(\ell_j + g_j), \qquad B_j(1-B_j) = \tfrac{1}{4}e^{-g_j}.$$

The function g is even and convex with $g(0) = g'(0) = 0$, so $g \ge 0$. The density of B_j is proportional to $(B_j(1-B_j))^{a-1}$, and $dB_j/d\ell_j = B_j(1-B_j)$. So ℓ_j has the density $e^{-ag(\ell_j)}/Z$ with $Z = 4^a\Gamma(a)^2/\Gamma(2a) = 2\sqrt{\pi}\,\Gamma(a)/\Gamma(a+\frac{1}{2})$, where the second form is the duplication formula for Γ.

By the proof of Proposition 6.4, D is the event that B_1B_2, $(1-B_1)B_3$ and $(1-B_2)(1-B_3)$ are all at most $\frac{1}{4}$. We multiply each product by 4 and take logarithms with the display above. For example the first condition becomes $\frac{1}{2}(\ell_1 - g_1) + \frac{1}{2}(\ell_2 - g_2) \le 0$. So D is the set where

$$M_1 \le g_1 + g_2, \qquad M_2 \le g_1 + g_3, \qquad M_3 \le g_2 + g_3, \tag{29}$$

with $M_1 = \ell_1 + \ell_2$, $M_2 = \ell_3 - \ell_1$ and $M_3 = -\ell_2 - \ell_3 = -M_1 - M_2$. Let $\psi = g_1 + g_2 + g_3$ (in this appendix ψ is not the Bessel function of Section 2.1). Then the joint density of $\ell = (\ell_1, \ell_2, \ell_3)$ is $Z^{-3}e^{-a\psi(\ell)}$, and

$$\mathbb{P}_\omega(D) = Z^{-3}\int_D e^{-a\psi(\ell)}d\ell. \tag{30}$$

Two bounds on g. For real y,

$$\frac{y^2}{4} - \frac{y^4}{96} \le g(y) \le \frac{y^2}{4}, \tag{g1}$$

$$g(y) \ge \frac{1}{10}\min(y^2, |y|). \tag{g2}$$

The first says that $g(y)$ is close to $y^2/4$ near 0, and we use it near the cyclic kernels. The second is a rough lower bound that we use far from 0. For (g1) put $x = y/2$, so that $g(y) = 2\log\cosh x$. The upper bound is $\cosh x \le e^{x^2/2}$, which holds term by term in the two power series. The lower bound says that $\chi(x) = \log\cosh x - \frac{x^2}{2} + \frac{x^4}{12}$ is nonnegative, and this holds as $\chi(0) = \chi'(0) = 0$ and $\chi'' = x^2 - \tanh^2 x \ge 0$. For (g2), $g'' = \frac{1}{2}\mathrm{sech}^2(y/2) > 0.2$ on $|y| \le 2$, so $g(y) \ge 0.1y^2$ there. For $|y| \ge 2$ convexity and $g(0) = 0$ give $g(y) \ge \frac{1}{2}|y|g(2) > 0.43|y|$.

Coordinates along and across the cyclic kernels. A cyclic kernel has $B_1 = 1 - B_2 = B_3$, so its logits are $\ell = tv$ with $v = (1, -1, 1)$ and $t \in \mathbb{R}$, and $M_1 = M_2 = M_3 = 0$ there. For general ℓ let $t = \langle \ell, v\rangle/3$. We use (t, M_1, M_2) as coordinates. The map $\ell \mapsto (t, M_1, M_2)$ is linear with determinant 1, and its inverse is $\ell = tv + w$ with $w = \frac{1}{3}(M_1 - M_2, 2M_1 + M_2, M_1 + 2M_2) \perp v$. So t measures the position along the line and w the displacement from it. Since $|v|^2 = 3$, $|\ell|^2 = 3t^2 + |w|^2$, and a direct computation gives $|w|^2 = \frac{1}{3}(M_1^2 + M_2^2 + M_3^2) = \frac{2}{3}(M_1^2 + M_1M_2 + M_2^2)$. Write $|M|$ and $|M|_\infty$ for the Euclidean norm and the maximum norm of (M_1, M_2). As $|M_1M_2| \le \frac{1}{2}|M|^2$,

$$\tfrac{1}{3}|M|^2 \le |w|^2 \le |M|^2 \le 2|M|_\infty^2. \tag{31}$$

Proof of Proposition 6.4(b). We compute the integral in (30) in four steps. Step 1 shows that on D the forms M_i are of second order in ℓ. Step 2 removes the part of D away from 0. Step 3 finds the section of D at t, and Step 4 integrates over t.

Step 1: an a priori bound. On D, each right side in (29) is at most ψ, so $M_1 \le \psi$, and $M_1 = -M_2 - M_3 \ge -2\psi$. The same holds for M_2. So by (g1), which gives $\psi \le \frac{1}{4}|\ell|^2$,

$$|M|_\infty \le 2\psi(\ell) \le \tfrac{1}{2}|\ell|^2 = \tfrac{1}{2}(3t^2 + |w|^2). \tag{32}$$

Step 2: localization. Put $\varepsilon = 1/12$ and let U be the set where $|t| \le \varepsilon$ and $|M|_\infty \le \varepsilon$. We show that the integral over $D \setminus U$ is exponentially small in a. Off U we have $|t| > \varepsilon$ or $|M| > \varepsilon$, so by (31) $|\ell| \ge \max(\sqrt{3}|t|, |M|/\sqrt{3}) \ge \varepsilon/\sqrt{3}$. Also some coordinate has $|\ell_j| \ge |\ell|/\sqrt{3}$. So (g2) gives

$$\psi \ge \gamma := \tfrac{1}{10}\min\big(|\ell|^2/3,\ |\ell|/\sqrt{3}\big) \ge \gamma_0 := \varepsilon^2/90.$$

Hence $a\psi \ge (a-1)\gamma_0 + \gamma$, that is, $e^{-a\psi} \le e^{-(a-1)\gamma_0}e^{-\gamma}$ off U. The function $e^{-\gamma(\ell)}$ is integrable over $\mathbb{R}^3$, so $\int_{D\setminus U} e^{-a\psi} = O(e^{-\gamma_0 a})$.

Step 3: the section of D at t. We show that the section of $D \cap U$ at t lies between two triangles of area close to $\frac{9}{8}t^4$. First, on $D \cap U$ the bounds (32) and (31) give

$$|M|_\infty \le \tfrac{3}{2}t^2 + \tfrac{1}{2}|w|^2 \le \tfrac{3}{2}t^2 + |M|_\infty^2 \le \tfrac{3}{2}t^2 + \varepsilon|M|_\infty,$$

so $|M|_\infty \le \frac{3}{2}t^2/(1-\varepsilon) \le 2t^2$. So $D \cap U$ lies in the set $E = \{|t| \le \varepsilon,\ |M|_\infty \le 2t^2\}$.

Next we estimate g_j and ψ on E. There $|w|^2 \le 8t^4$ by (31), and $\ell_j = \pm t + w_j$ with $|w_j| \le |w|$. So $|\ell_j| \le 2|t|$ and $|\ell_j^2 - t^2| \le 7|t|^3$. Then (g1) gives $|g_j - \frac{1}{4}t^2| \le 2|t|^3$, and with $|\ell|^2 = 3t^2 + |w|^2$ it bounds ψ. Hence on E

$$\big|g_i + g_j - \tfrac{1}{2}t^2\big| \le 4|t|^3 \quad (i \ne j), \qquad \big|\psi - \tfrac{3}{4}t^2\big| \le 2t^4. \tag{33}$$

So on E each right side of (29) is $\frac{1}{2}t^2$ up to $4|t|^3$. Now let $\mathcal{T}(\delta) = \{M : M_1, M_2 \le \delta,\ M_1+M_2 \ge -\delta\}$. This is the set where $M_1, M_2, M_3 \le \delta$. It is a triangle with corners (δ,δ), $(\delta,-2\delta)$ and $(-2\delta,\delta)$, so it has area $\frac{9}{2}\delta^2$ and lies in $\{|M|_\infty \le 2\delta\}$. Let D_t be the section of $D \cap U$ at t and put $\delta_\pm = \frac{1}{2}t^2(1 \pm 8|t|) \ge 0$, which are $\frac{1}{2}t^2 \pm 4|t|^3$. If $M \in D_t$, then $(t, M) \in E$, and (29) and (33) give $M_1, M_2, M_3 \le \delta_+$. Conversely, if $M \in \mathcal{T}(\delta_-)$, then $|M|_\infty \le 2\delta_- \le \min(t^2, \varepsilon)$, so $(t, M) \in E \cap U$, and (33) gives (29). So $\mathcal{T}(\delta_-) \subseteq D_t \subseteq \mathcal{T}(\delta_+)$, and

$$\tfrac{9}{8}t^4(1-8|t|)^2 \le |D_t| \le \tfrac{9}{8}t^4(1+8|t|)^2. \tag{34}$$

Step 4: integration. Let $J = \int_{D\cap U} e^{-a\psi}$. We integrate first over the section and then over t (the change of coordinates has Jacobian 1). By (33) and (34), J lies between the integrals over $|t| \le \varepsilon$ of $\frac{9}{8}t^4(1 \mp 8|t|)^2 e^{-a(\frac{3}{4}t^2 \pm 2t^4)}$. We substitute $t = y/\sqrt{a}$. Then these are $a^{-5/2}$ times integrals over $|y| \le \varepsilon\sqrt{a}$ whose integrands tend to $\frac{9}{8}y^4 e^{-3y^2/4}$ as $a \to \infty$ and are at most $\frac{9}{2}y^4 e^{-y^2/2}$, since $8|t| < 1$ and $2y^4/a = 2t^2y^2 \le 2\varepsilon^2 y^2$. By dominated convergence and $\int_{\mathbb{R}} y^4 e^{-3y^2/4} dy = \frac{3}{4}\sqrt{\pi}(\frac{4}{3})^{5/2}$,

$$J = a^{-5/2}\big(\sqrt{3\pi} + o(1)\big).$$

By Step 2 the same holds for the integral over D. It remains to find Z. As Γ is log-convex, $\Gamma(a+\frac{1}{2})^2 \le a\Gamma(a)^2$ and $\Gamma(a+1)^2 \le (a+\frac{1}{2})\Gamma(a+\frac{1}{2})^2$, so $Z = 2\sqrt{\pi/a}\,(1 + O(1/a))$. Then (30) gives

$$\mathbb{P}_\omega(D) = \frac{a^{3/2}}{8\pi^{3/2}} \cdot a^{-5/2}\sqrt{3\pi}\,(1+o(1)) = \frac{\sqrt{3}}{8\pi a}(1+o(1)) = \frac{3\sqrt{3}}{8\pi\omega}(1+o(1)). \qquad \square$$

Statements and declarations

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Data and code availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of `https://github.com/apemm/Kagey-Problems`, at the release `paperC-arxiv-v1`. The scripts are in the subfolder `paperC` and need numpy, scipy, sympy and mpmath. To rerun the checks, run `python -B verify_all.py --paper c` in the folder `problem131`. The largest cases run only with the option `--full`. Section 7 describes what each check does.

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